

ON THE INDUCTIVE BLOCKWISE ALPERIN WEIGHT CONDITION FOR TYPE B AND TYPE C

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ABSTRACT. The purpose of this paper is to prove that every finite simple group of type B or C satisfies the inductive blockwise Alperin weight condition at every prime ℓ dividing its order. For $\mathrm{PSp}_{2n}(q)$, where q is odd and $n \geq 3$, a permutation lattice argument based on Conlon's induction theorem removes the unitriangularity assumption from earlier results on the inductive condition at odd nondefining primes. At $\ell = 2$ in odd defining characteristic, errors and omissions in the underlying classification of radical 2-subgroups affect an earlier parametrisation of weights for arbitrary blocks. We verify the required weight parametrisation for principal blocks and use Jordan reduction to establish the inductive condition for all blocks. At odd ℓ in even defining characteristic and rank at least four, we prove the inductive condition using generic weights and Jordan reduction, without assuming unitriangularity. For $\Omega_{2n+1}(q)$, where q is odd and $n \geq 3$, we remove the unitriangularity assumption at odd nondefining primes using Conlon's induction theorem. At $\ell = 2$, we prove the stabiliser and extension property for Brauer characters of $\mathrm{Spin}_{2n+1}(q)$ that was assumed in earlier work on the inductive condition. We also give proofs of the inductive condition for the remaining sporadic groups J_4 , F'_{24} , the Baby Monster and the Monster, without making a priority claim. Together with the previously established cases, these results imply that the blockwise Alperin weight conjecture holds at ℓ for every finite group each of whose nonabelian simple sections of order divisible by ℓ is of type B, of type C, or sporadic.

1. INTRODUCTION

Let b be an ℓ -block of a finite group G . Alperin's weight conjecture asserts that the number of isomorphism classes of simple modules in b equals the number of G -conjugacy classes of weights belonging to b [1]. The inductive blockwise Alperin weight condition (*iBAW condition*), introduced by Späth [65, Definition 4.1], refines this numerical equality by requiring compatible equivariant bijections. The purpose of this paper is to prove the inductive blockwise Alperin weight condition for the finite simple groups of types B and C.

Combining the results proved here with the previously established cases, we obtain the following theorem.

Theorem 1.1. *Let S be a nonabelian finite simple group of type B, of type C, or sporadic. Then S satisfies the inductive blockwise Alperin weight condition at every prime dividing its order.*

Let $q = p^f$ be odd and let ℓ be an odd prime not dividing q . Feng, Li, and Zhang [30, Theorem 1.5] proved the inductive condition for unipotent blocks of $\mathrm{Sp}_{2n}(q)$ and $\mathrm{Spin}_{2n+1}(q)$, where $n \geq 2$. When f is odd and ℓ is *linear*, meaning that the multiplicative order of q modulo ℓ is odd, they also proved the inductive condition for $\mathrm{PSp}_{2n}(q)$ and for the blocks of $\mathrm{Spin}_{2n+1}(q)$ that dominate blocks of $\Omega_{2n+1}(q)$ [30, Theorems 1.2 and 1.4]. Li's result for type C [52, Theorem 2] and Feng–Li–Zhang's result for type B [28, Theorem 1] treat arbitrary blocks under unitriangularity assumptions. We remove these assumptions in rank at least three. Integral basic sets for the conformal symplectic and finite special Clifford groups identify the permutation lattices spanned by these basic sets with those spanned by the irreducible Brauer characters of the corresponding blocks. Conlon's induction theorem [25], applied at the prime

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2, then yields equivariant bijections between irreducible Brauer characters and the ordinary characters in these basic sets. Applying the same argument to the symplectic and spin groups gives the same stabiliser factorisations for Brauer characters as for ordinary characters, allowing us to verify the inductive condition.

For $\mathrm{Sp}_{2n}(q)$ with even q and odd ℓ , Li [52, Section 6] obtains the required descriptions of blocks and weights by adapting arguments for odd characteristic of An [3] and Fong–Srinivasan [37]. Li proves the inductive condition under a unitriangularity assumption [52, Theorem 3]. For $n \geq 4$, we give an independent proof of the inductive condition without assuming unitriangularity, using Geck’s basic set theorem [39] and the generic weight correspondences of Feng, Malle, and Zhang [34]. These give bijections between the irreducible Brauer characters and weights of unipotent blocks. The resulting bijections are equivariant because field automorphisms fix the unipotent characters and the conjugacy classes of generic weights involved, as explained in Lemma 3.6. After verifying the remaining parts of the inductive condition for unipotent blocks, we obtain the result for all blocks by Jordan reduction [31].

For odd q at the prime 2, we give an alternative proof of Feng and Malle’s result for $\mathrm{PSp}_{2n}(q)$ [33, Theorem 1]. The classification of radical 2-subgroups [2, p. 190] used in Feng and Malle’s weight parametrisation omits a family with even multiplicity, as noted by Feng, Yu, and Zhang [36, Remark 3.36]. Using the corrected classification, we verify the principal block weight calculation needed for Feng and Malle’s argument and obtain the result for all blocks by Jordan reduction [31].

For type B at the prime 2, Feng, Yu, and Zhang’s proof assumes a Brauer character stabiliser and extension property [36, Theorem 1 and Assumption 5.1]. We establish this property for $\mathrm{Spin}_{2n+1}(q)$, where q is odd and $n \geq 3$. For the principal block, we use generalised Gelfand–Graev characters to prove that field automorphisms fix every irreducible Brauer character. For nonprincipal blocks, we obtain the required stabiliser factorisations and extensions through the equivariant Jordan correspondence and proper Levi subgroups. Jordan reduction then gives the inductive condition for $\Omega_{2n+1}(q)$ when $(n, q) \neq (3, 3)$. We treat the exceptional covering group $3.\Omega_7(3)$ separately by explicit calculations.

We also give proofs of the inductive condition for J_4 , Fi'_{24} , $\mathbb{B}$ and $\mathbb{M}$. The proofs combine published weight counts with character table calculations and, for Fi'_{24} , a comparison of the actions of the outer automorphism on Brauer characters and weights. As of 19 September 2026, the website of the GAP package CTBlocks [13] still listed the verification for these four groups as incomplete. In a 2023 paper, Feng, Li, and Zhang cite Breuer and Feng’s work on sporadic simple groups as *in preparation* [32, p. 6518]. We have not found a publicly available manuscript of that project and therefore cannot compare its scope with the present work. Accordingly, we make no priority claim for the sporadic theorem.

Späth’s reduction theorem [65, Theorem A] gives the following consequence. A group K is said to be *involved* in a finite group H if K is isomorphic to a section of H .

Corollary 1.2. *Let H be a finite group and let ℓ be a prime. Suppose that every nonabelian simple group involved in H whose order is divisible by ℓ is of type B, of type C, or sporadic. Then the blockwise Alperin weight conjecture holds for H at ℓ .*

2. PRELIMINARIES

For modular representations at a prime ℓ , we use a splitting ℓ -modular system for the finite groups under consideration, with residue field k . If b is an ℓ -block, then $\mathrm{Irr}(b)$ and $\mathrm{IBr}(b)$ denote its sets of ordinary irreducible characters and irreducible Brauer characters, respectively. For an ordinary irreducible character or irreducible Brauer character ψ , let $\mathrm{bl}(\psi)$ denote the block containing ψ , and let $\mathrm{dz}(G)$ denote the set of ordinary irreducible characters of G of defect zero, namely those whose degrees have the same ℓ -part as $|G|$. For $\chi \in \mathrm{Irr}(G)$, χ^0 denotes its restriction to the ℓ -regular elements, namely the elements of order prime to ℓ . The *decomposition*

map $d_G : \mathbb{Z}[\text{Irr}(G)] \rightarrow \mathbb{Z}[\text{IBr}(G)]$ is the $\mathbb{Z}$ -linear map determined by $d_G(\chi) = \chi^0$. For an ℓ -block b of G , write d_b for the restriction of d_G from $\mathbb{Z}[\text{Irr}(b)]$ to $\mathbb{Z}[\text{IBr}(b)]$. We write automorphisms on the right. For a group H , a subgroup $Q \leq H$, and $\alpha \in \text{Aut}(H)$, set $Q^\alpha = \alpha^{-1}(Q)$. If Q is finite and χ is a class function of Q , define the class function χ^α of Q^α by $\chi^\alpha(h) = \chi(\alpha(h))$ for $h \in Q^\alpha$. For Brauer characters, this formula is evaluated on the ℓ -regular elements of Q^α . If a group A acts on sets $\Omega_1, \dots, \Omega_m$ and $x_i \in \Omega_i$, we write A_{x_i} for the stabiliser of x_i and $A_{x_1, \dots, x_m} = A_{x_1} \cap \dots \cap A_{x_m}$. For a finite group G , let $O_\ell(G)$ denote its ℓ -core, namely its largest normal ℓ -subgroup. A subgroup $Q \leq G$ is ℓ -radical if $Q = O_\ell(N_G(Q))$, and $\text{Rad}_\ell(G)$ denotes the set of ℓ -radical subgroups of G . An ℓ -weight of G is a pair (Q, φ) , where $Q \in \text{Rad}_\ell(G)$ and $\varphi \in \text{Irr}(N_G(Q)/Q)$ has defect zero. We call φ the *weight character* and, when convenient, identify it with its inflation to $N_G(Q)$. The weight *belongs to* an ℓ -block b of G if the block of $N_G(Q)$ containing the inflated weight character induces to b . We write $\text{Alp}(b)$ for the set of G -conjugacy classes of weights belonging to b , and $\text{Alp}(G)$ for the set of G -conjugacy classes of all ℓ -weights of G . Blocks and weights carry the induced right actions of automorphisms. In particular, $(Q, \varphi)^\alpha = (Q^\alpha, \varphi^\alpha)$.

Let S be nonabelian simple and let $\widehat{S}$ be its *universal central extension*. For a prime ℓ , set $X = X_\ell(S) = \widehat{S}/O_\ell(Z(\widehat{S}))$ and $Z = Z(X)$. The group X is the *universal ℓ' -covering group* used in Sp  th's inductive condition [65, Definition 4.1]. Its centre Z is an ℓ' -group and $X/Z \cong S$. The quotient map $X \rightarrow S$ induces an isomorphism $\text{Aut}(X) \rightarrow \text{Aut}(S)$ [65, p. 184], and consequently $\text{Out}(X) \cong \text{Out}(S)$.

Since Z is an ℓ' -group, the decomposition map d_Z identifies $\text{Irr}(Z)$ with $\text{IBr}(Z)$. We use *sector* as shorthand in the calculations for covering groups, where we compare Brauer character and weight counts over each central character and describe the action of outer automorphisms.

Definition 2.1. For $\nu \in \text{Irr}(Z)$, an ordinary irreducible or irreducible Brauer character ψ of X *lies over* ν if its restriction to Z is a positive integral multiple of ν . A weight (Q, θ) lies over ν if the inflation of θ to $N_X(Q)$ does. We say that such characters and X -conjugacy classes of weights belong to the ν -*sector*. The sector is called *trivial* when $\nu = 1_Z$, and *faithful* when ν is faithful.

Under the action defined above, $\alpha \in \text{Aut}(X)$ sends the ν -sector onto the ν^α -sector. Since Z is a central ℓ' -subgroup, each block b lies over a unique $\nu \in \text{Irr}(Z)$: its block idempotent lies under exactly one primitive idempotent of kZ . All ordinary and Brauer characters in b lie over ν . Block induction preserves this central character [65, Definition 2.1], so every weight belonging to b also lies over ν . We say that b lies in the ν -sector.

Lemma 2.2. *Let $N \triangleleft G$ be finite groups, let b be an ℓ -block of G , and let C be the group of ordinary linear characters of G/N of ℓ' -order, acting on blocks by tensoring with their inflations. Then every character in C_b factors through $G/NZ(G)$, and $|C_b| \leq |G : NZ(G)|$. The same conclusions hold when C is identified by reduction with the group of linear Brauer characters of G/N .*

Proof. Let $Z_{\ell'}$ be the subgroup of $Z(G)$ consisting of its ℓ -regular elements. Since $Z_{\ell'}$ is a central ℓ' -subgroup, all characters in $\text{IBr}(b)$ lie over the same character $\nu \in \text{IBr}(Z_{\ell'})$. For $\lambda \in C_b$, tensoring preserves b , so $(\lambda^0|_{Z_{\ell'}})\nu = \nu$. Thus λ is trivial on $Z_{\ell'}$. Its ℓ' -order also makes it trivial on $O_\ell(Z(G))$, and hence on $Z(G)$. Since λ is already trivial on N , it factors through $G/NZ(G)$. Thus $|C_b| \leq |G : NZ(G)|$. Reduction identifies ordinary linear characters of ℓ' -order with linear Brauer characters and commutes with tensoring, so the same argument applies to their action on blocks. $\square$

In Sp  th's iBAW condition [65, Definition 4.1(i)–(iv)], the local bijections give an $\text{Aut}(X)_b$ -equivariant bijection $\text{IBr}(b) \rightarrow \text{Alp}(b)$ for each ℓ -block b of X . The condition also requires compatibility with central characters, character extensions and blocks of intermediate groups.

For weights with subgroup $Q = 1$, the character χ^0 is matched with $\chi \in \text{dz}(X)$, with identical global and local extensions. We say that S satisfies iBAW at ℓ when the full condition holds.

Let $G \triangleleft A$ be finite groups, and let b be an A -stable ℓ -block of G . An A -equivariant bijection $\Omega : \text{IBr}(b) \rightarrow \text{Alp}(b)$ is an *iBAW bijection compatible with A* if, for every $\chi \in \text{IBr}(b)$ and every representative (Q, φ) of $\Omega(\chi)$,

$$(A_\chi, G, \chi) \succeq_b (N_A(Q)_\varphi, N_G(Q), \varphi^0).$$

Here $\succeq_b$ denotes block isomorphism of modular character triples [57, Definition 3.4], and φ^0 is the Brauer reduction of the weight character, inflated to $N_G(Q)$. Taking $A = G \rtimes \text{Aut}(G)_b$ gives an *iBAW bijection* for b in the sense of [31, Definition 3.5]. For the block form of the inductive BAW condition, let b be an ℓ -block of $X = X_\ell(S)$ with central character ν , and let $\bar{b}$ be the block it dominates on $X/\ker(\nu)$. The characters in $\text{IBr}(\bar{b})$ are faithful on the centre of this quotient. Following [50, Definition 3.2], we say that b satisfies the inductive BAW condition when $\bar{b}$ does.

We recall Sp  th's character triple formulation of the extension and block compatibility requirements in the inductive BAW condition.

Theorem 2.3 (Sp  th [66, Theorem 4.4]). *Let $\widehat{G}$ be the universal covering group of a nonabelian simple group S , and let ℓ be a prime. Let $\widehat{\theta} \in \text{IBr}(\widehat{G})$, $\widehat{Q} \in \text{Rad}_\ell(\widehat{G})$ and $\widehat{M} = N_{\widehat{G}}(\widehat{Q})$. Suppose that $\widehat{\theta}'$ is the inflation to $\widehat{M}$ of a character in $\text{dz}(\widehat{M}/\widehat{Q})$ and that $\text{bl}(\widehat{\theta}')^{\widehat{G}} = \text{bl}(\widehat{\theta})$. Then the following are equivalent.*

- (i) *The characters $\widehat{\theta}$ and $\widehat{\theta}'$ satisfy part (iii) of the inductive blockwise Alperin weight condition in [65, Definition 4.1].*
- (ii) *The modular character triples satisfy*

$$(2.1) \quad (G \rtimes \text{Aut}(G)_{\widehat{\theta}}, G, \widehat{\theta}) \succ_{F,b} (M(G \rtimes \text{Aut}(G))_{\widehat{M}, \widehat{\theta}'}, M, (\widehat{\theta}')^0),$$

where $G = \widehat{G}/\widehat{Z}$ and $M = \widehat{M}/\widehat{Z}$ for $\widehat{Z} = \ker(\widehat{\theta}_{Z(\widehat{G})})$.

Here F is an algebraically closed field of characteristic ℓ . Sp  th writes $\succ_{F,b}$ for block isomorphism of modular character triples, denoted by $\succeq_b$ elsewhere in this paper. In (ii), $\widehat{\theta}$ and $\widehat{\theta}'$ are regarded as characters of G and M , respectively.

Assume that S satisfies iBAW at ℓ , and let $\Omega : \text{IBr}(b) \rightarrow \text{Alp}(b)$ be the resulting bijection for a block b of $X = X_\ell(S)$. For each $\psi \in \text{IBr}(b)$, choose a representative (Q, φ) of $\Omega(\psi)$. Replace Q by its full preimage in the universal covering group, and inflate ψ and φ to the cover and the corresponding normaliser, respectively. The normaliser quotient is unchanged, and the lifted weight belongs to the block of the inflated Brauer character. The full inductive condition supplies part (i) of Theorem 2.3, so part (ii) gives the character triple condition on $X/(\ker(\psi) \cap Z(X))$. Since $\ker(\psi) \cap Z(X)$ is a central ℓ' -subgroup, inflation gives the condition on X by [19, Remark 4.4]. Thus Ω is an iBAW bijection for b .

Remark 2.4. (a) By [50, Lemma 3.3] and [65, Remark 5.18], it is enough to verify the inductive BAW condition for one representative of each $\text{Aut}(X_\ell(S))$ -orbit of ℓ -blocks of $X_\ell(S)$ in order to prove that S satisfies iBAW at ℓ .

(b) We will also combine bijections on individual blocks by the following elementary observation. Suppose that a group A acts on $\mathcal{B}$ and on disjoint unions $\mathcal{U} = \coprod_{b \in \mathcal{B}} \mathcal{U}_b$ and $\mathcal{V} = \coprod_{b \in \mathcal{B}} \mathcal{V}_b$, with $\mathcal{U}_b^a = \mathcal{U}_{b^a}$ and $\mathcal{V}_b^a = \mathcal{V}_{b^a}$ for $a \in A$. For one representative b of each A -orbit, choose an A_b -equivariant bijection $\Phi_b : \mathcal{U}_b \rightarrow \mathcal{V}_b$. Then $\Phi(x^a) = \Phi_b(x)^a$, for $x \in \mathcal{U}_b$, defines an A -equivariant bijection $\Phi : \mathcal{U} \rightarrow \mathcal{V}$, with $\Phi(\mathcal{U}_b) = \mathcal{V}_b$ for every $b \in \mathcal{B}$. The A_b -equivariance makes this definition independent of the choice of a .

Corollary 2.5 (Sp  th). *Let S be a nonabelian simple group. If S satisfies iBAW at ℓ , then every ℓ -block of S admits an iBAW bijection.*

Proof. Apply [65, Proposition 4.6(a)–(c)] with $K = S$, $G = \text{Aut}(S)$ and $r = 1$. The resulting blockwise bijections satisfy the character triple condition by [66, Proposition 3.6(b) and Theorem 3.5(b)]. $\square$

The following lemma uses the argument in the proof of [57, Lemma 6.3] for a central ℓ -subgroup.

Lemma 2.6. *Let $G \triangleleft A$ be finite groups, and suppose that $P = O_\ell(G) \leq Z(G)$. Set $\overline{G} = G/P$ and $\overline{A} = A/P$. Let b be an A -stable ℓ -block of G , and let $\overline{b}$ be the block of $\overline{G}$ dominated by b . If $\overline{b}$ admits an iBAW bijection compatible with $\overline{A}$, then b admits an iBAW bijection compatible with A .*

Proof. Since P is central in G , the block correspondence for $G \rightarrow \overline{G}$ identifies the irreducible Brauer characters of b and $\overline{b}$ under inflation [58, Theorem 9.10]. Every ℓ -radical subgroup Q of G contains P . Writing $\overline{Q} = Q/P$, the map $Q \mapsto \overline{Q}$ gives a bijection between the ℓ -radical subgroups of G and $\overline{G}$, with

$$N_G(Q)/Q \cong N_{\overline{G}}(\overline{Q})/\overline{Q}.$$

By [30, Lemma 2.1] and the block correspondence above, these identifications also give a bijection between $\text{Alp}(b)$ and $\text{Alp}(\overline{b})$. All these maps commute with the action of A .

Composing these correspondences with the given iBAW bijection for $\overline{b}$ gives an A -equivariant bijection $\text{IBr}(b) \rightarrow \text{Alp}(b)$. For $\chi \in \text{IBr}(b)$, let (Q, φ) represent its image. Then P is normal in A_χ , lies in $N_G(Q)$, and is contained in the kernels of χ and φ^0 . The quotient maps identify A_χ/P and $N_A(Q)_\varphi/P$ with the corresponding stabilisers in $\overline{A}$. Hence [57, Lemma 3.14] lifts the required block isomorphism of modular character triples, proving the assertion. $\square$

We use *BAW-good* in the sense of [31, Section 3.5] for blocks of universal ℓ' -covering groups. For these groups, Theorem 2.3 identifies BAW-goodness with the inductive BAW condition, which supplies an iBAW bijection. For a full universal covering group G , set $P = O_\ell(G) = O_\ell(Z(G))$. Following [31, Theorems 3.18 and 5.7], we also call a block b of G BAW-good if the block $\overline{b}$ it dominates on G/P is BAW-good. Such a block b also admits an iBAW bijection. Set $\overline{G} = G/P$. The quotient map induces an isomorphism $\text{Aut}(G) \rightarrow \text{Aut}(\overline{G})$, which identifies $\text{Aut}(G)_b$ with $\text{Aut}(\overline{G})_{\overline{b}}$. The block $\overline{b}$ admits an iBAW bijection, so Lemma 2.6, applied with $A = G \rtimes \text{Aut}(G)_b$, gives an iBAW bijection for b .

For the permutation arguments below, actions on A -sets and $\mathbb{Z}A$ -lattices are written on the left. If a set Y already carries a right action, as do the character sets above, we pass to the left action $a \cdot y = y^{a^{-1}}$.

For a finite group A , let $B(A)$ denote its Burnside ring, and let $\mathcal{S}$ be a set of representatives of the A -conjugacy classes of subgroups of A . For a finite A -set Y , write $[Y]$ for the element of $B(A)$ represented by Y , and, for $U \leq A$, let Y^U be the set of U -fixed points of Y . The *mark homomorphism*

$$B(A) \rightarrow \prod_{U \in \mathcal{S}} \mathbb{Z}, \quad [Y] \mapsto (|Y^U|)_{U \in \mathcal{S}},$$

is injective [12, Theorem 2.3.2 and Section 3.2].

Let r be a prime. A finite group A is *r -hypoelementary* if $A/O_r(A)$ is cyclic. Every subgroup of an r -hypoelementary group is r -hypoelementary.

Let $\mathbb{Z}_r$ denote the ring of r -adic integers. For a finite A -set Y , write $\mathbb{Z}_r[Y]$ for the *permutation lattice* with basis Y . Conlon's theorem [25], in the form stated in [12, Theorem 3.5.5], says that $\mathbb{Z}_r[Y] \cong \mathbb{Z}_r[Y']$ as $\mathbb{Z}_r A$ -lattices if and only if $|Y^U| = |Y'^U|$ for every r -hypoelementary subgroup $U \leq A$. We will use the following consequence with $r = 2$.

Corollary 2.7 (Conlon). *Let A be a finite r -hypoelementary group and let Y, Y' be finite A -sets. If the permutation lattices $\mathbb{Z}_r[Y]$ and $\mathbb{Z}_r[Y']$ are isomorphic as $\mathbb{Z}_r A$ -lattices, then $Y \cong Y'$ as A -sets.*

Proof. Every subgroup of A is r -hypoelementary. Conlon's theorem therefore gives $|Y^U| = |Y'^U|$ for every $U \leq A$. The injectivity of the mark homomorphism gives $Y \cong Y'$ as A -sets. $\square$

For an ℓ -block b , a *basic set* of Brauer characters is a $\mathbb{Z}$ -basis of $\mathbb{Z}[\text{IBr}(b)]$ consisting of Brauer characters [43, (1.1)–(1.2)]. A subset $\mathcal{X} \subseteq \text{Irr}(b)$ is an *integral basic set* for b if the decomposition map restricts to an isomorphism

$$d_b : \mathbb{Z}[\mathcal{X}] \xrightarrow{\sim} \mathbb{Z}[\text{IBr}(b)].$$

Lemma 2.8. *Let G be a finite group, let b be an ℓ -block of G , and let A be a finite group acting on $\text{Irr}(b)$ and $\text{IBr}(b)$ such that d_b is A -equivariant. Let $\mathcal{X} \subseteq \text{Irr}(b)$ be an A -stable integral basic set. Then d_b restricts to an isomorphism $\mathbb{Z}[\mathcal{X}] \cong \mathbb{Z}[\text{IBr}(b)]$ of $\mathbb{Z}A$ -permutation lattices. Consequently the underlying finite sets are A -isomorphic if A is 2-hypoelementary.*

Proof. By the definition of an integral basic set, this restriction is an isomorphism of $\mathbb{Z}$ -modules. Its A -equivariance makes it an isomorphism of $\mathbb{Z}A$ -permutation lattices. If A is 2-hypoelementary, extend scalars to $\mathbb{Z}_2$ and apply Corollary 2.7. $\square$

We shall repeatedly use the following standard result.

Lemma 2.9. *Let H be a finite group, let $N \triangleleft H$, and suppose that H/N is cyclic. Then every H -invariant character $\theta \in \text{Irr}(N) \cup \text{IBr}(N)$ extends to H .*

Proof. Apply [47, Corollary 11.22] for ordinary characters and [58, Theorem 8.12] for Brauer characters. $\square$

The following lemma is a special case of Brough–Späth's criterion [31, Theorem 3.18], with the required extensions supplied by Lemma 2.9.

Lemma 2.10. *Let G be a nonabelian simple group that is its own universal covering group, let ℓ be a prime dividing $|G|$, and suppose that $\text{Aut}(G) = G \rtimes E$ for a cyclic group E . If b is an E -stable ℓ -block of G and there is an E -equivariant bijection*

$$\Omega_b : \text{IBr}(b) \longrightarrow \text{Alp}(b),$$

then b is BAW-good.

Proof. Apply Brough–Späth's criterion [31, Theorem 3.18] with $\tilde{G} = G$, $\mathcal{B} = \{b\}$, and $\Omega = \tilde{\Omega} = \Omega_b$. The group G is perfect, E is abelian, and $C_{G \rtimes E}(G) = 1 = Z(G)$, so conditions (i)(a) and (i)(b) of the criterion hold. Inner automorphisms act trivially on $\text{IBr}(b)$ and on the G -classes of weights, so Ω_b is $G \rtimes E$ -equivariant. For $\varphi \in \text{IBr}(b)$, the quotient $(G \rtimes E)_\varphi / G$ is cyclic, so Lemma 2.9 extends φ to its stabiliser. Choose a representative (Q, θ) of $\Omega_b(\varphi)$. Identifying θ with its inflation to $N_G(Q)$, we have $\text{bl}(\varphi) = \text{bl}(\theta)^G = b$ because (Q, θ) belongs to b . This verifies the block induction condition (iii)(c) of the criterion.

Set $D = (G \rtimes E)_{Q, \theta}$. Then $D \cap G = N_G(Q)$ and $D/N_G(Q)$ embeds in E . The irreducible Brauer reduction θ^0 of θ , viewed as a character of $N_G(Q)/Q$, is D/Q -invariant. Lemma 2.9 extends it to D/Q , and inflation gives the required extension to D . Since $G = \tilde{G}$ and $Z(G) = 1$, the remaining conditions on restriction, covering, central characters and stabiliser products are immediate, and the extensions within $\tilde{G}$ are given by the characters themselves. Hence b is BAW-good. $\square$

Let $\mathbf{Y}$ be a connected reductive algebraic group with Frobenius endomorphism F_Y , and set $Y = \mathbf{Y}^{F_Y}$. Let $\mathbf{Y}^*$ be the dual group, equipped with the dual Frobenius endomorphism, also denoted by F_Y , and set $Y^* = (\mathbf{Y}^*)^{F_Y}$. For a semisimple element $u \in Y^*$, write $\mathcal{E}(Y, u)$ for the *rational Lusztig series* indexed by the Y^* -conjugacy class of u . For $\chi \in \text{Irr}(Y)$, write χ^* for the irreducible character equal, up to sign, to its Alvis–Curtis dual $D_Y(\chi)$.

For an irreducible character χ of a finite reductive group, let n_χ denote its *generic denominator*, as defined in [44, Section 3(B), p. 443]. Equivalently, n_χ is the reciprocal of the leading coefficient of its degree polynomial [45, Remark 2.3.28].

The proofs of the following two lemmas are given in Appendix B.

Lemma 2.11. *Let $\mathbf{H}$ be a connected reductive algebraic group with connected centre and simply connected derived subgroup of type $\mathbf{B}_n$, where $n \geq 3$. Suppose that $\mathbf{H}$ is defined over $\mathbb{F}_q$, where q is a power of an odd prime p , and let F be the corresponding Frobenius endomorphism. Set $H = \mathbf{H}^F$. Choose an F -stable Borel subgroup $\mathbf{B}^*$ of $\mathbf{H}^*$ and an F -stable maximal torus $\mathbf{T}^* \subseteq \mathbf{B}^*$. Let $s \in (\mathbf{T}^*)^F$ be an isolated semisimple element. Let W_s be the Weyl group of $C_{\mathbf{H}^*}(s)$ relative to $\mathbf{T}^*$, and let $\mathcal{F} \subseteq \text{Irr}(W_s)$ be a Lusztig family [45, Definition 4.1.17]. Denote by $\mathcal{G}_{\mathcal{F}}$ the finite group attached to $\mathcal{F}$ in Lusztig's parametrisation, and by $\text{Irr}_{s,\mathcal{F}}(H)$ the corresponding family of irreducible characters in $\mathcal{E}(H, s)$. Let $\mathcal{C}$ be the F -stable unipotent class associated with $(s, \mathcal{F})$. Suppose that, for some $u_0 \in \mathcal{C}^F$, the component group $A_{\mathbf{H}}(u_0) = C_{\mathbf{H}}(u_0)/C_{\mathbf{H}}(u_0)^\circ$ and the group $\mathcal{G}_{\mathcal{F}}$ are elementary abelian of the same order, and F acts trivially on $A_{\mathbf{H}}(u_0)$. Let $u_1, \dots, u_d$ represent the H -conjugacy classes in $\mathcal{C}^F$, and let Γ_{u_j} be the corresponding generalised Gelfand–Graev characters. Then there are $\rho_1, \dots, \rho_d \in \text{Irr}_{s,\mathcal{F}}(H)$ such that*

$$(2.2) \quad \langle \rho_i^*, \Gamma_{u_j} \rangle_H = \delta_{ij} \quad (1 \leq i, j \leq d).$$

Lemma 2.12. *Let $m \geq 3$, let q_0 be an odd prime power, and let $\mathbf{G}_0 = \text{Sp}_{2m} \subseteq \mathbf{H} = \text{CSp}_{2m}$ be defined over $\mathbb{F}_{q_0}$ with the standard Frobenius endomorphism F_0 . Set $H = \mathbf{H}^{F_0}$. Let $\mathcal{C}$ be a unipotent class of $\mathbf{H}$ stable under F_0 . Fix a maximal torus $\mathbf{T}^*$ of $\mathbf{H}^*$ split over $\mathbb{F}_{q_0}$. For $s \in \mathbf{T}^*$, let W_s denote the Weyl group of $C_{\mathbf{H}^*}(s)$ relative to $\mathbf{T}^*$. For the corresponding geometric unipotent class of $\mathbf{G}_0$, let $s_0 \in \mathbf{T}^*$ and $\mathcal{F} \subseteq \text{Irr}(W_{s_0})$ be the semisimple element and family selected in [69, 10.1–10.3]. Choose $z \in Z(\mathbf{H}^*)$ with $(zs_0)^2 = 1$ and set $\tilde{s} = zs_0$. Denote by $\mathcal{G}_{\mathcal{F}}$ the finite group attached to $\mathcal{F}$ in Lusztig's parametrisation, and by $\text{Irr}_{\tilde{s},\mathcal{F}}(H) \subseteq \mathcal{E}(H, \tilde{s})$ the corresponding family of ordinary irreducible characters. Let $u_1, \dots, u_d$ represent the H -conjugacy classes in $\mathcal{C}^{F_0}$, and let Γ_{u_j} be the corresponding generalised Gelfand–Graev characters. Then there are $\rho_1, \dots, \rho_d \in \text{Irr}_{\tilde{s},\mathcal{F}}(H)$ such that*

$$\langle \rho_i^*, \Gamma_{u_j} \rangle_H = \delta_{ij} \quad (1 \leq i, j \leq d).$$

3. SYMPLECTIC GROUPS

Let $q = p^f$ and $n \geq 2$. Throughout this section, $S = \text{PSp}_{2n}(q)$ denotes the simple group under consideration. We write $G = \text{Sp}_{2n}(q)$ for the finite symplectic group used in the blockwise arguments for S . We assume that $(n, q) \neq (2, 2)$, so that S is nonabelian simple. The purpose of this section is to prove the following result.

Theorem 3.1. *Let $S = \text{PSp}_{2n}(q)$. Then, for every prime $\ell \mid |S|$, S satisfies iBAW at ℓ on the universal ℓ' -covering group $X_\ell(S)$.*

3.1. Covering groups and some reductions. The centre and Schur multiplier tables [56, Tables 24.2–24.3 and Remark 24.19] determine the universal ℓ' -covering groups in Table 3.1.

TABLE 3.1. Universal ℓ' -covering groups in type C.

Parameters	Prime	$X_\ell(S)$
q odd	$\ell = 2$	$\text{PSp}_{2n}(q)$
q odd	ℓ odd	$\text{Sp}_{2n}(q)$
q even, $(n, q) \neq (3, 2)$	every ℓ	$\text{Sp}_{2n}(q)$
$(n, q) = (3, 2)$	$\ell = 2$	$\text{Sp}_6(2)$
$(n, q) = (3, 2)$	$\ell \in \{3, 5, 7\}$	$2 \cdot \text{Sp}_6(2)$

The case $\ell = p$ of Theorem 3.1 follows from [65, Theorem C]. For odd q and $\ell = 2$, we use the notation ε , ε_Γ , d_Γ , $\mathcal{F}_1$, $\mathcal{F}_2$, m_Γ , w_Γ , and $\mathcal{P}(w_\Gamma)$ from [33, Sections 2 and 5].

As noted by Feng, Yu, and Zhang [36], An's list of basic subgroups used in the weight parametrisation [2, p. 190] omits the subgroups denoted $R_{m,1,\gamma}^0$ in their notation when m is even. Their calculation of the centralisers distinguishes the two discriminants of the multiplicity space [36, proof of Proposition 3.48]. In the notation of [33, p. 11], when $\varepsilon_\Gamma = -\varepsilon$, $\gamma = 0$, and $c_1 = 1$, the correction in [36, Remark 3.49] replaces $E_{d_\Gamma,1,1} \wr A_{c'}$ by

$$R_{d_\Gamma,1,1}^{0,-} \wr A_{c'}, \quad c' = (c_2, \dots, c_t).$$

The corrected subgroup has a different centraliser in the symplectic group on the corresponding summand [36, proof of Proposition 3.48]. Although [36, Remark 3.49] states that the remaining arguments continue to apply, it does not supply the corresponding calculation of the weight characters underlying the blockwise weight labelling. For principal blocks, the omitted family is excluded by [36, Remark 3.50]. We check that no parameters indexed by $\Gamma \in \mathcal{F}_1 \cup \mathcal{F}_2$ occur and that the remaining R^0 factors have multiplicity spaces of dimension one. These facts allow us to use the principal block weight calculation in [33, Section 5]. Together with the type A cases, the Jordan reduction described in [31, Remark 5.8] yields the result for all blocks.

Let $\mathbf{Y}$ be a connected reductive algebraic group with Frobenius endomorphism F_Y , and set $Y = \mathbf{Y}^{F_Y}$. Let ℓ be a prime different from the defining characteristic. Write $\mathcal{E}(Y, \ell')$ for the union of the rational Lusztig series $\mathcal{E}(Y, u)$, where u runs through the semisimple ℓ' -elements of Y^* . For a semisimple ℓ' -element $s \in Y^*$, let $\mathcal{E}_\ell(Y, s)$ denote the set $\mathcal{E}_\pi(Y, (s))$ defined by Broué and Michel [17, Section 2] with $\pi = \{\ell\}$, using the notation of [30, Section 2.3]. Explicitly, $\mathcal{E}_\ell(Y, s) = \bigcup_t \mathcal{E}(Y, st)$, where t runs through the semisimple ℓ -elements of Y^* commuting with s . By [17, Theorem 2.2], this set is a union of ℓ -blocks. Let e_s^Y be the sum of the block idempotents $e_b \in kY$ for which $\text{Irr}(b) \subseteq \mathcal{E}_\ell(Y, s)$. We say that b belongs to e_s^Y if $e_b e_s^Y = e_b$, and write $\text{IBr}(Y, e_s^Y)$ for the union of $\text{IBr}(b)$ over these blocks. As s runs through representatives of the semisimple ℓ' -classes of Y^* , the idempotents e_s^Y are pairwise orthogonal and sum to 1.

We shall use the following form of the Jordan reduction of Feng, Li, and Zhang.

Lemma 3.2. *Let $\mathbf{G}$ be a simple, simply connected algebraic group in characteristic p , let F be a Steinberg endomorphism of $\mathbf{G}$, and set $G = \mathbf{G}^F$. Assume that $G/Z(G)$ is simple and that G is its universal covering group. Let $\ell \neq p$ be a prime. Fix a regular embedding $\mathbf{G} \hookrightarrow \tilde{\mathbf{G}}$ as in [31, Section 5], and set $\tilde{G} = \tilde{\mathbf{G}}^F$. Suppose that the following conditions hold.*

- (i) *For every semisimple ℓ' -element $s \in (\mathbf{G}^*)^F$, there is a choice of the automorphism group A_s used for s in [31, Section 5] such that every $\tilde{G}$ -orbit in $\text{IBr}(G)$ contains a character ψ satisfying*

$$(\tilde{G}A_s)_\psi = \tilde{G}_\psi(A_s)_\psi$$

and extending to $G(A_s)_\psi$.

- (ii) *Let $\mathbf{K}$ be a simple, simply connected algebraic group in characteristic p whose Dynkin diagram is isomorphic to a subdiagram of that of $\mathbf{G}$. For every Steinberg endomorphism F' of $\mathbf{K}$ such that $\mathbf{K}^{F'}/Z(\mathbf{K}^{F'})$ is simple, every strictly quasi-isolated ℓ -block of $\mathbf{K}^{F'}$, in the sense of [31, Section 4.3], admits an iBAW bijection.*

Then every ℓ -block of G is BAW-good.

Proof. In [31, Section 5], the group A is chosen after fixing the semisimple element s . Condition (i) supplies [31, Assumption 5.3] for this choice A_s , for each s , as required by [31, Hypothesis 5.5(a)]. Condition (ii) is [31, Hypothesis 5.5(b)]. The conclusion follows from [31, Theorem 5.7]. $\square$

To apply condition (ii) of Lemma 3.2, we also need the following consequence of the type A results [29, 32]. Let q_0 be a prime power, let $d \geq 2$, and let $H = \text{SL}_d(q_0)$ or $\text{SU}_d(q_0)$. If $S_0 = H/Z(H)$ is nonabelian simple and ℓ is a prime not dividing q_0 , then every ℓ -block of H admits an iBAW bijection. If H is the universal covering group of S_0 , its blocks are BAW-good

by the proof of [32, Theorem 1], and the passage from BAW-goodness to iBAW bijections in Section 2 gives the assertion.

The remaining possibilities for H , up to isomorphism, are $\mathrm{SL}_2(4)$, $\mathrm{SL}_2(9)$, $\mathrm{SL}_3(2)$, $\mathrm{SL}_3(4)$, $\mathrm{SL}_4(2)$, $\mathrm{SU}_4(2)$, $\mathrm{SU}_4(3)$ and $\mathrm{SU}_6(2)$ [56, Remark 24.19 and Table 24.3]. Each corresponding simple group S_0 satisfies iBAW at ℓ by [29, Proposition 4.6]. When $Z(H) = 1$, Corollary 2.5 gives the assertion directly.

It remains to consider $H = \mathrm{SL}_2(9)$, $\mathrm{SU}_4(3)$, $\mathrm{SL}_3(4)$ or $\mathrm{SU}_6(2)$. Suppose first that $\ell \mid |Z(H)|$. In these cases, $P = Z(H) = O_\ell(H)$ and $H/P = S_0$. Since H is perfect, the natural map $\mathrm{Aut}(H) \rightarrow \mathrm{Aut}(S_0)$ is injective. For an ℓ -block b of H , Corollary 2.5 gives an iBAW bijection for the block of S_0 dominated by b . By [57, Lemma 3.6(i)], restricting the automorphism group makes this bijection compatible with $S_0 \rtimes \mathrm{Aut}(H)_b$. Lemma 2.6, applied with $A = H \rtimes \mathrm{Aut}(H)_b$, therefore gives an iBAW bijection for b . For every other prime $\ell \nmid q_0$, the Sylow ℓ -subgroups of these four groups are cyclic, so [57, Proposition 8.5] gives the assertion.

We give an alternative proof of the following result of Feng and Malle [33, Theorem 1]. Our proof uses their principal block calculation and Jordan reduction, without relying on the weight parametrisation for arbitrary 2-blocks affected by the classification errors discussed above.

Proposition 3.3. *Let q be odd and $n \geq 2$. Then $S = \mathrm{PSp}_{2n}(q)$ satisfies iBAW at 2 on $X_2(S) = S$.*

Proof. By [53, Theorem 1.1], $\mathrm{PSp}_4(q_0)$ satisfies iBAW at 2 for every odd prime power q_0 . This proves the assertion when $n = 2$. For later use in condition (ii) of Lemma 3.2, we also need an iBAW bijection for the principal 2-block of $\mathrm{Sp}_4(q_0)$. Let $K = \mathrm{Sp}_4(q_0)$ and $\bar{K} = K/Z(K)$. Corollary 2.5, applied to $\bar{K}$, gives an iBAW bijection for its principal 2-block. Since $O_2(K) = Z(K)$ and $\mathrm{Aut}(K) \cong \mathrm{Aut}(\bar{K})$, Lemma 2.6, applied with $G = K$, $P = Z(K)$ and $A = K \rtimes \mathrm{Aut}(K)$, lifts this bijection to the principal 2-block of K .

Assume henceforth that $n \geq 3$. We now establish the principal block result in higher rank. Let $G_0 = \mathrm{Sp}_{2m}(q_0)$, where q_0 is an odd prime power and $m \geq 3$, and let B_1 be its principal 2-block. In the notation of [33, Section 5], this is the block labelled by $s = 1$. Let (R, φ) be a weight belonging to B_1 . By [36, Lemma 2.3], $Z(R) \in \mathrm{Syl}_2(C_{G_0}(R))$. By [36, Theorem 3.37], choose a decomposition of R as a direct product of basic subgroups in the classification used there, each acting on its corresponding summand in an orthogonal decomposition of the natural module. The omitted family does not occur in a principal weight [36, Remark 3.50]. Indeed, suppose that a factor from this family occurs in the decomposition. By [36, Proposition 3.48(4)], it has the form $R_{d,1,1,c}^{0,-}$ with d even. Let

$$R_0 = I_W \otimes E_-^3 \leq \mathrm{Sp}(W \otimes U),$$

where W is a nondegenerate orthogonal space over $\mathbb{F}_{q_0}$ of even dimension $d \geq 2$ and nonsquare discriminant, and $E_-^3 \cong Q_8$ acts on its symplectic module U . The group E_-^3 acts trivially on W . Thus, as an E_-^3 -module, $W \otimes U$ is a direct sum of d copies of U , with multiplicity space W . The omitted factor is a wreath product whose base group is a direct product of copies of R_0 . Choose a nonscalar orthogonal reflection t of W , acting as -1 on a nonsingular line and as 1 on its orthogonal complement. On the summand corresponding to this factor, set $\hat{t} = \mathrm{diag}(t \otimes I_U, \dots, t \otimes I_U)$. This is a symplectic involution which centralises each copy of R_0 and the coordinate permutations. Extending it by the identity on the other summands gives $\hat{t} \in C_{G_0}(R)$. It does not lie in R : its permutation part is trivial, while its action on the multiplicity space W is nonscalar. Thus $\langle Z(R), \hat{t} \rangle$ is a 2-subgroup of $C_{G_0}(R)$ properly containing $Z(R)$, a contradiction. Together with the other exclusions in [36, Remark 3.50], this leaves cases (3) and (6) of [36, Proposition 3.48]. Up to conjugacy, the basic subgroups in these cases are those attached to the elementary divisor $\Gamma = x - 1$ in [33, Section 5.1]. That section also describes the irreducible characters of 2-defect zero of their normaliser quotients, with normalisers taken in the symplectic groups on the corresponding summands. To specialise this parametrisation to B_1 , recall that its semisimple label is $s = 1$. Consequently, $m_\Gamma(1) = w_\Gamma(1) = 0$ for every elementary

divisor $\Gamma \in \mathcal{F}_1 \cup \mathcal{F}_2$. Thus the product of the sets $\mathcal{P}(w_\Gamma)$ over these elementary divisors in [33, Proposition 5.4] is a singleton, and only the parameters associated with the elementary divisor $\Gamma = x - 1$ occur. In particular, the cases involving the corrected $E_{dr,1,1}$ factor do not enter the calculation for the principal block. By [36, Proposition 3.48], every remaining factor of type R^0 has multiplicity space of dimension one. Its isometry group is therefore $\{I, -I\}$, as in the centraliser calculation in [36, proof of Proposition 3.48]. Equal factors may still occur repeatedly in the direct product decomposition of R . For these repeated factors, we use the construction of characters of wreath products in [2, (6D)], recalled after [33, Proposition 5.4]. This construction assigns to each irreducible character of 2-defect zero of the corresponding normaliser quotient a partition of the form $(r, r-1, \dots, 1)$ for some $r \geq 0$, with the empty partition understood when $r = 0$. Its size is $r(r+1)/2$, a triangular number. The sizes of these partitions sum to the number of copies of the corresponding basic subgroup. This sum need not be triangular, as the example in Remark 3.4 shows.

Let (R, φ) be a pair produced by this construction, with $\varphi \in \text{dz}(N_{G_0}(R)/R)$. In each of the two remaining families, a basic subgroup, including its wreath extension, has centraliser $\{I, -I\}$ on its corresponding summand [2, p. 198]. The central elements acting as $-I$ on one summand and as I on all the others force every element of $C_{G_0}(R)$ to preserve each summand. Thus $C_{G_0}(R) = Z(R)$, so $Z(R)$ is a Sylow 2-subgroup of $C_{G_0}(R)$ and $RC_{G_0}(R) = R$. The inflation of φ to $N_{G_0}(R)$ has R in its kernel and has defect zero on $N_{G_0}(R)/(RC_{G_0}(R))$. Hence [36, Lemma 2.3] shows that (R, φ) is a weight belonging to B_1 . The construction following [33, Proposition 5.4] recovers from each $x - 1$ parameter the conjugacy types and multiplicities of the basic factors and the character φ . The G_0 -conjugacy class of (R, φ) also determines its parameter. Thus this construction identifies $\text{Alp}(B_1)$ with the $x - 1$ parameter set.

For the principal block Brauer character calculation, we use [33, Corollary 4.3]. The required unitriangularity is stated in [24, Theorem 2.9] for every odd q_0 . Its proof uses the character selection in [24, Proposition 2.8], based on [42, Proposition 4.3] and [68, Proposition 5.4]. We verify this character selection for the families used in the automorphism argument in [33, Section 4.1]. Let $\mathbf{G}_0 = \text{Sp}_{2m} \subseteq \mathbf{H} = \text{CSp}_{2m}$, with the split Frobenius endomorphism F_0 defining the given group G_0 , and set $H = \mathbf{H}^{F_0}$. The partition parametrisation shows that every geometric unipotent class of $\mathbf{H}$ is stable under F_0 . For each such class $\mathcal{C}$, Lemma 2.12 supplies the characters of H with the required Kronecker delta scalar products. For unipotent elements $u \in \mathbf{G}_0$, the component groups $A_{\mathbf{G}_0}(u)$ are elementary abelian 2-groups [68, proof of Proposition 2.4]. For each geometric unipotent class, choose a representative as in [68, Definition 2.1], which is possible by Proposition 2.8 of the same paper. For the split pair $(\tilde{s}, \mathcal{F})$ selected in Lemma 2.12, the symbols in any type D factor of the selected family are nondegenerate. Hence the argument in [69, 10.3] gives condition (P5) of [68, Theorem 3.2].

We follow the restriction argument in the proof of [68, Proposition 5.4], using Lemma 2.12 for the selection supplied there by [42, Proposition 4.3]. The induction identity for generalised Gelfand–Graev characters [70, proof of Lemma 14.12], compatibility of Alvis–Curtis duality with restriction [45, Proposition 3.4.3], Frobenius reciprocity and condition (P5) of [68, Theorem 3.2] give the required scalar products on restriction to G_0 . The class count and Clifford theory argument in the proof of [68, Proposition 5.4] match the irreducible constituents with the G_0 -classes contained in each H -class.

The selected characters have unipotent support $\mathcal{C}$ by the proof of Lemma 2.12 in Appendix B, and so do their restriction constituents [45, Example 2.7.18]. These constituents lie in $\mathcal{E}(G_0, s)$, where s is the image of $\tilde{s}$ in $(\mathbf{G}_0^*)^{F_0}$, by [45, Proposition 2.6.16]. Since $s^2 = 1$ and Alvis–Curtis duality preserves rational Lusztig series [21, Proposition 9.8(iv)], these constituents and their duals belong to B_1 by [21, Theorem 21.14]. Together with the vanishing results for wave front sets in [70, Lemma 14.15 and Proposition 15.2], this gives the identity diagonal blocks and the vanishing entries used in [24, Theorem 2.9]. Thus, for $m \geq 3$ and every odd q_0 , this

establishes the character selection and unitriangularity used in the principal block argument of [33, Section 4.1].

For the principal $x - 1$ parameters obtained above, [33, Lemma 5.1] gives the field and diagonal actions on the characters of the normaliser quotients of the basic subgroups. Applying these actions to the wreath product construction, as in the proof of [33, Proposition 5.5], gives the weight actions described in [33, Proposition 5.9]. Together with [33, Corollary 4.3], the counting arguments in [33, Proposition 6.1 and the proof of Theorem 6.2] show that $\text{IBr}(B_1)$ and $\text{Alp}(B_1)$ have equal cardinalities and equal numbers of elements fixed by the nontrivial diagonal outer automorphism, while field automorphisms fix both sets pointwise. Matching the diagonal orbits of sizes one and two gives an $\text{Aut}(G_0)$ -equivariant bijection from $\text{IBr}(B_1)$ to $\text{Alp}(B_1)$. Set $Z_0 = Z(G_0)$ and $\overline{G}_0 = G_0/Z_0$. The correspondences in the proof of Lemma 2.6, with $P = Z_0$, give a corresponding $\text{Aut}(\overline{G}_0)$ -equivariant bijection for the principal 2-block of $\overline{G}_0$. For B_1 , the equivariant bijection gives condition (1) of [33, Proposition 3.4], and the trivial action of field automorphisms gives conditions (2) and (3). The blockwise argument in the proof of that proposition therefore verifies the inductive BAW condition for the principal block of $\overline{G}_0$. Apply Lemma 2.6 with $G = G_0$, $P = Z_0 = O_2(G_0)$ and $A = G_0 \rtimes \text{Aut}(G_0)$. The required iBAW bijection for the principal block of $\overline{G}_0$ follows from Theorem 2.3, using the natural isomorphism $\text{Aut}(G_0) \cong \text{Aut}(\overline{G}_0)$. Thus B_1 admits the required iBAW bijection.

Let $\mathbf{G}$ be a simply connected algebraic group of type $\mathbf{C}_n$ with Frobenius endomorphism F such that $\mathbf{G}^F = G = \text{Sp}_{2n}(q)$. We verify the two assumptions of Lemma 3.2 at $\ell = 2$. For each semisimple $2'$ -element $s \in (\mathbf{G}^*)^F$, conjugate the Levi subgroup associated with s and its series idempotent by a suitable element of G . By [60, Corollary 4.2 and Lemmas 4.3 and 4.5], the group A_s in condition (i) of Lemma 3.2 can then be chosen inside the field automorphism group. By [33, Corollary 4.6], if a diagonal automorphism and a field automorphism have the same action on a Brauer character, then both fix it. Thus, for $\tilde{G} = \text{CSp}_{2n}(q)$,

$$(\tilde{G}A_s)_\psi = \tilde{G}_\psi(A_s)_\psi \quad (\psi \in \text{IBr}(G)).$$

Since $(A_s)_\psi$ is cyclic, Lemma 2.9 gives the required extension. A connected Dynkin subdiagram of $\mathbf{C}_n$ has type $\mathbf{A}_r$ or $\mathbf{C}_r$. Let $\mathbf{K}$ and F' be as in condition (ii) of Lemma 3.2. If $\mathbf{K}$ has type $\mathbf{A}_r$, then $\mathbf{K}^{F'}$ is a special linear or unitary group. The result for special linear and unitary groups established after Lemma 3.2 gives the required iBAW bijections for its 2-blocks. This also covers type $\mathbf{C}_1$, since $\mathbf{C}_1 = \mathbf{A}_1$.

Suppose now that $\mathbf{K}$ has type $\mathbf{C}_r$, where $r \geq 2$, and let c be a strictly quasi-isolated 2-block of $\mathbf{K}^{F'}$. Choose a strictly quasi-isolated semisimple $2'$ -element $s \in \mathbf{K}^{*F'}$ such that $\text{Irr}(c) \subseteq \mathcal{E}_2(\mathbf{K}^{F'}, s)$, where $\mathbf{K}^* \cong \text{SO}_{2r+1}$. Since s is quasi-isolated and the defining characteristic is odd, [10, Proposition 4.11(a)] gives $s^2 = 1$. The element s has odd order, so $s = 1$. Thus c is the principal 2-block, the unique unipotent 2-block of $\mathbf{K}^{F'}$ [21, Theorem 21.14]. We have $\mathbf{K}^{F'} \cong \text{Sp}_{2r}(q_0)$ for an odd prime power q_0 . If $r = 2$, the rank two construction at the start of this proof gives an iBAW bijection for c . If $r \geq 3$, the principal block result proved above, applied with $m = r$, gives such a bijection. Hence both hypotheses of Lemma 3.2 hold.

Since G is the full universal covering group of S , Lemma 3.2 shows that every 2-block of G is BAW-good. Applying Remark 2.4(b) to the blockwise equivariant bijections gives an $\text{Aut}(G)$ -equivariant bijection $\text{IBr}(G) \rightarrow \text{Alp}(G)$ that preserves blocks. This verifies condition (1) of Feng and Malle's criterion [33, Proposition 3.4]. Condition (2) follows from [33, Corollary 4.6], and [33, Remark 3.5] then gives condition (3). The criterion therefore shows that S satisfies iBAW at 2 on $X_2(S) = S$. $\square$

Remark 3.4. The final sentence of [36, Remark 3.50] extends the triangularity assertion of [36, Remark 3.47(2)] to symplectic groups. The following principal 2-weight of $\text{Sp}_4(3)$ shows that this extension fails for the total multiplicity of a basic subgroup, although the individual partitions in its parametrisation have triangular sizes. Let $V = V_1 \perp V_2$ be an orthogonal direct sum of two symplectic planes over $\mathbb{F}_3$, and let $R = Q_8 \times Q_8$ act with one factor on each plane. Each factor is a

basic subgroup in the construction of [36, Section 3]. Set $G = \mathrm{Sp}(V) \cong \mathrm{Sp}_4(3)$. The summands are nonisomorphic absolutely irreducible R -modules, so $C_G(R) = \{\pm I_{V_1}\} \times \{\pm I_{V_2}\} = Z(R)$, and every element of $N_G(R)$ permutes the two summands. Since $Q_8 \trianglelefteq \mathrm{SL}_2(3) = \mathrm{Sp}(V_i)$, we have $N_G(R) = (\mathrm{SL}_2(3) \times \mathrm{SL}_2(3)) \rtimes C_2$, where the involution interchanges the factors. Hence $N_G(R)/R \cong C_3 \times S_3$, which has no nontrivial normal 2-subgroup, so R is radical. This quotient has irreducible characters of degree 2 and 2-defect zero. By [36, Lemma 2.3], these give principal 2-weights. Here the same basic subgroup occurs twice, so its total multiplicity is 2, which is not triangular. In the labelling by assignments of 2-cores recalled from [2, (6D)] after [33, Proposition 5.4], the three linear characters of $N_{\mathrm{Sp}(V_i)}(Q_8)/Q_8 \cong C_3$ are assigned the partitions $(1), (1), \emptyset$, up to permutation. Their sizes are 1, 1, 0, which are individually triangular, while their sum is 2. The proof of Proposition 3.3 uses the individual partition sizes and their sum, and does not require that sum to be triangular.

Proposition 3.5. *Let $q > 2$ be a prime power. Then $S = \mathrm{PSp}_4(q)$ satisfies iBAW at every prime dividing its order.*

Proof. Write $q = p^f$, and let ℓ be a prime dividing $|S|$. If $\ell = p$, the result follows from [65, Theorem C]. If q is odd, the result for every $\ell \neq p$ is proved in [53, Theorem 1.1], with the cases $\ell \mid q^2 - 1$ also proved by Brough and Schaeffer Fry [18, Theorem 1.1]. If $\ell \neq p$ and q is even, it follows from [63, Theorem 5.5]. $\square$

It remains to treat rank at least three at odd nondefining primes. For even q and $n \geq 4$, we give an independent proof. For odd q , we replace the uses of unitriangularity in Li's proof of [52, Theorem 2]. Neither argument assumes unitriangularity.

3.2. Even fields. Let $\mathbf{H}$ be a simply connected algebraic group of type C_r , where $r \geq 4$, over $\mathbb{F}_2$, and let F_2 be the Frobenius endomorphism induced by the squaring map on $\mathbb{F}_2$. For a positive integer a , set $F' = F_2^a$ and $H = \mathbf{H}^{F'} = \mathrm{Sp}_{2r}(2^a)$. The map F_2 restricts to a field automorphism of H of order a . Let $\mathbf{H}^*$ be the dual group of $\mathbf{H}$. Then $\mathbf{H}^*$ is adjoint of type B_r , and the Frobenius endomorphism dual to F' is also denoted by F' .

Let ℓ be an odd prime dividing $|H|$, let e be the multiplicative order of 2^a modulo ℓ , and let $E_H = \langle F_2|_H \rangle \cong C_a$. For an ℓ -block C of H , let $\mathcal{X}_C = \mathrm{Irr}(C) \cap \mathcal{E}(H, \ell')$. We follow [34, Definition 3.5] in writing e -JGC for (e, ℓ) -Jordan-generalised-cuspidal when ℓ is clear. For an F' -stable torus $\mathbf{T}$, we use the notation $\mathbf{T}_{\phi_e}$ of [34, Section 3.2] for its Sylow e -torus. In the notation of [34, Definitions 3.17–3.18], $\mathcal{W}(C)$ denotes the set of H -conjugacy classes of generic (e, ℓ) -weights belonging to C . For an F' -stable Levi subgroup $\mathbf{L}$ and $\lambda \in \mathrm{Irr}(\mathbf{L}^{F'})$, write $W_H(\mathbf{L}, \lambda) = N_H(\mathbf{L}, \lambda)/\mathbf{L}^{F'}$. The prime ℓ is good for $\mathbf{H}$, and $Z(\mathbf{H}) = Z(\mathbf{H}^*) = 1$, so [34, Condition 3.23] holds. In particular, if $(\mathbf{T}, \eta)$ represents an element of $\mathcal{W}(C)$, then $\mathrm{bl}(\eta)^H = C$ by [34, Proposition 3.24(b)].

Lemma 3.6. *If C is a unipotent ℓ -block of $H = \mathrm{Sp}_{2r}(2^a)$, then E_H fixes every element of $\mathcal{X}_C$ and $\mathcal{W}(C)$.*

Proof. By [20, Theorem 4.4(iii)], the characters in $\mathrm{Irr}(C)$ lie in rational Lusztig series labelled by semisimple ℓ -elements. The disjointness of these series therefore gives $\mathcal{X}_C = \mathrm{Irr}(C) \cap \mathcal{E}(H, 1)$. By [20, Theorem 4.4(i)], this is the generalised e -Harish-Chandra series belonging to C . Since field automorphisms fix the unipotent characters of H [22, p. 700], the group E_H fixes $\mathcal{X}_C$ pointwise.

Let $(\mathbf{T}, \eta)$ represent an element of $\mathcal{W}(C)$ and set $\mathbf{M} = C_{\mathbf{H}}(\mathbf{T})$. By the definition of a generic weight, $\mathbf{T} = Z^\circ(\mathbf{M})_{\phi_e}$ and there is an e -JGC character $\lambda \in \mathcal{E}(\mathbf{M}^{F'}, \ell')$ such that $\eta \in \mathrm{Irr}(N_H(\mathbf{T}) \mid \lambda)$ and some $\chi \in \mathrm{Irr}(C)$ occurs in the Lusztig induction of λ to H . Since $\mathbf{M} = C_{\mathbf{H}}(\mathbf{T})$ and $\mathbf{T} = Z^\circ(\mathbf{M})_{\phi_e}$, we have $N_H(\mathbf{T}) = N_H(\mathbf{M})$. By compatibility of Lusztig induction with rational Lusztig series [21, Proposition 15.7], every constituent of the Lusztig induction of λ lies in a series labelled by an ℓ' -element. Hence $\chi \in \mathcal{X}_C$. Since $\mathcal{X}_C \subseteq \mathcal{E}(H, 1)$, the same compatibility forces the semisimple label of λ to be trivial, so λ is unipotent.

Let $\sigma \in E_H$ be arbitrary. Choose $0 \leq j < a$ such that $\sigma = F_2^j|_H$, and use the same symbol σ for the endomorphism F_2^j of $\mathbf{H}$. Let $\mathbf{T}_0 \leq \mathbf{B}_0$ be an F_2 -stable maximal torus and Borel subgroup, and let Π be the corresponding set of simple roots. Let $W = N_{\mathbf{H}}(\mathbf{T}_0)/\mathbf{T}_0$ be the Weyl group. For $I \subseteq \Pi$, let $\mathbf{L}_I$ be the corresponding standard Levi subgroup. By the rational parametrisation of F' -stable Levi subgroups in [22, Section 2.3], choose a pair (I, w) representing the H -conjugacy class of $\mathbf{M}$, where $w \in W$ satisfies $w(I) = I$. Choose a representative $\dot{w} \in N_{\mathbf{H}}(\mathbf{T}_0)$ of w such that $F_2(\dot{w}) = \dot{w}$. By Lang's theorem, there is $g \in \mathbf{H}$ such that $g^{-1}F'(g) = \dot{w}$. After replacing $(\mathbf{T}, \eta)$ by an H -conjugate and λ by the corresponding conjugate, we may assume $\mathbf{M} = g\mathbf{L}_I g^{-1}$. Since $\mathbf{L}_I$ and $\dot{w}$ are defined over $\mathbb{F}_2$, $\sigma(\mathbf{L}_I) = \mathbf{L}_I$ and $\sigma(\dot{w}) = \dot{w}$. Define $h_\sigma = g\sigma(g)^{-1}$ and set $\tau = \text{Int}(h_\sigma) \circ \sigma$, where $\text{Int}(x)$ denotes conjugation by x . Since $F'(g) = g\dot{w}$ and F' commutes with σ , we have

$$F'(h_\sigma) = g\dot{w}\sigma(g\dot{w})^{-1} = g\sigma(g)^{-1} = h_\sigma.$$

Hence $h_\sigma \in H$, and τ commutes with F' . Since w stabilises I , the endomorphism $F_w = \text{Int}(\dot{w}) \circ F'$ restricts to a Frobenius endomorphism of $\mathbf{L}_I$. Moreover, σ commutes with F_w . Conjugation by g maps $\mathbf{L}_I^{F_w}$ onto $\mathbf{M}^{F'}$ and intertwines σ with τ :

$$(3.1) \quad \tau(gxg^{-1}) = g\sigma(x)g^{-1} \quad (x \in \mathbf{H}).$$

Equation (3.1) shows that τ stabilises $\mathbf{M}$. Since τ commutes with F' , it also stabilises $\mathbf{T} = Z^\circ(\mathbf{M})_{\phi_e}$. Define

$$\lambda_I(x) = \lambda(gxg^{-1}) \quad (x \in \mathbf{L}_I^{F_w}).$$

Then λ_I is a unipotent character of $\mathbf{L}_I^{F_w}$. By (3.1), it suffices to prove $\lambda_I^\sigma = \lambda_I$ to obtain $\lambda^\tau = \lambda$. The map σ preserves each simple component of $[\mathbf{L}_I, \mathbf{L}_I]$ and commutes with F_w , so it preserves each F_w -orbit of components. Projection onto one component in each orbit identifies the corresponding finite factor with the fixed points of a power of F_w . Since $\sigma = F_2^j$ preserves each component, it commutes with this projection. Thus its action on each finite factor is induced by F_2^j and is a field automorphism. These factors have type A or C, and their unipotent characters are fixed by field automorphisms [22, p. 700]. The standard parametrisation of unipotent characters under central isogenies and direct products identifies the unipotent characters of $\mathbf{L}_I^{F_w}$ with tuples of unipotent characters of these finite factors [45, Remark 4.2.1]. This identification commutes with σ , and the torus contributes only its trivial character. Hence $\lambda_I^\sigma = \lambda_I$, and (3.1) gives $\lambda^\tau = \lambda$.

We next show that τ acts trivially on $N_H(\mathbf{M})/\mathbf{M}^{F'}$. Let $W_I = N_{\mathbf{L}_I}(\mathbf{T}_0)/\mathbf{T}_0$. The quotient $N_{\mathbf{H}}(\mathbf{L}_I)/\mathbf{L}_I$ identifies with $N_W(W_I)/W_I$. Every element of W has a representative fixed by F_2 , so $\sigma = F_2^j$ acts trivially on W and on this quotient. For $y \in N_{\mathbf{H}}(\mathbf{L}_I)^{F_w}$, the element $y^{-1}\sigma(y)$ belongs to $\mathbf{L}_I$ and is fixed by F_w , since σ commutes with F_w . Thus σ acts trivially on $N_{\mathbf{H}}(\mathbf{L}_I)^{F_w}/\mathbf{L}_I^{F_w}$. Conjugation by g identifies this quotient with $N_H(\mathbf{M})/\mathbf{M}^{F'}$. By (3.1), τ acts trivially on $N_H(\mathbf{M})/\mathbf{M}^{F'}$.

Set $N = N_H(\mathbf{T}) = N_H(\mathbf{M})$ and $N_\lambda = N_H(\mathbf{M}, \lambda)$. To obtain a τ -invariant extension of λ to N_λ , embed $\mathbf{H}$ in $\mathbf{K} = \text{PGL}_{2r+1}$ by $A \mapsto \pi(\text{diag}(A, 1))$, where $\pi: \text{GL}_{2r+1} \rightarrow \text{PGL}_{2r+1}$ is the quotient map. This embedding is defined over $\mathbb{F}_2$ and commutes with F' and the standard field endomorphisms. We identify $\mathbf{H}$, and hence $\mathbf{M}$, with their images in $\mathbf{K}$. The group $\mathbf{K}$ is simple of adjoint type, and $\mathbf{M}$ is a closed connected F' -stable reductive subgroup of $\mathbf{K}$. Let $E(\mathbf{K}^{F'})$ denote the subgroup of $\text{Aut}(\mathbf{K}^{F'})$ generated by the standard field and graph automorphisms. The element $h_\sigma \sigma \in \mathbf{K}^{F'} E(\mathbf{K}^{F'})$ induces τ on H by conjugation. We use the same symbol τ for this element. Define $\widehat{N}_\lambda = (\mathbf{K}^{F'} E(\mathbf{K}^{F'}))_{\mathbf{M}, \lambda}$, where the subscript denotes the stabiliser of the pair $(\mathbf{M}, \lambda)$. Since λ is unipotent, Späth's theorem [67, Theorem 3.3] shows that λ extends to $\widehat{N}_\lambda$. Both N_λ and τ lie in $\widehat{N}_\lambda$. Choose $\widetilde{\lambda} \in \text{Irr}(\widehat{N}_\lambda)$ extending λ . Since τ normalises N and N_λ , the restriction $\widehat{\lambda} = \text{Res}_{N_\lambda}^{\widehat{N}_\lambda} \widetilde{\lambda}$ is a τ -invariant extension of λ .

Since $\eta \in \text{Irr}(N \mid \lambda)$ and N_λ is the inertia group of λ in N , Clifford theory and Gallagher's theorem give

$$\eta = \text{Ind}_{N_\lambda}^N(\widehat{\lambda}\xi) \quad \text{for some } \xi \in \text{Irr}(N_\lambda/\mathbf{M}^{F'}),$$

where ξ is inflated to N_λ . The automorphism τ fixes $\widehat{\lambda}$ and acts trivially on $N_\lambda/\mathbf{M}^{F'}$, so it fixes ξ and hence η . Thus τ fixes $(\mathbf{T}, \eta)$. Because $h_\sigma \in H$ and $\tau = \text{Int}(h_\sigma) \circ \sigma$, the automorphisms τ and σ induce the same action on H -conjugacy classes of generic weights. Hence σ fixes the H -conjugacy class of $(\mathbf{T}, \eta)$. This proves that E_H fixes $\mathcal{W}(C)$ pointwise. $\square$

The following lemma combines the unipotent character parametrisation with Feng–Malle–Zhang's correspondence for relative Weyl groups [34] and Lemma 3.6.

Lemma 3.7. *If C is a unipotent ℓ -block of $H = \text{Sp}_{2r}(2^a)$, then there is an E_H -equivariant bijection $\mathcal{X}_C \rightarrow \mathcal{W}(C)$.*

Proof. Let $(\mathbf{L}_0, \lambda_0)$ be the unipotent e -cuspidal pair labelling C . As shown in the proof of Lemma 3.6, the set $\mathcal{X}_C = \text{Irr}(C) \cap \mathcal{E}(H, 1)$ is the generalised e -Harish-Chandra series attached to $(\mathbf{L}_0, \lambda_0)$. Taking the inverse of the parametrisation in [22, Theorem 2.8] gives a bijection $\mathcal{X}_C \rightarrow \text{Irr}(W_H(\mathbf{L}_0, \lambda_0))$. For unipotent characters, e -cuspidality and e -Jordan-cuspidality coincide [49, Definition 2.1 and Remark 2.2]. The block labellings in [20, Theorem 4.4(i)] and [49, Theorem A(e)] therefore assign C to the same pair $(\mathbf{L}_0, \lambda_0)$. Thus [34, Theorem 7.5] gives a bijection from $\text{Irr}(W_H(\mathbf{L}_0, \lambda_0))$ to

$$\bigsqcup_{(\mathbf{M}, \lambda)} \text{dz}(W_H(\mathbf{M}, \lambda)),$$

where the pairs run through the H -classes of e -JGC pairs with $\lambda \in \mathcal{E}(\mathbf{M}^{F'}, \ell')$ whose induced block is C . The maximal extendibility condition holds here by the type C case of [34, Proposition 3.20]. By [34, Lemma 3.21] and the definition of $\mathcal{W}(C)$, this disjoint union is in bijection with $\mathcal{W}(C)$. Composing these maps gives a bijection $\mathcal{X}_C \rightarrow \mathcal{W}(C)$. By Lemma 3.6, E_H fixes both sets pointwise, so this bijection is E_H -equivariant. $\square$

The BAW-goodness assertion below appears in [31, Remark 5.9]. Here we derive it from the preceding correspondence for generic weights and an integral basic set.

Proposition 3.8. *Every unipotent ℓ -block of $H = \text{Sp}_{2r}(2^a)$ is BAW-good and admits an iBAW bijection.*

Proof. Let C be a unipotent ℓ -block of H . The hypotheses of [34, Condition 6.1] hold: $\mathbf{H}$ is simple and simply connected, ℓ is odd and good, and $\ell \nmid 2^{a+1}|Z(\mathbf{H})^{F'}|$. The group $\mathcal{B}$ of field and graph automorphisms in [34, Condition 6.1] is E_H , since type C_r has no nontrivial graph automorphisms.

Since $Z(\mathbf{H})/Z^\circ(\mathbf{H}) = 1$, the hypothesis of [39, Theorem A] holds. The blockwise consequence stated immediately after that theorem, together with the equality $\mathcal{X}_C = \text{Irr}(C) \cap \mathcal{E}(H, 1)$ established in the proof of Lemma 3.6, shows that $\mathcal{X}_C$ is an integral basic set for C . By Lemma 3.6, E_H fixes $\mathcal{X}_C$ pointwise, so C is E_H -stable. The decomposition map is E_H -equivariant and maps $\mathcal{X}_C$ to a basis of $\mathbb{Z}\text{IBr}(C)$. Hence E_H also fixes $\text{IBr}(C)$ pointwise, and any bijection $\text{IBr}(C) \rightarrow \mathcal{X}_C$ is E_H -equivariant. Since $\mathcal{B} = E_H$, [34, Theorem 6.2] gives an E_H -equivariant bijection $\mathcal{W}(C) \rightarrow \text{Alp}(C)$. Composing this map with the preceding bijection $\text{IBr}(C) \rightarrow \mathcal{X}_C$ and the bijection $\mathcal{X}_C \rightarrow \mathcal{W}(C)$ from Lemma 3.7 gives an E_H -equivariant bijection

$$\Omega_C : \text{IBr}(C) \rightarrow \text{Alp}(C).$$

For $r \geq 4$, the group $H = \text{Sp}_{2r}(2^a)$ is nonabelian simple, with $\text{Aut}(H) = H \rtimes E_H$. By Table 3.1, H is its own universal covering group. Hence Lemma 2.10 proves the result. $\square$

We now combine Proposition 3.8 with Jordan reduction to obtain the result for even q and $n \geq 4$. This gives an independent proof of the inductive conclusion in [52, Theorem 3] in these

ranks, without its unitriangularity assumption. At linear primes, the unitriangularity assumed in [52, Theorem 3] follows from [46, Theorem 8.2].

Proposition 3.9. *Let $G = \mathrm{Sp}_{2n}(q)$ and $S = \mathrm{PSp}_{2n}(q)$, where $q = 2^f$, $n \geq 4$, and ℓ is an odd prime dividing $|G|$. Then every ℓ -block of G is BAW-good. Consequently, S satisfies iBAW at ℓ on $G = X_\ell(S)$.*

Proof. Let $\mathbf{G}$ be a simple, simply connected algebraic group of type C_n in characteristic 2, and let F be a Frobenius endomorphism such that $\mathbf{G}^F = G$. We verify the two hypotheses of Lemma 3.2. Choose the standard regular embedding $\mathbf{G} = \mathrm{Sp}_{2n} \hookrightarrow \tilde{\mathbf{G}} = \mathrm{CSp}_{2n}$, and set $\tilde{G} = \tilde{\mathbf{G}}^F = \mathrm{CSp}_{2n}(q)$. Let $\mu : \tilde{G} \rightarrow \mathbb{F}_q^\times$ be the multiplier homomorphism, characterised by $\langle gv, gw \rangle = \mu(g)\langle v, w \rangle$ for $g \in \tilde{G}$ and $v, w \in \mathbb{F}_q^{2n}$, where $\langle \cdot, \cdot \rangle$ is the defining symplectic form. Since q is even, for each $g \in \tilde{G}$ there is $c \in \mathbb{F}_q^\times$ with $c^2 = \mu(g)$. Then $c^{-1}g \in G$, so $\tilde{G} = GZ(\tilde{G})$ and $\tilde{G}$ acts on G by inner automorphisms. Let $E \cong C_f$ act on G and $\tilde{G}$ by the field automorphisms induced by entrywise squaring. Then $\mathrm{Aut}(G) = G \rtimes E$. For every $\psi \in \mathrm{IBr}(G)$, the group $\tilde{G}$ fixes ψ , and hence

$$(\tilde{G} \rtimes E)_\psi = \tilde{G} \rtimes E_\psi.$$

The group E_ψ is cyclic, so by Lemma 2.9, ψ extends to $G \rtimes E_\psi$. For every semisimple ℓ' -element $s \in (\mathbf{G}^*)^F$, the construction in [60, Corollary 4.2 and Lemmas 4.3 and 4.5] allows us, after conjugating the Levi subgroup associated with s and its series idempotent by an element of G , to choose the group A_s in Lemma 3.2 inside E , since type C_n with $n \geq 4$ has no graph automorphism. Restricting the preceding factorisation and extensions to A_s verifies condition (i) of that lemma.

To verify condition (ii) of Lemma 3.2, let $(\mathbf{K}, F')$ be a pair as in that condition. If $\ell \nmid |\mathbf{K}^{F'}|$, then every ℓ -block c has defect zero. Writing $\mathrm{Irr}(c) = \{\chi\}$, the map $\chi^0 \mapsto (1, \chi)$ is an iBAW bijection for c , since the global and local modular character triples are identical. We may therefore assume that $\ell \mid |\mathbf{K}^{F'}|$.

A connected Dynkin subdiagram of C_n has type A_m or C_m . If $\mathbf{K}$ has type A_m (including $C_1 = A_1$), then $\mathbf{K}^{F'}$ is a special linear or unitary group. The result for these groups established after Lemma 3.2 gives the required iBAW bijections for its ℓ -blocks.

Suppose that $\mathbf{K}$ has type C_m , where $m \geq 2$, and let c be a strictly quasi-isolated ℓ -block of $\mathbf{K}^{F'}$. Choose a strictly quasi-isolated semisimple ℓ' -element $s \in \mathbf{K}^{*F'}$ such that $\mathrm{Irr}(c) \subseteq \mathcal{E}_\ell(\mathbf{K}^{F'}, s)$. Since $\mathbf{K}^*$ has type B_m in characteristic 2 and s is quasi-isolated, [10, Example 4.8] gives $s = 1$. Hence c is unipotent.

First suppose that F' is split, and write $\mathbf{K}^{F'} = \mathrm{Sp}_{2m}(2^a)$. For $m \geq 4$, Proposition 3.8 gives the required iBAW bijection for c . The groups $\mathrm{Sp}_4(2^a)$ with $a \geq 2$ and $\mathrm{Sp}_6(2^a)$ with $a \geq 1$ satisfy iBAW at ℓ by [63, Theorem 5.5]. Corollary 2.5 therefore gives the required iBAW bijections for their ℓ -blocks. The simplicity assumption in Lemma 3.2(ii) excludes $\mathrm{Sp}_4(2)$. The program `sp6.g`, described in Section A.2, supplies the numerical details of the calculation on $2.\mathrm{Sp}_6(2)$ at $\ell = 3$ in the proof of [63, Theorem 5.5]. If F' is not split, then $m = 2$ and $\mathbf{K}^{F'}$ is a Suzuki group. Each simple Suzuki group satisfies iBAW at ℓ by [65, Corollary 6.3], so Corollary 2.5 gives the required iBAW bijections for its ℓ -blocks.

This proves condition (ii) of Lemma 3.2. Condition (i) was verified above. Since $G = S$ is its own universal covering group by Table 3.1, Lemma 3.2 shows that every ℓ -block of G is BAW-good. Since $G = X_\ell(S)$, Remark 2.4(a) gives the final assertion. $\square$

3.3. Odd fields. Throughout this subsection, assume that $q = p^f$ is odd, $n \geq 3$, and that $\ell \mid |S|$ is an odd prime distinct from p . For linear primes, the unitriangularity assumed in [52, Theorem 2] follows from [46, Theorem 8.2 and Corollary 8.7]. For $\mathrm{Sp}_{2n}(q)$ and $\mathrm{Spin}_{2n+1}(q)$, Feng and Späth proved unitriangularity for the sums of ℓ -blocks associated with semisimple ℓ' -elements whose connected centralisers are Levi subgroups of the dual algebraic group [35, Proposition 7.3].

Set

$$G = \mathrm{Sp}_{2n}(q), \quad \tilde{G} = \mathrm{CSp}_{2n}(q).$$

Let F_p be the automorphism of $\tilde{G}$ obtained by raising every matrix entry to the p th power, and set $E = \langle F_p \rangle \cong C_f$. The group E acts on G by restriction. Here $G = X_\ell(S)$. We use the conformal group to describe the diagonal and field actions. Let N be the group of linear Brauer characters of $\tilde{G}/G$, and set $A = N \rtimes E$. The group N is cyclic of order equal to the ℓ' -part of $q - 1$. The group A acts on the ℓ -blocks and irreducible Brauer characters of $\tilde{G}$, with N acting by tensoring and E by field automorphisms. For a block $\tilde{B}$ of $\tilde{G}$, let $J = A_{\tilde{B}}$ be its stabiliser in A . Write $D = J \cap N$.

Lemma 3.10. *For every ℓ -block $\tilde{B}$ of $\tilde{G} = \mathrm{CSp}_{2n}(q)$, D is a normal subgroup of J of order at most 2, and J/D is cyclic. Consequently, every subgroup of J is 2-hypoelementary.*

Proof. The multiplier homomorphism $\mu : \tilde{G} \rightarrow \mathbb{F}_q^\times$ is surjective with kernel G , and $\mu(cI) = c^2$ for $c \in \mathbb{F}_q^\times$. Thus it identifies $\tilde{G}/G$ with $\mathbb{F}_q^\times$ and sends $Z(\tilde{G})$ onto $(\mathbb{F}_q^\times)^2$. Hence $|\tilde{G} : GZ(\tilde{G})| = 2$. Applying Lemma 2.2 to $G \triangleleft \tilde{G}$ and $\tilde{B}$, using reduction to identify the linear characters, gives $|D| \leq 2$. Since $N \triangleleft A$, we have $D \triangleleft J$. The projection $A \rightarrow E$ embeds J/D in E , so J/D is cyclic. As $|D| \leq 2$, we have $D \leq O_2(J)$, and $J/O_2(J)$ is cyclic. Thus J and every subgroup of J are 2-hypoelementary. $\square$

Write $Z_{\tilde{B}} = \mathrm{Irr}(\tilde{B}) \cap \mathcal{E}(\tilde{G}, \ell')$. The set $Z_{\tilde{B}}$ is stable under J . Indeed, $\mathcal{E}(\tilde{G}, \ell')$ is preserved by tensoring with linear ℓ' -characters [52, Remark 2.5] and by field automorphisms [22, Theorem 3.1].

Proposition 3.11. *For every ℓ -block $\tilde{B}$ of $\tilde{G} = \mathrm{CSp}_{2n}(q)$, there is a J -equivariant bijection $\mathrm{IBr}(\tilde{B}) \rightarrow Z_{\tilde{B}}$.*

Proof. The prime ℓ is odd and hence good for type C. The conformal algebraic group has connected centre. By [43, Theorem 5.1], $\mathcal{E}(\tilde{G}, \ell')$ is an integral basic set for $\tilde{G}$. Since the decomposition map respects the block decomposition, $Z_{\tilde{B}}$ is an integral basic set for $\tilde{B}$.

Identify N with the group of ordinary linear ℓ' -characters of $\tilde{G}/G$ via reduction. This identification is E -equivariant, and the decomposition map commutes with tensoring and field automorphisms. Hence $d_{\tilde{B}}$ is J -equivariant. The set $Z_{\tilde{B}}$ is J -stable, and J is 2-hypoelementary by Lemma 3.10. Lemma 2.8 gives the asserted bijection. $\square$

Lemma 3.12. *Let $G = \mathrm{Sp}_{2n}(q)$ and $\tilde{G} = \mathrm{CSp}_{2n}(q)$. Then there is an $\mathrm{Aut}(G)$ -equivariant bijection*

$$\beta : \mathrm{IBr}(G) \rightarrow \mathcal{E}(G, \ell')$$

such that

$$\beta(\mathrm{IBr}(B)) = \mathrm{Irr}(B) \cap \mathcal{E}(G, \ell')$$

for every ℓ -block B of G . Moreover, for every $\varphi \in \mathrm{IBr}(G)$,

$$(\tilde{G} \rtimes E)_\varphi = \tilde{G}_\varphi \rtimes E_\varphi,$$

and φ extends to $G \rtimes E_\varphi$.

Proof. Let $\mathbf{G}$ be the simply connected symplectic algebraic group and let F be its defining Frobenius endomorphism, so that $\mathbf{G}^F = G$. Since ℓ is odd and $Z(\mathbf{G}) = \{\pm I\}$, [39, Theorem A] shows that $\mathcal{E}(G, \ell')$ is an integral basic set for G . Fix an ℓ -block B of G and set $\mathcal{X}_B = \mathrm{Irr}(B) \cap \mathcal{E}(G, \ell')$. Since the decomposition map respects the block decomposition, $\mathcal{X}_B$ is an integral basic set for B .

Diagonal automorphisms preserve each rational Lusztig series [21, Proposition 15.6(i)]. Field automorphisms send a series $\mathcal{E}(G, s)$ to a series $\mathcal{E}(G, s')$ with $|s'| = |s|$ [71, Proposition 7.2]. Thus $\mathcal{E}(G, \ell')$ is $\mathrm{Aut}(G)$ -stable, and $\mathcal{X}_B$ is $\mathrm{Aut}(G)_B$ -stable. The actions on $\mathcal{X}_B$ and $\mathrm{IBr}(B)$ factor through $L = \mathrm{Aut}(G)_B / \mathrm{Inn}(G)$, and the decomposition map is L -equivariant. The group

$\text{Out}(G) \cong C_2 \times C_f$ is 2-hypoelementary, so the same holds for L . Lemma 2.8 gives an $\text{Aut}(G)_B$ -equivariant bijection $\beta_B : \text{IBr}(B) \rightarrow \mathcal{X}_B$. Applying Remark 2.4(b) to these blockwise bijections gives the required map β .

Let $\varphi \in \text{IBr}(G)$ and write $\chi = \beta(\varphi)$. Applying [23, Theorem 3.1] to the ordinary character χ gives $(\tilde{G} \rtimes E)_\chi = \tilde{G}_\chi \rtimes E_\chi$. Equivariance and bijectivity of β therefore give

$$(\tilde{G} \rtimes E)_\varphi = (\tilde{G} \rtimes E)_\chi = \tilde{G}_\chi \rtimes E_\chi = \tilde{G}_\varphi \rtimes E_\varphi.$$

Finally, E_φ is cyclic, so Lemma 2.9 gives the required extension. $\square$

Remark 3.13. The proofs of Proposition 3.11 and Lemma 3.12 apply Conlon's theorem over the 2-adic integers through Lemma 2.8. Here ℓ is odd. These arguments give neither a canonical Brauer labelling nor a unitriangular decomposition matrix.

For odd q , the following proposition removes the unitriangularity assumptions from Li's result [52, Theorem 2]. We use the ordinary character and weight calculations in that paper, together with its stabiliser and extension results for weights. Proposition 3.11 replaces the use of unitriangularity for $\tilde{G}$. Lemma 3.12 gives the conclusions of [52, Lemma 5.4] without the unitriangularity assumption for G .

Proposition 3.14. *Let $G = \text{Sp}_{2n}(q)$ and $S = \text{PSp}_{2n}(q)$. Then every ℓ -block of G satisfies the inductive BAW condition. Consequently, S satisfies iBAW at ℓ on $G = X_\ell(S)$.*

Proof. We apply [19, Theorem 4.5] with $\mathcal{B}$ equal to the set of all ℓ -blocks of G and with $\tilde{G}$ and E as above. The group E is cyclic, $G = [\tilde{G}, \tilde{G}]$, and

$$C_{\tilde{G} \rtimes E}(G) = Z(\tilde{G}), \quad (\tilde{G} \rtimes E)/Z(\tilde{G}) \cong \text{Aut}(G).$$

Since $\tilde{G}/G$ is cyclic, every $\varphi \in \text{IBr}(G)$ extends to $\tilde{G}_\varphi$ by Lemma 2.9. For a weight (Q, θ) of G , the quotient $(N_{\tilde{G}}(Q)/Q)_\theta / (N_G(Q)/Q)$ is also cyclic, so the same lemma extends θ to $(N_{\tilde{G}}(Q)/Q)_\theta$. This verifies condition (i) of [19, Theorem 4.5].

The parametrisations of ordinary characters in [52, Lemma 3.13] and of weights in [52, Theorem 4.13 and Remarks 4.7–4.8] give a bijection

$$\rho : \mathcal{E}(\tilde{G}, \ell') \rightarrow \text{Alp}(\tilde{G})$$

that preserves blocks. The proofs of [52, Lemmas 5.1 and 5.3] show that field automorphisms and tensoring by N induce the same changes of labels on the ordinary characters and on the corresponding weights. In the proof of [52, Lemma 5.1], unitriangularity identifies the field action on Brauer character labels with that on ordinary character labels. In the proof of [52, Lemma 5.3], it identifies the corresponding actions of tensoring with linear characters. The comparison between ordinary characters and weights uses neither identification. Thus ρ is A -equivariant.

Choose one block $\tilde{B}$ in each A -orbit and let $J = A_{\tilde{B}}$. Proposition 3.11 gives a J -equivariant bijection

$$\alpha_{\tilde{B}} : \text{IBr}(\tilde{B}) \rightarrow Z_{\tilde{B}}.$$

Composing with ρ and applying Remark 2.4(b) to the A -orbits gives an A -equivariant bijection

$$\tilde{\Omega} : \text{IBr}(\tilde{G}) \rightarrow \text{Alp}(\tilde{G})$$

that preserves blocks. In the notation of [19, Theorem 4.5(ii)],

$$J_G(\tilde{\varphi}) = J_G(\tilde{\Omega}(\tilde{\varphi})) = \tilde{G} \quad (\tilde{\varphi} \in \text{IBr}(\tilde{G}))$$

by [52, Remark 2.2(1)]. This verifies condition (ii) of [19, Theorem 4.5]. For every $\varphi \in \text{IBr}(G)$, Lemma 3.12 gives the stabiliser factorisation and extension required by condition (iii) of [19, Theorem 4.5].

Let $\omega \in \text{Alp}(G)$. By [52, Section 2.C], choose a representative (R, ψ) with R in the standard family used in [52, Lemma 5.5]. By [52, Lemma 5.5(1)–(2)],

$$(\tilde{G}E)_{R,\psi} = \tilde{G}_{R,\psi}(GE)_{R,\psi},$$

and ψ extends to $(GE)_{R,\psi}/R$. For $H \in \{\tilde{G}E, \tilde{G}, GE\}$, the stabiliser of the conjugacy class ω satisfies $H_\omega = GH_{R,\psi}$. Hence

$$(\tilde{G}E)_\omega = \tilde{G}_\omega(GE)_\omega = \tilde{G}_\omega E_\omega.$$

This verifies condition (iv) of [19, Theorem 4.5].

Since $\text{Out}(G)$ is abelian, [19, Theorem 4.5] proves the inductive BAW condition for every ℓ -block of G . As $G = X_\ell(S)$, Remark 2.4(a) shows that S satisfies iBAW at ℓ . $\square$

3.4. Proof of Theorem 3.1.

Proof. If $n = 2$, Proposition 3.5 gives the result. We may therefore assume that $n \geq 3$. The case $\ell = p$ follows from [65, Theorem C], so assume that $\ell \neq p$.

If q is odd, Proposition 3.3 applies when $\ell = 2$, and Proposition 3.14 applies when ℓ is odd. If q is even, then ℓ is odd. For $n = 3$, the result follows from [63, Theorem 5.5], and for $n \geq 4$ it follows from Proposition 3.9. $\square$

4. ODD-DIMENSIONAL ORTHOGONAL GROUPS

In even characteristic, the finite simple groups of types B_n and C_n are isomorphic. In every characteristic, the identifications $B_1 = A_1$ and $B_2 = C_2$ reduce the cases of ranks one and two to the corresponding results for types A and C. Unless stated otherwise, throughout this section $q = p^f$ is an odd prime power, $n \geq 3$, and $S = \Omega_{2n+1}(q)$ denotes the nonabelian simple group under consideration. Our arguments use special Clifford groups, reduction to proper Levi subgroups, and generalised Gelfand–Graev characters, and include a separate treatment of the exceptional universal $2'$ -covering group $3.\Omega_7(3)$.

Theorem 4.1. *Let $S = \Omega_{2n+1}(q)$, and let ℓ be a prime dividing $|S|$. Then every ℓ -block of $X_\ell(S)$ satisfies the inductive BAW condition. Consequently, S satisfies iBAW at ℓ on the universal ℓ' -covering group $X_\ell(S)$.*

4.1. Covering groups. By [56, Tables 24.2–24.3 and Remark 24.19], the universal covering group of $\Omega_{2n+1}(q)$ has kernel C_2 , except that the multiplier of $\Omega_7(3)$ is C_6 . The required covering groups are listed in Table 4.1.

TABLE 4.1. Universal ℓ' -covering groups of type B in odd characteristic.

Parameters	Prime	$X_\ell(\Omega_{2n+1}(q))$
$(n, q) \neq (3, 3)$	$\ell = 2$	$\Omega_{2n+1}(q)$
$(n, q) \neq (3, 3)$	ℓ odd	$\text{Spin}_{2n+1}(q)$
$(n, q) = (3, 3)$	$\ell = 2$	$3.\Omega_7(3)$
$(n, q) = (3, 3)$	$\ell = 3$	$2.\Omega_7(3) = \text{Spin}_7(3)$
$(n, q) = (3, 3)$	$\ell \notin \{2, 3\}$	$6.\Omega_7(3)$

The last line is used only for $\ell \in \{5, 7, 13\}$. In particular, the universal $2'$ -covering group of $\Omega_7(3)$ is $3.\Omega_7(3)$.

We have $\text{Out}(S) \cong C_2 \times C_f$, where the two factors are induced by diagonal and field automorphisms, respectively. See, for example, [56, Proposition 24.21 and Theorem 24.24].

4.2. Odd nondefining primes. For odd nondefining primes, Feng, Li, and Zhang proved the result for $\Omega_7(3)$ without a unitriangularity assumption and for the remaining groups under their unitriangularity assumption [28, Theorem 1 and Assumption 3.11]. We remove this assumption from the bijection and stabiliser arguments using permutation lattices and Lemma 2.2.

Throughout this subsection, ℓ is an odd prime not dividing q . Let $G = \text{Spin}_{2n+1}(q)$, and let $G \triangleleft \tilde{G}$ be its embedding in the corresponding finite special Clifford group described in [28, Section 3.1]. Let F_p be the field automorphism of $\tilde{G}$ induced by $x \mapsto x^p$ on $\mathbb{F}_q$, and set $E = \langle F_p \rangle$. The group E acts on G by restriction. Let C be the group of ordinary linear characters of $\tilde{G}/G$ of ℓ' -order, and set $A = C \rtimes E$. The group A acts on the ℓ -blocks and ordinary and Brauer characters of $\tilde{G}$. Here C acts by tensoring with the inflated linear characters, using their reductions on Brauer characters, and E acts by field automorphisms. The natural actions of $\text{Aut}(G)$ on blocks and characters factor through $\text{Out}(G)$. For $M \in \{G, \tilde{G}\}$ and an ℓ -block B of M , set

$$\mathcal{X}_B = \text{Irr}(B) \cap \mathcal{E}(M, \ell').$$

Lemma 4.2. *Let $G = \text{Spin}_{2n+1}(q)$, and let $\tilde{G}$ be the corresponding finite special Clifford group. Let $M \in \{G, \tilde{G}\}$, let B be an ℓ -block of M , and set*

$$J_B = \begin{cases} \text{Aut}(G)_B / \text{Inn}(G), & M = G, \\ A_B, & M = \tilde{G}. \end{cases}$$

Then J_B is 2-hypoelementary, $\mathcal{X}_B$ is J_B -stable, and the decomposition map restricts to an isomorphism

$$d_B : \mathbb{Z}[\mathcal{X}_B] \xrightarrow{\sim} \mathbb{Z}[\text{IBr}(B)]$$

of $\mathbb{Z}J_B$ -permutation lattices. Consequently, there exists a J_B -equivariant bijection

$$\text{IBr}(B) \longrightarrow \mathcal{X}_B.$$

Proof. The connected reductive algebraic groups defining G and $\tilde{G}$ have type B_n . Their centres have component groups $Z(\mathbf{H})/Z^\circ(\mathbf{H})$ of orders 2 and 1, respectively, where $\mathbf{H}$ denotes the corresponding algebraic group. Since the odd prime ℓ is good for type B_n and divides the order of neither component group, the hypotheses of [28, Theorem 2.3] hold for both groups. Applying that theorem to all rational ℓ' -series shows that $\mathcal{E}(G, \ell')$ and $\mathcal{E}(\tilde{G}, \ell')$ are integral basic sets. Since the decomposition map is block diagonal, $\mathcal{X}_B$ is an integral basic set for B .

Diagonal automorphisms preserve each rational Lusztig series [21, Proposition 15.6(i)]. For $M \in \{G, \tilde{G}\}$, field automorphisms send a series $\mathcal{E}(M, s)$ to a series $\mathcal{E}(M, s')$ with $|s'| = |s|$ [71, Proposition 7.2]. Thus field and diagonal automorphisms preserve $\mathcal{E}(G, \ell')$, and field automorphisms preserve $\mathcal{E}(\tilde{G}, \ell')$. Tensoring with the inflation of an ordinary linear character of $\tilde{G}/G$ of ℓ' -order also preserves $\mathcal{E}(\tilde{G}, \ell')$, since it multiplies the semisimple element labelling the series by a central ℓ' -element [45, Theorem 4.7.1(3)]. Hence $\mathcal{X}_B$ is J_B -stable. The decomposition map commutes with automorphisms and with tensoring by linear characters, so its restriction to $\mathbb{Z}[\mathcal{X}_B]$ is J_B -equivariant. For $M = G$, we have $\text{Out}(G) \cong C_2 \times E$. This group is 2-hypoelementary, and therefore so is its subgroup J_B .

Now let $M = \tilde{G}$, write $J = A_B$, and set $D = J \cap C$. We have $|\tilde{G} : GZ(\tilde{G})| = 2$ [28, proof of Lemma 5.16]. Applying Lemma 2.2 to $G \triangleleft \tilde{G}$ and B gives $|D| \leq 2$. Since $C \triangleleft A$, we have $D \triangleleft J$, and hence $D \leq O_2(J)$. Projection to E embeds J/D in the cyclic group E . Therefore $J/O_2(J)$ is cyclic, so $J = J_B$ is 2-hypoelementary. Lemma 2.8 now gives the asserted lattice isomorphism and equivariant bijection in both cases. $\square$

Proposition 4.3. *Let $S = \Omega_{2n+1}(q)$, and let ℓ be an odd prime dividing $|S|$ with $\ell \neq p$. Then every ℓ -block of $X_\ell(S)$ satisfies the inductive BAW condition.*

Proof. First suppose $(n, q) \neq (3, 3)$. Then $G = X_\ell(S)$ by Table 4.1. We apply Brough–Späth’s criterion in the form of [28, Theorem 2.1] and verify its conditions (1)–(4). The structural hypotheses in condition (1) hold for G , $\tilde{G}$, and E , as noted in the proof of [28, Theorem 1].

For each A -orbit of blocks of $\tilde{G}$, choose a representative $\tilde{B}$. Compose the $A_{\tilde{B}}$ -equivariant bijection $\text{IBr}(\tilde{B}) \rightarrow \mathcal{X}_{\tilde{B}}$ of Lemma 4.2 with the restriction $\mathcal{X}_{\tilde{B}} \rightarrow \text{Alp}(\tilde{B})$ of the blockwise A -equivariant bijection in [28, Proposition 7.2]. Extending these bijections to the remaining blocks by Remark 2.4(b) gives the blockwise A -equivariant bijection required by condition (2)(i) of the criterion. Condition (2)(ii) holds by [28, Remark 2.2], since ℓ is odd.

Let B be an ℓ -block of G , let $\varphi \in \text{IBr}(B)$, and let $\chi \in \mathcal{X}_B$ be its image under the bijection of Lemma 4.2. Equivariance and the fact that each character determines its block give $\text{Aut}(G)_{\varphi} = \text{Aut}(G)_{\chi}$. The ordinary character stabiliser theorem for type B due to Cabanes and Späth, stated in [28, Theorem 3.10], gives $(\tilde{G} \rtimes E)_{\chi} = \tilde{G}_{\chi} \rtimes E_{\chi}$. The equality of the automorphism stabilisers identifies the stabilisers of φ and χ in $\tilde{G} \rtimes E$, $\tilde{G}$ and E . Hence

$$(\tilde{G} \rtimes E)_{\varphi} = \tilde{G}_{\varphi} \rtimes E_{\varphi}.$$

This verifies condition (3) of the criterion. The weight stabiliser factorisation required by condition (4) follows from [28, Proposition 7.5]. The criterion therefore proves the inductive BAW condition for every ℓ -block of G .

If $(n, q) = (3, 3)$, then $\ell \in \{5, 7, 13\}$ and $X_{\ell}(S) = 6.\Omega_7(3)$ by Table 4.1. Its Sylow ℓ -subgroups have order ℓ , so every ℓ -block has cyclic defect. As in the proof of [28, Theorem 1], the result follows from [50, Theorem 1.1]. $\square$

4.3. Character representatives for the Levi reduction. The reductions for nonprincipal 2-blocks require compatible character representatives on a Levi subgroup and its derived subgroup. Lemma 4.4 combines suitable representatives on direct factors. Lemma 4.5 describes the product decomposition of the derived subgroup and the action induced by the regular embedding. Lemma 4.6 deduces the stabiliser factorisation for a Brauer character of the Levi subgroup. The required representatives and stabiliser factorisations on the factors are obtained in the proof of Proposition 4.13.

Lemma 4.4. *Let $I = \{1, \dots, r\}$, and for each $i \in I$ let H_i be a finite group with $\Delta_i \leq \text{Aut}(H_i)$. Set*

$$N = H_1 \times \dots \times H_r, \quad \Delta = \prod_{i \in I} \Delta_i,$$

with Δ acting componentwise on N . Let a cyclic group E act on N by automorphisms, normalise Δ , and permute the pairs (H_i, Δ_i) . Choose $\chi_i \in \text{IBr}(H_i)$ for each $i \in I$. Write $\mathcal{O}$ for the Δ -orbit of $\chi_1 \times \dots \times \chi_r$ and $\mathcal{O}_i$ for the Δ_i -orbit of χ_i . Let $E_{\mathcal{O}}$ be the setwise stabiliser of $\mathcal{O}$ in E , and choose a generator τ of $E_{\mathcal{O}}$. Suppose that for each cycle C of the permutation of I induced by τ , there are $i \in C$ and $\theta_i \in \mathcal{O}_i$ such that

$$(\Delta_i \rtimes \langle \tau^{|C|} \rangle)_{\theta_i} = (\Delta_i)_{\theta_i} \langle \tau^{|C|} \rangle_{\theta_i}.$$

Then $\mathcal{O}$ contains a character θ fixed by $E_{\mathcal{O}}$. In particular,

$$(\Delta \rtimes E_{\mathcal{O}})_{\theta} = \Delta_{\theta} \rtimes E_{\mathcal{O}}.$$

Proof. Since Δ acts componentwise, $\mathcal{O}$ consists of the products of characters in the orbits $\mathcal{O}_i$, and τ sends $\mathcal{O}_i$ to $\mathcal{O}_{i\tau}$. For a cycle C of length t , choose i and θ_i as in the hypothesis. Some element of $\Delta_i \tau^t$ fixes θ_i , and projection of the assumed stabiliser equality onto $\langle \tau^t \rangle$ shows that τ^t fixes θ_i . Set $\theta_{i\tau^j} = \theta_i^{\tau^j}$ for $0 \leq j < t$ on each cycle. Their product belongs to $\mathcal{O}$ and is fixed by τ , hence by $E_{\mathcal{O}}$, which gives the stabiliser equality. $\square$

Lemma 4.5. *Let $\mathbf{G} \hookrightarrow \tilde{\mathbf{G}}$ be a regular embedding, where $\mathbf{G}$ is simply connected of type $\mathbf{B}_n$. Let F be a Steinberg endomorphism of $\tilde{\mathbf{G}}$ stabilising $\mathbf{G}$, and let $\mathbf{L}$ be an F -stable Levi subgroup of $\mathbf{G}$. Set*

$$\mathbf{H} = [\mathbf{L}, \mathbf{L}], \quad L_0 = \mathbf{H}^F, \quad L = \mathbf{L}^F, \quad \tilde{L} = (\mathbf{L}Z(\tilde{\mathbf{G}}))^F.$$

Write $\mathbf{H} = \prod_{j \in I} \mathbf{H}_j$, where I is a finite index set for its simple components. Denote the F -orbits on I by $C_1, \dots, C_r$. Choose $x_i \in C_i$, and set $n_i = |C_i|$ and $H_i = \mathbf{H}_{x_i}^{F^{n_i}}$. Then the following hold.

- (1) *There is an isomorphism*

$$(4.5.1) \quad L_0 \cong \prod_{i=1}^r H_i,$$

where conjugation by $\tilde{L}$ preserves each factor H_i . Let $\rho_i : \tilde{L} \rightarrow \text{Aut}(H_i)$ be the conjugation homomorphism on the i th factor of (4.5.1), and set $\Delta_i = \rho_i(\tilde{L})$. Then Δ_i is the full group of inner and diagonal automorphisms of H_i , and the homomorphism

$$\tilde{L} \longrightarrow \prod_{i=1}^r \Delta_i, \quad g \longmapsto (\rho_1(g), \dots, \rho_r(g))$$

is surjective.

- (2) For $\chi_i \in \text{IBr}(H_i)$, let $\mathcal{O}_i$ be the Δ_i -orbit of χ_i . Using (4.5.1), the $\tilde{L}$ -orbit of $\chi_1 \times \dots \times \chi_r$ is

$$\{\psi_1 \times \dots \times \psi_r \mid \psi_i \in \mathcal{O}_i \text{ for } 1 \leq i \leq r\}.$$

- (3) We have $L_0 \triangleleft L \triangleleft \tilde{L}$, and the quotient $\tilde{L}/L_0$ is abelian. If σ is a field endomorphism of $\tilde{\mathbf{G}}$ commuting with F and stabilising $\mathbf{G}$ and $\mathbf{L}$, then the automorphism induced by σ on $\tilde{\mathbf{G}}^F$ normalises this chain.

Proof. (1) Since $\mathbf{G}$ is simply connected, [56, Proposition 12.14] shows that $\mathbf{H} = [\mathbf{L}, \mathbf{L}]$ is simply connected. It is therefore a direct product of its simple components. For $1 \leq i \leq r$, set

$$\mathbf{K}_i = \prod_{j \in C_i} \mathbf{H}_j.$$

Each $\mathbf{K}_i$ is F -stable, and projection onto the component $\mathbf{H}_{x_i}$ gives an isomorphism

$$\mathbf{K}_i^F \longrightarrow \mathbf{H}_{x_i}^{F^{n_i}} = H_i.$$

Taking the product over the F -orbits gives (4.5.1).

Set $\mathbf{M} = \mathbf{L}Z(\tilde{\mathbf{G}})$. This is an F -stable Levi subgroup of $\tilde{\mathbf{G}}$. Its centre is connected because $Z(\tilde{\mathbf{G}})$ is connected. We have $[\mathbf{M}, \mathbf{M}] = \mathbf{H}$ and $\mathbf{M} = \mathbf{H}Z(\mathbf{M})$. Thus conjugation by $\mathbf{M}$ preserves each direct factor of $\mathbf{H}$, and

$$\mathbf{M}/Z(\mathbf{M}) \cong \mathbf{H}/Z(\mathbf{H}) \cong \prod_i \mathbf{K}_i/Z(\mathbf{K}_i).$$

Since $Z(\mathbf{M})$ is connected, Lang's theorem makes the natural homomorphism

$$(4.5.2) \quad \tilde{L} = \mathbf{M}^F \longrightarrow (\mathbf{M}/Z(\mathbf{M}))^F \cong \prod_i (\mathbf{K}_i/Z(\mathbf{K}_i))^F$$

surjective. Conjugation by $(\mathbf{K}_i/Z(\mathbf{K}_i))^F$ induces the full group of inner and diagonal automorphisms of $\mathbf{K}_i^F \cong H_i$ (see, for example, [45, Remark 1.5.12]). Hence Δ_i is this group, and the homomorphism $\tilde{L} \rightarrow \prod_i \Delta_i$ is surjective.

(2) Products of characters identify $\text{IBr}(L_0)$ with $\prod_i \text{IBr}(H_i)$. The orbit description follows from (1).

(3) The normal inclusions follow from $\mathbf{H} \triangleleft \mathbf{L} \triangleleft \mathbf{M}$, and $[\tilde{L}, \tilde{L}] \leq [\mathbf{M}, \mathbf{M}]^F = L_0$. Hence $\tilde{L}/L_0$ is abelian. Since σ preserves $\mathbf{L}$, $[\mathbf{L}, \mathbf{L}]$ and $Z(\tilde{\mathbf{G}})$, and commutes with F , it also preserves L_0 , L and $\tilde{L}$. $\square$

Lemma 4.6. *Let ℓ be a prime and $N \leq H \leq G$ be finite groups. Assume that N is normal in G and that the quotient group G/N is abelian. Let a finite group E act on G by automorphisms stabilising N and H . For $\psi \in \text{IBr}(H)$, let θ be an irreducible constituent of ψ_N . Suppose that θ extends to G_θ and that*

$$(G \rtimes E)_\theta = G_\theta \rtimes E_\theta.$$

Then

$$(G \rtimes E)_\psi = G_\psi \rtimes E_\psi.$$

Proof. Set $I = H_\theta$, choose an extension $\hat{\theta} \in \text{IBr}(G_\theta)$ of θ , and let $\eta \in \text{IBr}(I \mid \theta)$ be the Clifford correspondent of ψ . By Gallagher's theorem for Brauer characters, we have $\eta = (\hat{\theta})_I \lambda$ for some $\lambda \in \text{IBr}(I/N)$, inflated to I . The character $(\hat{\theta})_I$ is fixed by G_θ , as it is the restriction of a character of that group. Since G/N is abelian, conjugation by G_θ acts trivially on I/N and therefore fixes λ . Thus G_θ fixes η and, by induction from I to H , also fixes ψ .

Let $g \in (G \rtimes E)_\psi$. The constituents of ψ_N form a single H -orbit, so there is $h \in H$ such that hg fixes θ . By hypothesis, write $hg = ae$ with $a \in G_\theta$ and $e \in E_\theta$. The elements h , g and a fix ψ , so e fixes ψ . Consequently $g = h^{-1}ae \in G_\psi E_\psi$. The reverse inclusion is immediate. $\square$

4.4. The principal 2-block in rank at least three. Throughout this subsection, let $\mathbf{G}$ be a simply connected algebraic group of type $\mathbf{B}_n$, and let F_0 be a standard Frobenius endomorphism of $\mathbf{G}$ such that $\mathbf{G}^{F_0} = \text{Spin}_{2n+1}(p)$. Set $F = F_0^f$ and $G = \mathbf{G}^F = \text{Spin}_{2n+1}(q)$. Choose a regular embedding $\mathbf{G} \hookrightarrow \tilde{\mathbf{G}}$ to which F_0 extends, and denote this extension also by F_0 . Set $\tilde{G} = \tilde{\mathbf{G}}^F$. For the remainder of this section, write $B_0(Y)$ for the principal 2-block of a finite group Y . For $\mathbf{H} \in \{\mathbf{G}, \tilde{\mathbf{G}}\}$ and $u \in \mathbf{H}$, write $A_{\mathbf{H}}(u) = C_{\mathbf{H}}(u)/C_{\mathbf{H}}(u)^\circ$ for the component group of the centraliser of u . For each unipotent element $u \in G$, let Γ_u denote the generalised Gelfand–Graev character of G associated with u , and let P_u be its principal block component.

The following lemma uses Taylor's analysis of the Frobenius action on the component groups of spin groups [68, Section 2], together with Taylor's equivariance formula for generalised Gelfand–Graev characters [71, Proposition 11.10].

Lemma 4.7. *Let $G = \text{Spin}_{2n+1}(q)$. Then every field automorphism of G induced by a power of F_0 fixes every unipotent conjugacy class of G and every character Γ_u .*

Proof. Let $\mathcal{C}$ be a geometric unipotent class of $\mathbf{G}$. Unipotent classes in type $\mathbf{B}$ are parametrised by the Jordan block sizes in the natural orthogonal representation. Since F_0 preserves these sizes, $\mathcal{C}$ is F_0 -stable. By the discussion of spin groups in [68, Section 2], choose $u_0 \in \mathcal{C}^{F_0}$ such that F_0 acts trivially on $A = A_{\mathbf{G}}(u_0)$. Then $F = F_0^f$ also acts trivially on A , so [56, Theorem 21.11] identifies the G -classes in $\mathcal{C}^F$ with the ordinary conjugacy classes of A . This identification commutes with F_0 , which acts trivially on A . Hence every power of F_0 fixes every G -class in $\mathcal{C}^F$.

The equivariance of generalised Gelfand–Graev characters [71, Proposition 11.10] now shows that every character Γ_u is fixed by the field automorphisms. $\square$

Lemma 4.8. *Let $H = \text{Spin}_{2m+1}(q_0)$, where $m \geq 2$ and q_0 is an odd prime power. Then*

$$\text{Irr}(B_0(H)) = \mathcal{E}_2(H, 1).$$

Moreover, every irreducible character of H whose semisimple label is quasi-isolated belongs to $B_0(H)$, and $B_0(H)$ is the only strictly quasi-isolated 2-block of H .

Proof. The equality is [21, Theorem 21.14]. The dual algebraic group is adjoint of type $\mathbf{C}_m$, and its quasi-isolated semisimple elements have order dividing 4 by [10, Proposition 5.3(a)]. Hence every rational Lusztig series with such a label is contained in $\mathcal{E}_2(H, 1)$, proving the assertion about ordinary characters. If a 2-block b is strictly quasi-isolated, its strictly quasi-isolated semisimple label s has odd order. Since s is also quasi-isolated, its order divides 4, so $s = 1$. The equality then gives $b = B_0(H)$. Conversely, the identity is strictly quasi-isolated, so $B_0(H)$ is strictly quasi-isolated. $\square$

For a Weyl group W and $E \in \text{Irr}(W)$, let $b(E)$ be the least nonnegative integer i for which E occurs in the i th symmetric power of the reflection representation [45, 4.1.1]. Truncated induction of Weyl group characters, also called j -induction, is denoted by j_W^W [69, p. 476].

Proposition 4.9. *Let $G = \text{Spin}_{2n+1}(q)$, and let $\mathcal{U}$ be an ordered set of representatives of the unipotent conjugacy classes of G . Assume that representatives in the same geometric class are consecutive and that the geometric classes are ordered by nondecreasing dimension. Then there are characters $\eta_u \in \text{Irr}(B_0(G))$, indexed by $u \in \mathcal{U}$, such that the matrix*

$$(\langle \eta_u, P_v \rangle_G)_{u,v \in \mathcal{U}}$$

is block lower triangular, with one diagonal block for each geometric class, and every diagonal block is a permutation matrix. Moreover, the P_u form a basis of the $\mathbb{Q}$ -span of the projective indecomposable characters of $B_0(G)$.

Proof. Fix a geometric unipotent class $\mathcal{C}$. As in the proof of Lemma 4.7, choose $u_0 \in \mathcal{C}^F$ such that F acts trivially on $A = A_{\mathbf{G}}(u_0)$. Let $Z_{\mathcal{C}}$ be the image of $Z(\mathbf{G})$ in A . The regular embedding gives

$$V = A/Z_{\mathcal{C}} \cong A_{\tilde{\mathbf{G}}}(u_0).$$

By [68, Section 2], V is an elementary abelian 2-group. Since F acts trivially on A , it also acts trivially on V . The parametrisation of rational classes identifies the map from G -classes to $\tilde{G}$ -classes in $\mathcal{C}^F$ with the map from conjugacy classes of A to elements of V . For each representative $\tilde{u}$ of a $\tilde{G}$ -class in $\mathcal{C}^F$, let $\tilde{\Gamma}_{\tilde{u}}$ denote the corresponding generalised Gelfand–Graev character.

We first choose an isolated family for $\tilde{\mathbf{G}}$. By [42, Proposition 2.4], there is a geometric isolated pair associated with $\mathcal{C}$ whose family group has order $|V|$. Let $\mathbf{D} = [\tilde{\mathbf{G}}^*, \tilde{\mathbf{G}}^*] \cong \text{Sp}_{2n}$. Write the semisimple element in the chosen pair as zt , where z is central and $t \in \mathbf{D}$. Multiplication by z leaves the centraliser unchanged, so t is isolated. Its eigenvalues belong to $\{1, -1\}$, since any other reciprocal pair of eigenvalues would give a general linear factor and place the centraliser in a proper Levi subgroup. Let $2a$ and $2b$ be the dimensions of its 1 and -1 eigenspaces, respectively, where $a + b = n$.

Choose an F -stable maximal torus $\tilde{\mathbf{T}}^*$ contained in an F -stable Borel subgroup, and set $\mathbf{T}_{\mathbf{D}} = \tilde{\mathbf{T}}^* \cap \mathbf{D}$. Relative to a symplectic basis diagonalising $\mathbf{T}_{\mathbf{D}}$, let $t' = \text{diag}(I_a, -I_b, I_a, -I_b)$. Its centraliser in $\mathbf{D}$ is $\text{Sp}_{2a} \times \text{Sp}_{2b}$, with root system $\mathbf{C}_a \sqcup \mathbf{C}_b$, omitting factors of rank zero. The eigenspace dimensions show that t' is conjugate to t in $\mathbf{D}$ and is therefore isolated. Since $\mathbf{C}_n$ has no nontrivial graph automorphism, F acts by the q -power map on $\mathbf{T}_{\mathbf{D}}$. Thus t' is fixed by F for odd q .

Let W be the Weyl group of $\tilde{\mathbf{G}}^*$, and let $W_t \leq W$ be the Weyl group of $C_{\tilde{\mathbf{G}}^*}(t)$. Removing the central factor and replacing t by t' replace W_t by a conjugate in W . Apply this conjugacy to the family and its unique special irreducible character. Conjugating W_t and its character leaves induction from W_t to W unchanged and preserves the b -invariant. Hence the constituent selected by j -induction is unchanged. The associated unipotent class is the class corresponding to this constituent under the Springer correspondence [42, p. 821], so it remains $\mathcal{C}$. The order of the group attached to the resulting family is also unchanged.

Set $\tilde{s}' = t'$, and denote the resulting family by $\mathcal{F}'$. Thus $\tilde{s}' \in (\tilde{\mathbf{T}}^*)^F$ is isolated and $|\mathcal{G}_{\mathcal{F}'}| = |V|$. The group $\mathcal{G}_{\mathcal{F}'}$ is elementary abelian [45, Remark 4.2.17]. Lemma 2.11 gives characters $\tilde{\rho}_{\tilde{u}}$ belonging to the family in $\mathcal{E}(\tilde{G}, \tilde{s}')$ corresponding to $\mathcal{F}'$ such that

$$\langle \tilde{\rho}_{\tilde{u}}, \tilde{\Gamma}_{\tilde{v}} \rangle_{\tilde{G}} = \delta_{\tilde{u}, \tilde{v}}$$

for all chosen representatives $\tilde{u}, \tilde{v}$.

Independently of this family, choose a pair $(\tilde{s}, \psi)$ associated with $\mathcal{C}$ as in [68, Theorem 3.2], with the image s of $\tilde{s}$ in $\mathbf{G}^*$ quasi-isolated. The character ψ is unipotent on $C_{\mathbf{G}^*}^{\mathbf{G}}(s)^F$. Let $\tilde{\psi} \in \mathcal{E}(\tilde{G}, \tilde{s})$ be the character corresponding to ψ under the bijections in [68, (3.2)]. Property (P3) of [68, Theorem 3.2] shows that $\tilde{\psi}$ has unipotent support $\mathcal{C}$. Property (P1) of the same theorem

and the degree calculation in the proof of [68, Theorem 3.1] give $n_{\tilde{\psi}} = n_{\psi} = |V|$. Also $n_{\tilde{\psi}^*} = n_{\tilde{\psi}}$ by [45, Proposition 3.4.21]. For each chosen representative $\tilde{u}$ of a $\tilde{G}$ -class in $\mathcal{C}^F$, the Frobenius action on its component group differs from that at u_0 by an inner automorphism of V . Since V is abelian, $[A_{\tilde{G}}(\tilde{u}) : A_{\tilde{G}}(\tilde{u})^F] = 1$. Choose the Frobenius structure on the constant local system on $\mathcal{C}$ for which the trace function takes the value 1 at every point of $\mathcal{C}^F$. The wave front set of $\tilde{\psi}^*$ is $\mathcal{C}$ by [70, Lemma 14.15]. Thus [70, Proposition 15.4], with the expansion in [70, (11.15)], gives

$$\sum_{\tilde{u}} \langle \tilde{\psi}^*, \tilde{\Gamma}_{\tilde{u}} \rangle_{\tilde{G}} = \frac{|A_{\tilde{G}}(u_0)|}{n_{\tilde{\psi}^*}} = 1,$$

where the sum ranges over the chosen representatives. Each summand is a nonnegative integer, so exactly one scalar product is 1 and all the others vanish.

Every irreducible character of $\tilde{G}$ has multiplicity free restriction to G [45, Theorem 1.7.15]. For $\tilde{\rho} = \tilde{\psi}$ or one of the characters $\tilde{\rho}_{\tilde{u}}$ just chosen, write

$$\text{Res}_{\tilde{G}}^{\tilde{G}}(\tilde{\rho}) = \rho_1 + \cdots + \rho_r.$$

Alvis–Curtis duality commutes with restriction from $\tilde{G}$ to G and is an involution [45, Proposition 3.4.3 and Corollary 3.4.5]. Consequently, $\text{Res}_{\tilde{G}}^{\tilde{G}}(\tilde{\rho}^*)$ is a sum of the distinct irreducible characters ρ_i^* with coefficients in $\{1, -1\}$. Since this restriction is a character, every coefficient is 1. Together with the identity $\tilde{\Gamma}_v = \text{Ind}_{\tilde{G}}^{\tilde{G}} \Gamma_v$ in the proof of [70, Lemma 14.12] and Frobenius reciprocity, this gives

$$(4.9.1) \quad \sum_{i=1}^r \langle \rho_i^*, \Gamma_v \rangle_G = \langle \tilde{\rho}^*, \tilde{\Gamma}_v \rangle_{\tilde{G}} \quad (v \in \mathcal{C}^F).$$

For these selected characters, the right hand side belongs to $\{0, 1\}$ by their construction above. We now choose characters ρ_u , indexed by $u \in \mathcal{U} \cap \mathcal{C}^F$, such that $(\langle \rho_u^*, \Gamma_v \rangle_G)_{u, v \in \mathcal{U} \cap \mathcal{C}^F}$ is a permutation matrix.

Whenever a $\tilde{G}$ -class in $\mathcal{C}^F$ is a single G -class, restrict the character $\tilde{\rho}_{\tilde{u}}$ chosen for that class. Conjugation by $\tilde{G}$ acts transitively on the restriction constituents and fixes the G -class. The equivariance of generalised Gelfand–Graev characters [39, Proposition 2.2] makes the summands in (4.9.1) equal for $v = \tilde{u}$. They are nonnegative integers with sum one, so $r = 1$. Equation (4.9.1) also shows that zero scalar products in $\tilde{G}$ remain zero for every restriction constituent.

If $Z_C = 1$, then $A \cong V$ and every $\tilde{G}$ -class in $\mathcal{C}^F$ is a single G -class. The irreducible restrictions just obtained therefore give the required characters ρ_u , indexed by $u \in \mathcal{U} \cap \mathcal{C}^F$, for this geometric class.

Suppose now that $Z_C \neq 1$, and denote its nonidentity element by ϑ . If A is abelian, Lusztig’s presentation of A , as recalled in [68, pp. 45–46], gives $A = Z_C \cong C_2$. Indeed, nontrivial central image requires each odd part of the orthogonal partition to occur once. The number of odd parts is odd. If there were at least three, the generators y_2, y_3 in that presentation would have nontrivial commutator ϑ , contradicting commutativity. Thus there is only one odd part and A is generated by ϑ . In this case $V = 1$, and the unique $\tilde{G}$ -class contains two G -classes.

If A is nonabelian, Lusztig’s presentation recalled in [68, pp. 45–46] shows that A is an extraspecial 2-group, with $Z(A) = Z_C = \langle \vartheta \rangle$. The $\tilde{G}$ -class corresponding to $0 \in V$ contains the two G -classes corresponding to 1 and ϑ . Every other $\tilde{G}$ -class is a single G -class, since the inverse image of a nonzero element of V is a conjugacy class $\{x, x\vartheta\}$ of A .

In both cases with $Z_C \neq 1$, there is exactly one $\tilde{G}$ -class containing two G -classes, and the required characters have already been chosen for every other $\tilde{G}$ -class. Property (P2) of [68, Theorem 3.2] and the restriction formula in the proof of [68, Theorem 3.1] give exactly $|Z_C| = 2$ irreducible constituents of $\text{Res}_{\tilde{G}}^{\tilde{G}} \tilde{\psi}$. On a $\tilde{G}$ -class consisting of one G -class, conjugation makes the two summands in (4.9.1) equal. They are nonnegative integers, so their sum cannot be one.

Hence the unique nonzero scalar product of $\tilde{\psi}^*$ occurs on the class containing two G -classes. Let u_1, u_2 represent those classes, and write $\text{Res}_{\tilde{G}} \tilde{\psi} = \rho_1 + \rho_2$. Equation (4.9.1) gives

$$\langle \rho_1^*, \Gamma_{u_j} \rangle_G + \langle \rho_2^*, \Gamma_{u_j} \rangle_G = 1 \quad (j = 1, 2).$$

An element interchanging the two constituents must interchange the two G -classes. Otherwise equivariance would make the two entries in either column equal, although they are nonnegative integers with sum one. The scalar product matrix on these classes is consequently

$$\begin{pmatrix} a & b \\ b & a \end{pmatrix}, \quad a, b \in \mathbb{Z}_{\geq 0}, \quad a + b = 1,$$

and is a permutation matrix. Together with the characters chosen for the $\tilde{G}$ -classes containing a single G -class, this gives characters ρ_u for which $(\langle \rho_u^*, \Gamma_v \rangle_G)_{u,v \in \mathcal{U} \cap \mathcal{C}^F}$ is a permutation matrix.

All the selected characters of $\tilde{G}$ have unipotent support $\mathcal{C}$. Compatibility with duality and [70, Lemmas 14.12 and 14.15] show that each restriction constituent ρ_u has the same support. Its semisimple label is the image of $\tilde{s}$ or $\tilde{s}'$ in $\mathbf{G}^*$, according to the character selected. The image of $\tilde{s}$ is quasi-isolated by its choice. The image of $\tilde{s}'$ is isolated by [10, Proposition 2.3(b)]. Both images are therefore quasi-isolated 2-elements [10, Proposition 5.3(a)].

For every $u \in \mathcal{U}$, set $\eta_u = \rho_u^*$. The semisimple label of ρ_u is quasi-isolated, so Lemma 4.8 gives $\rho_u \in \text{Irr}(B_0(G))$. Alvis–Curtis duality preserves rational Lusztig series [21, Proposition 9.8(iv)], and therefore η_u lies in the same series and also belongs to $B_0(G)$. Hence $\langle \eta_u, P_v \rangle_G = \langle \eta_u, \Gamma_v \rangle_G$ for all $u, v \in \mathcal{U}$.

Let $\mathcal{C}_u$ be the geometric class containing u . By [70, Lemma 14.15], $\mathcal{C}_u$ is the wave front set of η_u . The result of Achar and Aubert in the form of [70, Proposition 15.2] shows that $\langle \eta_u, \Gamma_v \rangle_G \neq 0$ only if $\mathcal{C}_v \subseteq \overline{\mathcal{C}_u}$, where $\overline{\mathcal{C}_u}$ denotes the Zariski closure of $\mathcal{C}_u$ in $\mathbf{G}$. For distinct classes, this inclusion implies $\dim \mathcal{C}_v < \dim \mathcal{C}_u$. The chosen ordering therefore makes the full scalar product matrix block lower triangular, with the permutation matrices obtained above on its diagonal.

By the equality $\mathcal{E}_2(G, 1) = \text{Irr}(B_0(G))$ in Lemma 4.8, [24, Proposition 2.7] gives $|\mathcal{U}| = |\text{IBr}(B_0(G))|$. Each Γ_u is induced from a linear character of a p -subgroup. Since p is odd, P_u is projective in characteristic 2. The scalar product matrix is nonsingular, so the P_u are linearly independent. Their number equals the dimension of the $\mathbb{Q}$ -span of the projective indecomposable characters of $B_0(G)$, and they therefore form a basis of that space. $\square$

Corollary 4.10. *Let $G = \text{Spin}_{2n+1}(q)$, and let E be the group of field automorphisms of G . Then every $\psi \in \text{IBr}(B_0(G))$ is fixed by E , satisfies*

$$(\tilde{G} \rtimes E)_\psi = \tilde{G}_\psi \rtimes E,$$

and extends to $G \rtimes E$. If $(n, q) \neq (3, 3)$, then the principal 2-block of $\Omega_{2n+1}(q)$ is BAW-good.

Proof. By Lemma 4.7, E fixes every Γ_u and therefore every principal block component P_u . Proposition 4.9 shows that the P_u form a basis over $\mathbb{Q}$ of the space spanned by the projective indecomposable characters of $B_0(G)$. Thus E acts trivially on this space. It fixes every projective indecomposable character and, by duality, every irreducible Brauer character of $B_0(G)$. In particular, E fixes every $\tilde{G}$ -conjugate of ψ , which gives the stabiliser equality. Since E is cyclic, Lemma 2.9 extends ψ to $G \rtimes E$.

Suppose that $(n, q) \neq (3, 3)$. By Table 4.1, $S = \Omega_{2n+1}(q)$ is its own universal 2'-covering group. Set $H = \text{SO}_{2n+1}(q)$. Since $Z(G)$ is a central 2-group, inflation identifies $\text{IBr}(B_0(S))$ with $\text{IBr}(B_0(G))$ [58, Theorem 9.10]. Thus field automorphisms fix every character in $\text{IBr}(B_0(S))$. The principal block of H covers $B_0(S)$. As $|H : S| = 2$, each irreducible Brauer character of S lies under a unique irreducible Brauer character of H [58, Theorem 8.11]. Consequently field automorphisms also fix every character in $\text{IBr}(B_0(H))$. This gives the instance of [36, Corollary 5.15] used in the proof of Theorem 5.16 of that paper. The preceding argument

verifies the field invariance hypothesis of [36, Theorem 5.16] and Assumption 5.1 of the same paper for $B_0(G)$. That theorem therefore gives the inductive BAW condition for the principal 2-block of S , which is BAW-good by Theorem 2.3. $\square$

By the equivalence stated after [36, Question 5.27], Corollary 4.10 gives a positive answer for $\text{Spin}_{2n+1}(q)$ with q odd and $n \geq 3$. Indeed, field automorphisms fix both the unipotent conjugacy classes and the principal block Brauer characters, so only the diagonal involution acts on these sets. Applying [24, Proposition 2.7] to $\text{Spin}_{2n+1}(q)$ and $\text{SO}_{2n+1}(q)$, together with [21, Theorem 21.14] and [58, Theorem 8.11], gives equal numbers of diagonal orbits of length two. For $n = 3$, the field invariance is already obtained in the proof of [36, Corollary 5.17].

4.5. The exceptional triple cover. For odd $q \neq 3$, Feng, Yu, and Zhang established the inductive BAW condition for the principal 2-block of $\Omega_7(q)$ in the proof of [36, Corollary 5.17]. This also follows from Corollary 4.10. We now treat the exceptional universal $2'$ -covering group $3.\Omega_7(3)$.

Proposition 4.11. *Let $X = 3.\Omega_7(3)$. Then every 2-block of X satisfies the inductive BAW condition.*

Proof. Let $S = \Omega_7(3)$ and $H = \text{SO}_7(3)$. Since $X = X_2(S)$ and $\text{Out}(S) \cong C_2$, [30, Corollary 2.13] shows that a 2-block B of X satisfies the inductive BAW condition whenever there is an $\text{Aut}(X)_B$ -equivariant bijection $\text{IBr}(B) \rightarrow \text{Alp}(B)$. The calculations in Section A.3 give the nine blocks $B_1, \dots, B_9$, with $B_1 = B_0(X)$, and the data in Table 4.2 below.

Conjugation by H induces $\text{Aut}(S)$. By [36, Lemma 5.7 and the proof of Proposition 5.11], the numbers of principal weight classes of H and S equal $|\text{IBr}(B_0(H))|$ and $|\text{IBr}(B_0(S))|$, respectively. The proof of [36, Proposition 5.11] applies to $\text{Spin}_7(3)$, whose principal Brauer characters and principal weight classes correspond bijectively to those of S under the central quotient with kernel of order 2. The calculation in Section A.3 gives $|\text{IBr}(B_0(H))| = 10$ and $|\text{IBr}(B_0(S))| = 12$. The proof of [36, Proposition 5.11] also shows that every principal weight class of S is covered by a unique principal weight class of H . By [36, Corollary 2.22(2)], the classes of S covered by each class of H form one orbit under H/S . Consequently, $H/S \cong C_2$ has eight singleton orbits and two orbits of size two on the principal weight classes of S . The calculation in `o7brauer.g`, described in Section A.3, shows that eight of the ten irreducible Brauer characters of $B_0(H)$ restrict irreducibly to S , while the other two each restrict to a sum of two distinct irreducible Brauer characters. The irreducible constituents of these ten restrictions are precisely the characters in $\text{IBr}(B_0(S))$. By Clifford theory, the action of H/S on this set has eight singleton orbits and two orbits of size two. Comparing this with the action on $\text{Alp}(B_0(S))$, we obtain an $\text{Aut}(S)$ -equivariant bijection

$$\text{IBr}(B_0(S)) \longrightarrow \text{Alp}(B_0(S)).$$

Since $X \rightarrow S$ has central $2'$ -kernel, [30, Lemma 2.5] lifts this to an $\text{Aut}(X)$ -equivariant bijection $\text{IBr}(B_1) \rightarrow \text{Alp}(B_1)$.

For B_2 , the character table calculation in Section A.3 gives defect order 8, five ordinary characters and two Brauer characters. One ordinary character has height 1, so Kessar–Malle’s theorem [48, Theorem 1.1] excludes abelian defect groups. The defect group of B_2 is therefore dihedral or quaternion. A block with quaternion defect group of order 8 has one or three irreducible Brauer characters [62, Theorem 8.1]. Hence the defect group of B_2 is D_8 . By the proof of [61, Theorem 4.1], B_2 has exactly two X -conjugacy classes of weights, represented by weights whose radical subgroups have orders 8 and 4, respectively. Since automorphisms preserve subgroup order, the outer involution fixes both classes. The program `o7block2.g` verifies that the outer involution also fixes both irreducible Brauer characters of B_2 . Hence any bijection $\text{IBr}(B_2) \rightarrow \text{Alp}(B_2)$ is $\text{Aut}(X)_{B_2}$ -equivariant. The block B_3 has defect group C_2 and is therefore nilpotent. It satisfies the inductive BAW condition by [50, Theorem 1.3]. The blocks B_4 and B_5 have defect zero and satisfy the condition trivially.

Table 4.2 below shows that B_8 and B_9 are the only blocks of defect 3 with nontrivial central characters. They are therefore the two faithful blocks with dihedral defect groups of order 8 described in [55, Section 5.5]. Each has two irreducible Brauer characters, so [61, Theorem 4.1] gives two weight classes in each. The exhaustive radical computation in Section A.3 gives eight weight classes in each faithful sector. Since B_8 and B_9 account for two in their respective sectors, B_6 and B_7 each have six, equal to their numbers of Brauer characters. The outer involution swaps B_6 with B_7 and B_8 with B_9 , so $\text{Aut}(X)_{B_i} = \text{Inn}(X)$ for $i \in \{6, 7, 8, 9\}$, so any bijection $\text{IBr}(B_i) \rightarrow \text{Alp}(B_i)$ is $\text{Aut}(X)_{B_i}$ -equivariant for each such i . The criterion of [30, Corollary 2.13] now proves the inductive BAW condition for B_1, B_2, B_6, B_7, B_8 and B_9 . $\square$

In Table 4.2 below, let ω be the nontrivial linear character of $Z(X) \cong C_3$ over which B_6 lies. Its square ω^2 is the other nontrivial linear character. For each block B , let D_B be a defect group of B .

TABLE 4.2. The 2-blocks of $3.\Omega_7(3)$ in the ordering of CTblLib 1.3.11 [14].

B	$ D_B $	$ \text{IBr}(B) $	central character	outer action
B_1	2^9	12	1	fixed
B_2	2^3	2	1	fixed
B_3	2	1	1	fixed
B_4	1	1	1	fixed
B_5	1	1	1	fixed
B_6	2^9	6	ω	B_7
B_7	2^9	6	ω^2	B_6
B_8	2^3	2	ω	B_9
B_9	2^3	2	ω^2	B_8

4.6. All blocks at the prime 2. The following lemma uses the reduction in the proof of [36, Theorem 1], specialised to type B. The proof includes the lower rank cases and uses Lemma 2.6 to pass from the orthogonal quotients to the spin groups.

Lemma 4.12. *Let $\mathbf{G}$ be a simple, simply connected algebraic group of type B_n , where $n \geq 3$, and let F be a split Frobenius endomorphism such that $\mathbf{G}^F = \text{Spin}_{2n+1}(q)$, with $(n, q) \neq (3, 3)$. Suppose that [31, Assumption 5.3] holds for $(\mathbf{G}, F)$ at $\ell = 2$. Then every 2-block of $\Omega_{2n+1}(q)$ is BAW-good.*

Proof. Set $G = \mathbf{G}^F$ and $S = \Omega_{2n+1}(q)$. By Section 4.1, G is the universal covering group of S . We apply Lemma 3.2 with $\ell = 2$. The hypothesis gives the character representatives, stabiliser factorisation and extensions required in part (i) of Lemma 3.2.

Let $\mathbf{K}$ and F' be as in Lemma 3.2(ii). A connected Dynkin subdiagram of B_n has type A_m or B_m . If $\mathbf{K}$ has type A_m , then $\mathbf{K}^{F'}$ is a special linear or unitary group. The consequence of the type A results stated after Lemma 3.2 gives the required iBAW bijection for every 2-block of $\mathbf{K}^{F'}$. This also covers $B_1 = A_1$.

Suppose that $\mathbf{K}$ has type B_m , where $m \geq 2$. Then $H = \mathbf{K}^{F'} \cong \text{Spin}_{2m+1}(q_0)$ for some power q_0 of p . By Lemma 4.8, the only strictly quasi-isolated 2-block of H is its principal block. For $m = 2$, we have $\text{Spin}_5(q_0) \cong \text{Sp}_4(q_0)$. The rank two construction at the start of the proof of Proposition 3.3 gives the required iBAW bijection for its principal block.

For $m \geq 3$, set $T = \Omega_{2m+1}(q_0)$. If $(m, q_0) \neq (3, 3)$, Corollary 4.10 shows that the principal 2-block of T is BAW-good and therefore admits an iBAW bijection. In the exceptional case, Proposition 4.11 and Remark 2.4(a) show that T satisfies iBAW at 2, so Corollary 2.5 gives the required principal block bijection on T . We have $Z(H) = O_2(H)$ and $H/Z(H) \cong T$. By Section 4.1, H is the universal covering group of T unless $(m, q_0) = (3, 3)$, in which case $H = \text{Spin}_7(3) = X_3(T)$. Thus the natural isomorphism $\text{Aut}(H) \cong \text{Aut}(T)$ identifies $(H \rtimes \text{Aut}(H))/Z(H)$ with $T \rtimes \text{Aut}(T)$. Lemma 2.6 therefore lifts this bijection to an iBAW bijection for the principal 2-block of H .

This verifies the second hypothesis of Lemma 3.2, which shows that every 2-block of G is BAW-good. Since $G/O_2(Z(G)) = S$, every 2-block of S is dominated by a 2-block of G and is consequently BAW-good. $\square$

Proposition 4.13. *Let $\mathbf{G}$ be a simple, simply connected algebraic group of type $\mathbf{B}_n$, where $n \geq 3$, in odd characteristic p . Let ϕ be a standard Frobenius endomorphism of $\mathbf{G}$ such that $\mathbf{G}^\phi = \text{Spin}_{2n+1}(p)$. Let $f \geq 1$, and set $q = p^f$, $F = \phi^f$ and $G = \mathbf{G}^F$. Choose a regular embedding $\mathbf{G} \hookrightarrow \tilde{\mathbf{G}}$ to which ϕ extends, set $\tilde{G} = \tilde{\mathbf{G}}^F$, and let E be the cyclic group generated by the restriction of ϕ to $\tilde{G}$. Then, at the prime 2, every $\tilde{G}$ -orbit in $\text{IBr}(G)$ contains a character ψ such that*

$$(\tilde{G} \rtimes E)_\psi = \tilde{G}_\psi \rtimes E_\psi$$

and ψ extends to $G \rtimes E_\psi$.

Proof. If $f = 1$, then $E = 1$ and both assertions are immediate. Assume that $f > 1$. Corollary 4.10 supplies the required representatives for the principal block of G .

Let b be a nonprincipal 2-block of G . We use the idempotents e_s^G defined in Section 3. By [21, Theorem 9.12], choose a semisimple element $s \in (\mathbf{G}^*)^F$ of odd order such that b belongs to e_s^G . Since b is nonprincipal, Lemma 4.8 shows that s is not strictly quasi-isolated.

As in the proof of [31, Theorem 5.7], choose a minimal Levi subgroup $\mathbf{L}^*$ of $\mathbf{G}^*$ containing $C_{\mathbf{G}^*}^\circ(s)C_{\mathbf{G}^*}(s)^F$. This product contains a maximal torus, and intersections of Levi subgroups containing a common maximal torus are Levi subgroups. Hence $\mathbf{L}^*$ is unique and therefore F -stable. It is proper because s is not strictly quasi-isolated.

The action of $\tilde{G}$ fixes e_s^G by [11, Lemma 7.4]. Set $D = E_{e_s^G}$. Choose a positive divisor c of f such that $F_0 = \phi^c$ induces a generator of D , and set $h = f/c$, so that $F = F_0^h$. By [60, Lemma 4.5(a)–(b)], there is a Frobenius endomorphism conjugate to F_0 by an element of G which stabilises a Levi subgroup $\mathbf{L}$ in duality with $\mathbf{L}^*$ and its idempotent e_s^L . Conjugating this Levi subgroup and making the corresponding rational conjugation of $(s, \mathbf{L}^*)$, we may therefore assume that F_0 itself stabilises $\mathbf{L}$ and e_s^L , where $L = \mathbf{L}^F$. Set

$$L_0 = [\mathbf{L}, \mathbf{L}]^F, \quad \tilde{L} = (\mathbf{L}Z(\tilde{\mathbf{G}}))^F.$$

Write $L_0 \cong \prod_i H_i$ as in Lemma 4.5. The factors arising from components of type $\mathbf{A}$ are special linear or unitary groups over extension fields. There is at most one component containing a short root. If such a component occurs, then F_0 preserves it because F_0 preserves root lengths. Its Dynkin diagram has no nontrivial automorphism preserving root lengths. The group of points fixed by F on this component is therefore $\text{Spin}_{2m+1}(q)$ for some $m < n$. If d is a block of L belonging to e_s^L and d_0 is a block of L_0 covered by d , then d_0 is strictly quasi-isolated by the proof of [31, Theorem 5.7]. Write $d_0 = d_1 \otimes \cdots \otimes d_r$, where d_i is a block of H_i . The proof of [31, Proposition 5.6] shows that each d_i is strictly quasi-isolated. If H_i has type $\mathbf{B}_m$ with $m \geq 2$, then d_i is principal by Lemma 4.8.

For a type $\mathbf{A}$ factor, the representatives and stabiliser factorisation for the diagonal, field and graph automorphisms are provided by [27, Theorem 8.1]. This includes $\mathbf{B}_1 = \mathbf{A}_1$ and the group $\text{SL}_2(3)$. For a factor H_i of type $\mathbf{B}_m$, let Δ_i be the group induced by $\tilde{L}$ on H_i , as in Lemma 4.5, and let E_i denote the group of field automorphisms of H_i . For $m = 2$, identify H_i with $\text{Sp}_4(q)$. Here $f > 1$, so $q \geq 9$. The verification following Lemma 4.5 of Brough and Schaeffer Fry [18, p. 1203] gives a representative in each diagonal orbit with the required stabiliser factorisation. Here $\Delta_i/\text{Inn}(H_i)$ has order two, so field automorphisms centralise this quotient and the factorisation is preserved by diagonal conjugation. Thus

$$(4.1) \quad (\Delta_i E_i)_\chi = (\Delta_i)_\chi (E_i)_\chi \quad (\chi \in \text{IBr}(B_0(H_i))).$$

For $m \geq 3$, Corollary 4.10 shows that E_i fixes every character in $\text{IBr}(B_0(H_i))$, so (4.1) also holds. In particular, it holds with E_i replaced by any subgroup of E_i .

Recall that $\text{IBr}(L, e_s^L)$ is the union of $\text{IBr}(d)$ over the 2-blocks d of L belonging to e_s^L . Fix an $\tilde{L}$ -orbit $\mathcal{P}$ in $\text{IBr}(L, e_s^L)$ and choose $\psi_0 \in \mathcal{P}$. Let θ_0 be an irreducible constituent of $(\psi_0)_{L_0}$, and let $\mathcal{O}$ be its $\tilde{L}$ -orbit. Let $d = \text{bl}(\psi_0)$, and let d_0 be the block containing θ_0 . Then d covers d_0 . Since d belongs to e_s^L , the argument above shows that every factor block of d_0 is strictly quasi-isolated. The type A factors therefore satisfy [27, Theorem 8.1], while the type B factors of rank at least two are principal and satisfy (4.1). By Lemma 4.5, the image of the conjugation action of $\tilde{L}$ on L_0 is $\prod_i \Delta_i$. If $\theta_0 = \theta_{0,1} \times \cdots \times \theta_{0,r}$, then $\mathcal{O}$ consists of the products of characters in the Δ_i -orbits of $\theta_{0,i}$.

Let $D_{\mathcal{O}}$ be the stabiliser of $\mathcal{O}$ in D , and choose a generator τ induced by F_0^a , with $a > 0$. Consider a cycle C of the permutation of the factors H_i induced by τ , and set $t = |C|$. Choose one index i from the cycle C , and let $\mathbf{H}_j$ be the algebraic component used to define H_i in Lemma 4.5. Thus $H_i = \mathbf{H}_j^{F^{n_i}}$, where n_i is the length of the F -orbit of $\mathbf{H}_j$. Since τ^t stabilises this factor, there is an integer k such that $F_0^{at}(\mathbf{H}_j) = F^k(\mathbf{H}_j)$. Replacing k by a congruent integer modulo n_i , choose $k \leq 0$, which gives $at - hk \geq 0$. Under the projection onto $\mathbf{H}_j$ in the proof of Lemma 4.5, the action of τ^t on $\mathbf{H}_j^{F^{n_i}}$ is induced by

$$F^{-k} F_0^{at} = F_0^{at-hk}.$$

If u is the F_0 -orbit length of $\mathbf{H}_j$, both $at - hk$ and hn_i are multiples of u . Thus F_0^{at-hk} and F^{n_i} are powers of F_0^u on $\mathbf{H}_j$. By the classification of Steinberg endomorphisms [56, Theorem 22.5] and Lang's theorem, we may conjugate F_0^u to a standard Frobenius endomorphism, possibly composed with a graph automorphism. The same conjugation identifies both powers with powers of this standard endomorphism. Consequently, τ^t acts on $\mathbf{H}_j^{F^{n_i}}$ as a field automorphism in type B, and as a field automorphism possibly composed with a graph automorphism in type A. Let $\Phi_i \leq \text{Aut}(H_i)$ be the group generated by the field automorphisms and, when H_i has type A, the graph automorphisms. Thus $\Phi_i = E_i$ in type B. The automorphism induced by τ^t belongs to Φ_i . In type B, $\Delta_i \cap E_i = 1$, so (4.1) holds in $\Delta_i \rtimes \Phi_i$. In type A, [27, Theorem 8.1] gives the corresponding factorisation in $\Delta_i \rtimes \Phi_i$ for a suitable representative of each Δ_i -orbit. The action just described defines a homomorphism $\alpha_i : \langle \tau^t \rangle \rightarrow \Phi_i$. Taking inverse images under the homomorphism $(\delta, \sigma) \mapsto (\delta, \alpha_i(\sigma))$ from $\Delta_i \rtimes \langle \tau^t \rangle$ to $\Delta_i \rtimes \Phi_i$ verifies the hypothesis of Lemma 4.4, which gives $\theta \in \mathcal{O}$ fixed by $D_{\mathcal{O}}$. Hence

$$(\tilde{L}D_{\mathcal{O}})_{\theta} = \tilde{L}_{\theta}D_{\mathcal{O}}.$$

Choose $y \in \tilde{L}$ such that $\theta_0^y = \theta$, and set $\psi = \psi_0^y$. Then $\psi \in \mathcal{P} \subseteq \text{IBr}(L, e_s^L)$, and θ is a constituent of ψ_{L_0} .

The proof of Lemma 4.5 shows that $[\mathbf{L}, \mathbf{L}] \hookrightarrow \mathbf{LZ}(\tilde{\mathbf{G}})$ is a regular embedding. Apply [45, Theorem 1.7.15] to this embedding. The restriction to L_0 of every irreducible 2-modular representation of $\tilde{L}$ is multiplicity free. Choosing a representation lying over θ , modular Clifford theory therefore shows that θ extends to $\tilde{L}_{\theta}$. Lemma 4.5 also shows that $L_0 \triangleleft L \triangleleft \tilde{L}$, that $\tilde{L}/L_0$ is abelian, and that $D_{\mathcal{O}}$ normalises this chain. Lemma 4.6 now gives

$$(\tilde{L}D_{\mathcal{O}})_{\psi} = \tilde{L}_{\psi}(D_{\mathcal{O}})_{\psi}.$$

If $x = \tilde{g}\delta \in (\tilde{L}D)_{\psi}$, with $\tilde{g} \in \tilde{L}$ and $\delta \in D$, then Clifford theory shows that θ^x is an L -conjugate of θ . Thus $\mathcal{O}^{\delta} \cap \mathcal{O} \neq \emptyset$, so $\delta \in D_{\mathcal{O}}$. Consequently,

$$(\tilde{L}D)_{\psi} = \tilde{L}_{\psi}D_{\psi}, \quad (D_{\mathcal{O}})_{\psi} = D_{\psi}.$$

Let Ψ be the image of ψ under the bijection $\text{IBr}(L, e_s^L) \rightarrow \text{IBr}(G, e_s^G)$ induced by Jordan decomposition. This bijection preserves the block correspondence and is $\tilde{L}D$ -equivariant by [31, Lemma 5.1 and the proof of Proposition 5.2]. Since $Z(\mathbf{G}) \leq \mathbf{L}$, the intersection $\mathbf{G} \cap \mathbf{LZ}(\tilde{\mathbf{G}}) = \mathbf{L}$ is connected. Lang's theorem applied to $\tilde{\mathbf{G}} = \mathbf{G}(\mathbf{LZ}(\tilde{\mathbf{G}}))$ therefore gives $\tilde{G} = G\tilde{L}$, with $G \cap \tilde{L} = L$. Inclusion induces an isomorphism $\tilde{L}/L \cong \tilde{G}/G$. Under the equivariant Jordan bijection, this

isomorphism identifies the diagonal actions on ψ and Ψ . Hence

$$(\tilde{G}D)_\Psi = \tilde{G}_\Psi D_\Psi, \quad D_\Psi = D_\psi.$$

The equivariance of rational Lusztig series [71, Proposition 7.2] also shows that field automorphisms permute the idempotents e_s^G . Since each block belongs to exactly one of these idempotents, every element of $(\tilde{G}E)_\Psi$ fixes e_s^G . The group $\tilde{G}$ fixes this idempotent, so the image of any such element under the projection $\tilde{G} \rtimes E \rightarrow E$ lies in D . Consequently,

$$(\tilde{G}E)_\Psi = (\tilde{G}D)_\Psi = \tilde{G}_\Psi E_\Psi, \quad E_\Psi = D_\Psi.$$

The group E_Ψ is cyclic, so Lemma 2.9 extends Ψ to $G \rtimes E_\Psi$.

Since $\mathcal{P}$ was arbitrary, the required representatives exist on every $\tilde{G}$ -orbit in $\text{IBr}(G, e_s^G)$. Varying s covers all nonprincipal orbits, while Corollary 4.10 covers the principal ones. This proves the assertion. $\square$

Proposition 4.14. *Let $S = \Omega_{2n+1}(q)$. Then every 2-block of $X_2(S)$ satisfies the inductive BAW condition.*

Proof. If $(n, q) = (3, 3)$, the result is Proposition 4.11. Otherwise, $X_2(S) = S$ by Table 4.1. Let $\mathbf{G}$ be simply connected of type $\mathbf{B}_n$, and choose F with $G = \mathbf{G}^F = \text{Spin}_{2n+1}(q)$. For every semisimple $2'$ -element $s \in (\mathbf{G}^*)^F$, [60, Corollary 4.2] describes the stabiliser of e_s^G in $\text{Out}(G)$ using diagonal automorphisms and a power of the standard field automorphism, since type $\mathbf{B}$ has no graph automorphisms. By [60, Lemma 4.5(a)–(b)], after conjugation by G , the associated Levi subgroup and its idempotent may be chosen stable under this field automorphism. Thus the group A_s in Lemma 3.2 may be chosen inside the field automorphism group E of Proposition 4.13. Restricting the stabiliser factorisation and extensions supplied by that proposition to A_s verifies [31, Assumption 5.3] at the prime 2. Lemma 4.12 therefore shows that every 2-block of S is BAW-good, and Theorem 2.3 gives the inductive BAW condition. $\square$

4.7. Proof of Theorem 4.1.

Proof. The defining prime case follows from [65, Theorem C]. Proposition 4.3 treats odd non-defining primes, and Proposition 4.14 treats the prime 2. Thus every ℓ -block of $X_\ell(S)$ satisfies the inductive BAW condition. The final assertion follows from Remark 2.4(a). $\square$

Corollary 4.15. *Every finite simple group of type $\mathbf{B}$ satisfies iBAW at every prime dividing its order.*

Proof. Theorem 4.1 treats odd characteristic in ranks at least three. In even characteristic, the isomorphisms between the finite simple groups of types $\mathbf{B}_n$ and $\mathbf{C}_n$ reduce the assertion to Theorem 3.1, which also covers rank two through $\mathbf{B}_2 = \mathbf{C}_2$. The case of rank one follows from [29, Proposition 4.6], since $\mathbf{B}_1 = \mathbf{A}_1$. The same result covers the derived subgroup $\text{Sp}_4(2)' \cong A_6 \cong \text{PSL}_2(9)$. $\square$

5. SPORADIC GROUPS

Let S be one of the 26 sporadic simple groups and let $\ell \mid |S|$. As before, $X = X_\ell(S)$ denotes the universal ℓ' -covering group.

Theorem 5.1. *Every sporadic simple group S satisfies iBAW at every prime $\ell \mid |S|$ on $X_\ell(S)$.*

The overview distributed with CTBlocks 0.9.5 [13] states that the inductive blockwise Alperin weight condition in [65, Definition 4.1] has been checked for every sporadic simple group other than J_4 , Fi'_{24} , the Baby Monster and the Monster. We use the CTBlocks verification for those 22 groups and prove the remaining four cases below.

5.1. Lemmas for the sporadic proof. When the outer automorphism group is trivial, the criterion in [30, Corollary 2.13] gives the following.

Lemma 5.2. *Let S be a nonabelian finite simple group and let $X = X_\ell(S)$ be its universal ℓ' -covering group. Suppose that $\text{Out}(X) = 1$ and $|\text{IBr}(b)| = |\text{Alp}(b)|$ for every ℓ -block b of X . Then S satisfies iBAW at ℓ on X .*

Proof. Since X is the universal ℓ' -covering group of S , we have $\text{Out}(S) \cong \text{Out}(X) = 1$. For each ℓ -block b of X , the cardinality assumption gives a bijection $\Omega_b : \text{IBr}(b) \rightarrow \text{Alp}(b)$. Every automorphism of X is inner and therefore fixes every irreducible Brauer character and every X -conjugacy class of weights. Thus Ω_b is $\text{Aut}(X)_b$ -equivariant. [30, Corollary 2.13] therefore shows that every ℓ -block of X satisfies the inductive BAW condition. Remark 2.4(a) gives the conclusion. $\square$

The following lemma is an elementary consequence of the orbit decomposition of finite C_2 -sets.

Lemma 5.3. *Let $\mathcal{B}$ be a finite C_2 -set with orbits $\mathcal{B}_0, \dots, \mathcal{B}_r$. Let Y and Y' be finite C_2 -sets with decompositions*

$$Y = \bigsqcup_{b \in \mathcal{B}} Y_b, \quad Y' = \bigsqcup_{b \in \mathcal{B}} Y'_b$$

such that $Y_b^a = Y_{b^a}$ and $(Y'_b)^a = Y'_{b^a}$ for $a \in C_2$ and $b \in \mathcal{B}$. Set $U_i = \bigsqcup_{b \in \mathcal{B}_i} Y_b$ and $U'_i = \bigsqcup_{b \in \mathcal{B}_i} Y'_b$. If $Y \cong Y'$ and $U_i \cong U'_i$ as C_2 -sets for $1 \leq i \leq r$, then $U_0 \cong U'_0$. Moreover, this isomorphism can be chosen to map Y_b onto Y'_b for every $b \in \mathcal{B}_0$.

Proof. A finite C_2 -set is determined up to isomorphism by its numbers of singleton orbits and orbits of size two. Cancelling the known parts therefore gives $U_0 \cong U'_0$. If $\mathcal{B}_0$ is a singleton, this isomorphism has the required property. Otherwise, write $\mathcal{B}_0 = \{b, b^t\}$, where t generates C_2 . The action of t exchanges the two parts on each side, so $|Y_b| = \frac{1}{2}|U_0| = \frac{1}{2}|U'_0| = |Y'_b|$. Choose a bijection $f_b : Y_b \rightarrow Y'_b$ and extend it by $f(y^t) = f_b(y)^t$ for $y \in Y_b$. Together with $f|_{Y_b} = f_b$, this defines the required C_2 -equivariant bijection. $\square$

5.2. The four remaining groups. The Schur multipliers and outer automorphism groups used below are taken from [26, Table 1].

Proposition 5.4. *For each of the 45 pairs (S, ℓ) listed below, S satisfies iBAW at ℓ on its universal ℓ' -covering group.*

S	primes ℓ
J_4	2, 3, 5, 7, 11, 23, 29, 31, 37, 43
Fi'_{24}	2, 3, 5, 7, 11, 13, 17, 23, 29
$\mathbb{B}$	2, 3, 5, 7, 11, 13, 17, 19, 23, 31, 47
$\mathbb{M}$	2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 41, 47, 59, 71

Proof. For J_4 the Schur multiplier and outer automorphism group are trivial. The blockwise Alperin weight conjecture for every block with noncyclic defect group is [7, Theorem 5.2]. Blocks with cyclic defect group, including blocks of defect zero, are covered by [65, Proposition 6.2]. Lemma 5.2 shows that J_4 satisfies iBAW at every prime dividing its order.

We next consider Fi'_{24} . The blockwise weight counts for the blocks of Fi'_{24} with noncyclic defect groups are established in [4, Theorem 4.3]. We also determine the actions of the outer involution needed for the equivariant bijections. Set $X = X_\ell(Fi'_{24})$, so that $X = Fi'_{24}$ at 3 and $X = 3.Fi'_{24}$ at every other prime. For $\ell \neq 3$, let $Z = Z(X)$ and let $\pi : X \rightarrow \overline{X} := X/Z = Fi'_{24}$ be the quotient map. For every ℓ -subgroup $\overline{Q}$ of $\overline{X}$, Schur–Zassenhaus gives a complement Q to Z in $\pi^{-1}(\overline{Q})$. Since Z is central, this complement is unique. Moreover, Q is radical if and only if $\overline{Q}$ is radical [59, Lemma 2.3(c)], and

$$(5.1) \quad \begin{aligned} \overline{N_X(Q)} &= N_{\overline{X}}(\overline{Q}), \\ N_X(Q)/(QZ) &\cong N_{\overline{X}}(\overline{Q})/\overline{Q}. \end{aligned}$$

By [58, Theorem 9.9(c)], the block correspondence for $X \rightarrow \overline{X}$ gives a bijection between the blocks of $\overline{X}$ and the blocks of X lying over 1_Z . The latter are precisely the blocks in the trivial sector of Definition 2.1. For corresponding blocks $\overline{b}$ and b , inflation gives a bijection

$$\mathrm{IBr}(\overline{b}) \rightarrow \mathrm{IBr}(b).$$

For a weight $(\overline{Q}, \overline{\theta})$ of $\overline{b}$, let Q be the unique lift described above, and let θ be the inflation of $\overline{\theta}$ along the quotient homomorphism $N_X(Q)/Q \rightarrow N_{\overline{X}}(\overline{Q})/\overline{Q}$, $nQ \mapsto n\overline{Q}$. Passing to conjugacy classes gives a bijection

$$\mathrm{Alp}(\overline{b}) \rightarrow \mathrm{Alp}(b)$$

by [30, Lemma 2.3(iii)]. These bijections commute with the induced automorphism actions. Thus an equivariant bijection $\mathrm{IBr}(\overline{b}) \rightarrow \mathrm{Alp}(\overline{b})$ gives an equivariant bijection $\mathrm{IBr}(b) \rightarrow \mathrm{Alp}(b)$ [30, Lemma 2.5].

For $\ell \neq 3$, the outer involution exchanges the two faithful sectors of $X = 3.Fi'_{24}$ [5, Lemma 4.7]. The stabiliser of a block in either sector is therefore $\mathrm{Inn}(X)$, so any bijection between its irreducible Brauer characters and weight classes is equivariant. Remark 2.4(b) extends this bijection to the other block in its orbit.

For a finite C_2 -set Y , we use the pair $(|Y|, |Y^{C_2}|)$, with C_2 acting through the nontrivial outer automorphism. We compare these pairs for $\mathrm{IBr}(b)$ and $\mathrm{Alp}(b)$ whenever the outer involution fixes the block b . Throughout the following argument, we use Table 8 of [5]. In the column headed $|\mathrm{Irr}^0|$, an entry a/b denotes $a + b$ weights, of which a are fixed by the outer involution. For the blocks specified below, the program `fi24blocks.g` computes the number of Brauer characters, the number fixed by the outer involution, and the action of the outer involution on blocks. It lists the block defects separately. At 2, the pairs of total and fixed Brauer character counts for the blocks in the trivial sector are

$$(5.2) \quad (33, 25), \quad (3, 1), \quad (3, 3), \quad (1, 1), \quad (1, 1).$$

The last two blocks have defect zero. For each of them we use the bijection $\chi^0 \mapsto (1, \chi)$, where χ is its unique ordinary irreducible character. The second and third blocks have defect groups of order 4 and 8 by [4, Lemma 4.2(d)]. The Alperin weight conjecture holds for both blocks by [62, Theorem 13.7]. Together with (5.2), this gives three weights for each block.

The radical subgroup of a weight belonging to either block is conjugate to a subgroup of its defect group [58, Theorem 4.14]. The row labelled 1 in [5, Table 8] accounts for the two blocks of defect zero, and the row labelled 2 has entry 0/0 in the trivial column. The classes $(2^2)_a$ and $(2^2)_b$ in [5, Table 8] are denoted by 2^2 and $(2^2)^*$, respectively, in [4, Table 2]. The defect groups of the block of defect 2 lie in the class $(2^2)_b$ by [4, Lemma 4.2(d)]. Its three weights therefore lie among the four weights in the $(2^2)_b$ row of [5, Table 8]. The entry 2/2 means that these four weights consist of two fixed weights and one orbit of length two. Since the block is fixed by the outer involution, its three weights consist of the orbit of length two and one fixed weight, giving the pair (3, 1).

For the block with dihedral defect group of order 8, the possible radical classes are $(2^2)_a$, $(2^2)_b$ and D_8 , with entries 1/0, 2/2 and 1/0 in the trivial column of [5, Table 8]. Three of the four $(2^2)_b$ -weights belong to the block of defect 2. Hence the only weights that can belong to the block with defect group D_8 are the $(2^2)_a$ -weight, the remaining fixed $(2^2)_b$ -weight and the D_8 -weight. Since this block has three weights, it contains all three. Each is fixed, giving the pair (3, 3). These pairs agree with the corresponding Brauer character counts in (5.2), so each of the two blocks b admits an $\mathrm{Aut}(X)_b$ -equivariant bijection $\mathrm{IBr}(b) \rightarrow \mathrm{Alp}(b)$. The program `fi24blocks.g`, described in Section A.4, independently confirms this distribution of the $(2^2)_a$ - and $(2^2)_b$ -weights for every candidate fusion.

Summing the entries in the trivial column of [5, Table 8] gives the pair (41, 31) for the union of the weight sets of all blocks in the trivial sector. Lemma 5.3, after removal of the two blocks

just treated and the two blocks of defect zero, gives an equivariant bijection for the principal block. Its pair of total and fixed weight counts is $(33, 25)$, in agreement with (5.2).

At 2, each faithful sector contains exactly two blocks. They have defects 21 and 3 and contain 23 and 2 Brauer characters, respectively. Summing the column labelled λ_3 in [5, Table 8] gives 25 weights in each faithful sector. The block of defect 3 has two weights by [62, Theorem 13.7]. The other block therefore has 23 weights.

At 3, set $G = Fi'_{24}$. The classification of radical subgroups and their normalisers in [4, Proposition 4.1 and Table 1] gives a radical subgroup $Q \cong C_3^2$ with $N_G(Q) \cong (3^2 : 2 \times G_2(3)).2$. By [4, Lemma 4.2(c) and Theorem 4.3], G has a unique nonprincipal block B_1 of positive defect, with defect group Q and four weights. In CTblLib 1.3.11 [14], the table **F3+** represents G . The table $(3^2 : 2 \times G_2(3)).2$ is the seventeenth entry in CTblLib's list of character tables of maximal subgroups of G , and represents $N_G(Q)$.

Let b be the 3-block of $N_G(Q)$ numbered 2 in the CTblLib calculation described in Section A.4. By [58, Theorem 4.14], the induced block b^G is defined. It can be identified from the reduced central characters of induced ordinary characters using [58, Lemma 6.1]. The programs **fi24p3.g** and **fi24blocks.g** select the four characters of $N_G(Q)/Q$ that have defect zero and whose inflations lie in b . The program **fi24blocks.g** applies the central character test to every class fusion returned by **PossibleClassFusions** for the indicated tables. For each of the six ordinary characters in b , every candidate identifies the block of G numbered 2. The three blocks of G have defects 16, 2, 0, respectively, and the first is principal. Thus the second is B_1 , and $b^G = B_1$. The four selected characters therefore give all four B_1 -weights.

The 3^2 entry of [5, Table 8] shows that two of these weights are fixed by the outer involution, giving the pair $(4, 2)$. The program **fi24blocks.g** gives the same pair of total and fixed Brauer character counts. Thus the two C_2 -sets are isomorphic, giving an equivariant bijection for B_1 . For the block of defect zero, use $\chi^0 \mapsto (1, \chi)$. Lemma 5.3, applied to the equivariant bijection constructed in [5, Section 4.3.4] between the irreducible Brauer characters and the conjugacy classes of weights in the entire sector, gives an equivariant bijection for the principal block. Its pair of total and fixed weight counts is $(25, 25)$, equal to the corresponding pair for Brauer characters computed by the program.

At 5, the nontrivial radical subgroups of Fi'_{24} have orders 5 and 25 [4, Proposition 4.1 and Table 1]. The central quotient correspondence in (5.1) gives their unique radical lifts to $3.Fi'_{24}$ and the corresponding normalisers. Weight characters in the trivial sector are obtained by the inflation described there. The program **fi24weights.g**, described in Section A.4, determines the block distributions and outer actions of these weights. It verifies that the resulting pairs of total and fixed weight counts are independent of the candidate class fusions and that every weight associated with a radical subgroup of order 5 belongs to a block with cyclic defect. The classification in [4, Proposition 4.1 and Table 1], together with the corresponding CTblLib character tables, accounts for the weights associated with all nontrivial radical subgroups. In the trivial sector the program gives the pairs $(16, 16)$, $(14, 6)$ and $(16, 16)$ for the three blocks with defect group C_5^2 , in the block order of the character table. The program **fi24blocks.g** gives the same pairs for Brauer characters, so these three blocks admit equivariant bijections. It also gives the pairs of blocks 45, 46 and 47, 48 interchanged by the outer involution. CTblLib's identifications of the character tables and the outer action show that these are the two pairs of blocks exchanged between the faithful sectors. The blocks in these pairs have 16 and 14 weights and Brauer characters, respectively. The blocks in the faithful sectors are treated by applying Remark 2.4(b) to their outer automorphism orbits. All remaining 5-blocks have cyclic defect or defect zero and are covered by [65, Proposition 6.2].

At 7, each of the three sectors has one block of noncyclic defect. Use the equivariant bijection for each entire sector constructed in [5, Section 4.3.4], with the counts in [5, Table 8]. The blocks with cyclic defect and the blocks of defect zero admit equivariant bijections by [65, Proposition 6.2]. In the trivial sector, apply Lemma 5.3 to the bijection on the entire sector,

using these blockwise bijections for the cyclic and defect zero blocks. This gives an equivariant bijection for the unique block of noncyclic defect in the trivial sector. Its pair of total and fixed weight counts is $(22, 12)$, equal to the corresponding pair for Brauer characters computed by `fi24blocks.g`. Each faithful sector has 63 weights and, by the sector bijection, 63 Brauer characters. The program `fi24blocks.g` gives 22 Brauer characters in its unique block of noncyclic defect. The remaining blocks therefore contain 41 Brauer characters and, by [65, Proposition 6.2], 41 weights. Thus the block of noncyclic defect has 22 weights. Choose a bijection for one of these two blocks and define the bijection for the other by conjugating with the outer involution.

At 11, 13, 17, 23, and 29, every block of positive defect has cyclic defect, as verified by `fi24blocks.g`. These blocks and the blocks of defect zero are covered by [65, Proposition 6.2]. We have therefore constructed an $\text{Aut}(X)_b$ -equivariant bijection $\text{IBr}(b) \rightarrow \text{Alp}(b)$ for every ℓ -block b of $X = X_\ell(Fi'_{24})$. Since $\text{Out}(Fi'_{24}) \cong C_2$, [30, Corollary 2.13] shows that every such block satisfies the inductive BAW condition. Remark 2.4(a) completes the proof for Fi'_{24} .

For $\mathbb{B}$, the universal ℓ' -covering group is $\mathbb{B}$ at 2 and $2.\mathbb{B}$ at odd primes. At 2 the ordinary character table has two blocks. They contain 179 and 5 ordinary irreducible characters and have defects 41 and 3, respectively. The correctness of the ordinary character table is proved in [15, Section 7]. The exact restriction matrices on the 2-regular classes have ranks 25 and 2. The calculation is given in Section A.5. By (A.1), it gives $|\text{IBr}(\mathbb{B})| = 27$. The weight calculation in [5, proof of Theorem 4.2 and Table 2], with the corrections in [6], gives 27 conjugacy classes of 2-weights. The nonprincipal block b of defect 3 has $|\text{IBr}(b)| = 2$ by (A.1), and $|\text{IBr}(b)| = |\text{Alp}(b)|$ by [62, Theorem 13.7]. Subtraction gives 25 weights for the principal block, which has 25 Brauer characters by (A.1). Thus $|\text{IBr}(b)| = |\text{Alp}(b)|$ for every 2-block b of $\mathbb{B}$. Since $\text{Out}(\mathbb{B}) = 1$, Lemma 5.2 shows that $\mathbb{B}$ satisfies iBAW at 2. At $\ell = 7$, the total 220 printed for the number of 7-regular conjugacy classes of $2.\mathbb{B}$ in the proof of [8, Lemma 5.2] should be 222. The supplementary calculation described in Section A.5 verifies the corrected total and the restriction ranks 24, 24, 21, 24 for the four blocks with noncyclic defect. The blockwise values used in [8, Theorem 5.3] are therefore unaffected. At odd primes, every block b of $2.\mathbb{B}$ with noncyclic defect satisfies $|\text{IBr}(b)| = |\text{Alp}(b)|$ by [8, Theorem 5.3]. Since $\text{Out}(\mathbb{B}) = 1$, [65, Proposition 6.2], applied to the simple group $\mathbb{B}$, gives the same equality for every block of $2.\mathbb{B}$ with cyclic defect, including defect zero. Since $\text{Out}(2.\mathbb{B}) \cong \text{Out}(\mathbb{B}) = 1$, Lemma 5.2 applies.

For $\mathbb{M}$ the universal ℓ' -covering group is $\mathbb{M}$ at every prime. At 2 there are five blocks, of defects 46, 0, 0, 4, 0 and numbers of ordinary irreducible characters 183, 1, 1, 8, 1. The correctness of the ordinary character table is proved in [16, Section 5]. The program in Section A.5 verifies that $\mathbb{M}$ has 61 2-regular conjugacy classes, so $|\text{IBr}(\mathbb{M})| = 61$ by [58, Corollary 2.10]. The weight calculation in [5, proof of Theorem 4.2 and Table 17], with the corrections in [6], gives 61 conjugacy classes of 2-weights. Each of the three blocks b of defect zero satisfies $|\text{IBr}(b)| = |\text{Alp}(b)| = 1$. The unique nonprincipal block b of defect 4 satisfies $|\text{IBr}(b)| = |\text{Alp}(b)|$ by [62, Theorem 13.7]. These four blocks are all the nonprincipal 2-blocks. Subtracting their Brauer character and weight counts from the respective totals gives $|\text{IBr}(B_0(\mathbb{M}))| = |\text{Alp}(B_0(\mathbb{M}))|$ for the principal block. Thus $|\text{IBr}(b)| = |\text{Alp}(b)|$ for every 2-block b of $\mathbb{M}$, and Lemma 5.2 shows that $\mathbb{M}$ satisfies iBAW at 2. For every odd $\ell \mid |\mathbb{M}|$, every ℓ -block b of $\mathbb{M}$ with noncyclic defect satisfies $|\text{IBr}(b)| = |\text{Alp}(b)|$ by [9, Theorem 4.4]. Since $\text{Out}(\mathbb{M}) = 1$, [65, Proposition 6.2], applied to the simple group $\mathbb{M}$, gives the same equality for every block with cyclic defect, including defect zero. Thus $|\text{IBr}(b)| = |\text{Alp}(b)|$ for every ℓ -block b of $\mathbb{M}$, and Lemma 5.2 shows that $\mathbb{M}$ satisfies iBAW at every odd prime dividing its order. $\square$

5.3. Proof of the sporadic theorem.

Proof of Theorem 5.1. The result follows from Proposition 5.4 and the CTBlocks verification [13] for the other 22 sporadic groups. $\square$

6. PROOF OF THEOREM 1.1 AND COROLLARY 1.2

Proof. Theorem 1.1 follows from Theorem 3.1, Corollary 4.15, and Theorem 5.1. For Corollary 1.2, let H be the finite group in its statement. Späth's reduction theorem [65, Theorem A] reduces the assertion to the nonabelian simple groups involved in H . Theorem 1.1 applies when ℓ divides the order of such a group S . Otherwise, every ℓ -block of its universal ℓ' -covering group has defect zero and satisfies the inductive BAW condition trivially. Remark 2.4(a) gives the conclusion. $\square$

APPENDIX A. FINITE COMPUTATIONS

The computational supplement contains eleven programs in the directory `code` and their complete transcripts in `outputs`. The programs were run with GAP 4.16.0 [38], CTblLib 1.3.11 [14], and AtlasRep 2.1.11 [72]. They are listed in Table A.1. The identifications of named character tables and stored fusions with the groups and inclusions below are taken from CTblLib and the classifications cited in each calculation. The computations with these tables do not establish those identifications.

A.1. Numbers of irreducible Brauer characters. For a block b , let V_b be the complex span of the restrictions to the ℓ -regular classes of the ordinary characters in b . The irreducible Brauer characters in b are linearly independent on the ℓ -regular classes. Since the decomposition matrix of b has full column rank, these characters span V_b . Consequently

$$(A.1) \quad \dim V_b = |\mathrm{IBr}(b)|.$$

A.2. $2.\mathrm{Sp}_6(2)$ at the prime 3. Let $X = 2.\mathrm{Sp}_6(2)$. The program `sp6.g` supplies the numerical details of the GAP calculation for X at 3 in [63, proof of Theorem 5.5]. It obtains the faithful permutation representation of degree 240 from AtlasRep. For a Sylow 3-subgroup P , it chooses representatives Q of the 20 P -conjugacy classes of subgroups, tests the condition

TABLE A.1. Programs in the computational supplement.

Program	Calculation
<code>sp6.g</code>	The 3-radical subgroups and blockwise weight counts for $2.\mathrm{Sp}_6(2)$.
<code>o7weights.g</code>	Two principal weight classes of $\mathrm{SO}_7(3)$ that each cover two principal weight classes of $\Omega_7(3)$.
<code>o7brauer.g</code>	Restrictions of principal block Brauer characters from $\mathrm{SO}_7(3)$ to $\Omega_7(3)$.
<code>o7blocks.g</code>	The nine 2-blocks of $3.\Omega_7(3)$ and their permutation under the outer involution.
<code>o7block2.g</code>	The decomposition matrix and outer action for the block B_2 .
<code>o7radical.g</code>	The 2-radical subgroups of $3.\Omega_7(3)$ and weight counts in its three central character sectors.
<code>fi24blocks.g</code>	Block data, Brauer character counts, outer actions, and block induction for Fi'_{24} and $3.Fi'_{24}$.
<code>fi24p3.g</code>	The quotient character table and inflated characters for the normaliser of a radical subgroup C_3^2 of Fi'_{24} .
<code>fi24weights.g</code>	Block distributions and outer actions of weights associated with radical subgroups of orders 5 and 25.
<code>baby.g</code>	Numbers of irreducible Brauer characters of the Baby Monster at 2 and its double cover at 7.
<code>monster.g</code>	The ordinary 2-blocks and number of 2-regular conjugacy classes of the Monster.

$Q = O_3(N_X(Q))$ for each representative, and then fuses the surviving classes in X . There are five conjugacy classes in X of 3-radical subgroups, including the trivial subgroup, of orders 1, 3, 27, 27, 81, whose respective normalisers have orders 2903040, 8640, 2592, 2592, 648. CTblLib supplies the named character tables used for these normalisers, together with stored fusions into the character table of X . The program verifies the orders of these tables, their 3-cores, and the corresponding quotient orders. For each named table, it selects the characters which are trivial on Q and have defect zero after passage to $N_X(Q)/Q$. It induces these characters to X , uses the resulting central characters to determine the blocks of X to which they belong, and checks that each induced character determines a unique block. In the block order of the character table, the resulting weight counts are 10, 2, 2, 1, 6. In the same order, the matrices of restrictions of the ordinary characters in these blocks to the 3-regular classes have ranks 10, 2, 2, 1, 6. By (A.1), these are the numbers of irreducible Brauer characters. The Brauer table supplied by CTblLib confirms these numbers. Thus the program verifies the numerical blockwise equalities in the exceptional calculation. The inductive BAW conclusion is supplied by [63, Theorem 5.5].

A.3. $3.\Omega_7(3)$ at the prime 2. Set $X = 3.\Omega_7(3)$. The programs `o7blocks.g` and `o7block2.g` use the CTblLib tables `3.07(3)` and `3.07(3).2` of X and its extension by the outer involution, together with the stored fusion for this inclusion, to determine the compatible table automorphism and its action on blocks and Brauer characters.

A.3.1. The principal block. Set $H = \text{SO}_7(3)$ and $S = \Omega_7(3)$. The program `o7weights.g` enumerates the P -conjugacy classes of subgroups of orders 2^7 and 2^8 in a Sylow 2-subgroup P of S . Among these classes, it retains one at each order whose representatives Q satisfy $C_P(Q) = Z(Q)$ and are 2-radical in H . For each retained representative Q , it verifies $C_H(Q) = Z(Q)$ and finds one ordinary character of 2-defect zero in $N_H(Q)/Q$. These give principal weights by [36, Lemma 2.3]. Since $Q \leq S$, [36, Corollary 2.22(2)] shows that each resulting principal H -weight class covers two S -weight classes.

The program `o7brauer.g` computes the restrictions of the ten principal block Brauer characters of H to S . Eight restrictions are irreducible and two have two distinct constituents. Modular Clifford theory identifies the two pairs of constituents as the two orbits of size two. Thus the principal Brauer characters and principal weights of S have isomorphic C_2 -actions.

A.3.2. The nine blocks of the triple cover. The program `o7blocks.g` determines the nine 2-blocks of X , their defects, central character sectors, Brauer characters, and their permutation under the outer involution. These data give Table 4.2.

For B_2 , `o7block2.g` gives five ordinary characters with respective heights $(0, 0, 1, 0, 0)$ and two Brauer characters. Its decomposition matrix is

$$\begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

It also shows that the outer involution fixes both Brauer characters. Proposition 4.11 uses these data to identify the defect group as D_8 and to show that the outer involution fixes both weight classes.

In the AtlasRep permutation representation of degree 1134, the program `o7radical.g` enumerates the 3021 conjugacy classes of subgroups of a Sylow 2-subgroup. It retains the 26 representatives satisfying $Q = O_2(N_X(Q))$ and fuses them into 12 X -classes. Every 2-radical subgroup of X is conjugate into the chosen Sylow subgroup, so the enumeration is exhaustive. For each fused class it forms $N_X(Q)/Q$, finds its ordinary characters of defect zero, and lists the restrictions of their inflations to $Z(X) \cong C_3$. The resulting totals in the three central character

sectors are $(17, 8, 8)$. The proof of Proposition 4.11 uses the two faithful totals to determine the weight counts of B_6 and B_7 .

A.4. The Fischer group. The program `fi24blocks.g` loads the named ordinary character tables `F3+`, `3.F3+`, and their outer extensions, with the corresponding maximal subgroup data from `CTblLib`. For a fixed character table and prime, block numbers refer to the numbering returned by `PrimeBlocks`, separately for each table. The program checks the covering group order and the set of prime divisors of the group order. At 2, 3, 5 and 7, it treats the blocks considered in Section 5. At 5 and 7, these are precisely the blocks of noncyclic defect. For an invariant selected block, it applies (A.1) to the restricted ordinary characters and to their sums with their images under the outer involution. The second rank is the dimension of the subspace fixed by the involution, so twice that rank minus the first is the number of fixed Brauer characters. For these blocks, the program checks the stated numbers of Brauer characters and fixed points, the central characters of the blocks, and the permutations of their labels under the outer involution. It separately checks the complete distributions of block defects at 5, 7, 11, 13, 17, 23, and 29. The weight counts and sector totals are supplied separately by `fi24weights.g` or the cited sources. At 2, the program `fi24blocks.g` also gives an independent check of the distribution among blocks of the $(2^2)_a$ - and $(2^2)_b$ -weights obtained in the proof of Proposition 5.4. It uses the identifications, supplied by [4] and `CTblLib`, of the tables $2\sim 2$. $U6(2).3.2$ and $(A4\times 08+(2).3).2$ with the normalisers in Fi'_{24} of representatives of these two radical classes, respectively. The selected characters are the inflations of the corresponding defect zero characters of the normaliser quotients. The program uses induced central characters to determine their blocks in Fi'_{24} . The fusions induced by the corresponding inclusions are among the candidates tested. For every fusion returned by `PossibleClassFusions`, the program verifies that each central character determines a unique block and that the resulting distribution is independent of the candidate.

For $G = Fi'_{24}$, the classification of radical subgroups and their normalisers in [4, Proposition 4.1 and Table 1] gives a radical subgroup $Q \cong C_3^2$ with $N_G(Q) \cong (3^2 : 2 \times G_2(3)).2$. The table $(3\sim 2 : 2 \times G_2(3)).2$ is the seventeenth entry in `CTblLib`'s list of character tables of maximal subgroups of G . `CTblLib` also supplies its fusion into `F3+`.

The program `fi24p3.g` performs a separate calculation using the character table of $N_G(Q)$. It forms the character table of the quotient by the computed 3-core and selects the four irreducible ordinary characters belonging to 3-blocks of defect zero. It inflates these characters to $N_G(Q)$, determines their block numbers, and verifies that each corresponding row of the decomposition matrix has exactly one nonzero entry. Character numbers refer to positions in the ordered lists of irreducible ordinary characters. Each list of six entries in the output records, in order, the character number in the quotient table, its degree and block number, the character number and block number of its inflation, and the column containing the nonzero decomposition number.

The program `fi24blocks.g` then considers each entry in the list of 32 class fusions returned by `PossibleClassFusions` between the character tables of $N_G(Q)$ and G . No assertion that the entries are distinct or form an orbit is used. The stored `CTblLib` fusion is one of these. For each of the six irreducible ordinary characters at positions 23, 24, 51, 52, 65, 76 in the character table of $N_G(Q)$, all belonging to block 2, the induced central character determines block 2 of the character table of G for every returned candidate. The three blocks of G have defects 16, 2, 0. By [58, Theorem 4.14 and Lemma 6.1], these calculations with central characters give the block induction used in Section 5. This calculation treats $Q \cong C_3^2$ and does not enumerate all nontrivial radical subgroups of G .

The program `fi24weights.g` uses the character tables of the normalisers of radical subgroups of orders 5 and 25 in Fi'_{24} and $3.Fi'_{24}$. It selects the characters inflated from characters of defect zero of the quotient by the radical subgroup. The classification in [4, Proposition 4.1 and Table 1] gives the normalisers, whose named character tables are supplied by `CTblLib`. For each entry

returned by `PossibleClassFusions` for the named tables, the program induces every selected character to the corresponding group, Fi'_{24} or $3.Fi'_{24}$, requires its central character to determine a unique block, and obtains the same block distribution. For each returned outer fusion entry, it verifies an involutory permutation preserving the set of selected characters. The resulting pairs of total and fixed weight counts are independent of the candidate choices. The program obtains the pairs (16, 16), (14, 6), and (16, 16) of the three invariant blocks and the counts in table positions 45–48. The program `fi24blocks.g` gives the pairs of blocks interchanged by the outer involution, $45 \leftrightarrow 46$ and $47 \leftrightarrow 48$. Using CTblLib's identification of the tables and the outer action, these are the two pairs of blocks exchanged between the faithful sectors, with 16 and 14 weights per block, respectively. For the radical subgroup of order 5, the same selection of quotient characters and central character test place every selected weight in a block of cyclic defect.

A.5. The Baby Monster and the Monster. For the Baby Monster, `baby.g` uses the ordinary character table in CTblLib, whose correctness is proved in [15, Section 7]. It uses `PrimeBlocks` to determine the block containing each ordinary irreducible character. For each block, it restricts its ordinary characters to the 2-regular classes and computes the rank of the resulting matrix. The two blocks have defects 41, 3, numbers of ordinary characters 179, 5, and ranks 25 and 2. Both the matrix formed by combining bases for the row spaces of these two matrices and the matrix of all the restricted characters have rank 27. The calculation does not use a Baby Monster 2-Brauer table from CTblLib. The program `baby.g` also loads the ordinary table of $2.B$ at 7 and verifies that it has 222 7-regular classes, rather than the 220 printed in the proof of [8, Lemma 5.2], and that its four blocks of defect 2 have restriction ranks 24, 24, 21, 24. Hence the blockwise values used in [8, Theorem 5.3] are unchanged.

For the Monster, `monster.g` uses the ordinary character table in CTblLib, whose correctness is proved in [16, Section 5]. It verifies that the ordinary irreducible characters form five blocks with respective defects (46, 0, 0, 4, 0) and numbers of ordinary irreducible characters (183, 1, 1, 8, 1). It also checks the three characters of defect zero, all heights in the block of defect 4, and that there are 61 2-regular classes. The total number of Brauer characters is obtained from these 61 classes, rather than from the number of ordinary characters. The verified block defects and numbers of ordinary characters are used in Proposition 5.4, together with the total number of Brauer characters and the result for the block of defect 4.

APPENDIX B. PROOFS OF LEMMAS 2.11 AND 2.12

Proof of Lemma 2.11. For sufficiently large p and q , Geck and Hézard prove the conclusion in [42, Proposition 4.3]. We adapt their proof to the present hypotheses. We first prove that the restrictions of $D_H(\Gamma_{u_1}), \dots, D_H(\Gamma_{u_d})$ to $\mathcal{C}^F$ are linearly independent. We then show that the restrictions of the characters in $\text{Irr}_{s, \mathcal{F}}(H)$ span the H -invariant functions on $\mathcal{C}^F$. We obtain characters satisfying (2.2) by combining these facts with the multiplicity sum formula (B.10).

Fix a prime $\ell_0 \neq p$. For each F -stable pair $\iota = (\mathcal{C}, \mathcal{E}_\iota)$, where $\mathcal{E}_\iota$ is an irreducible $\mathbf{H}$ -equivariant $\overline{\mathbb{Q}_{\ell_0}}$ -local system on $\mathcal{C}$, choose a Frobenius isomorphism with the normalisation of [70, 10.1–10.8]. Let Y_ι be the trace function associated with $\mathcal{E}_\iota$, and let X_ι be the trace function associated with the corresponding intersection cohomology complex. Extend them by zero outside $\mathcal{C}^F$ and $\overline{\mathcal{C}}^F$, respectively. Following [70, 11.12], we obtain these functions from those of [70, 10.8] by composition with the Springer isomorphism chosen for the construction of the generalised Gelfand–Graev characters [70, Proposition 4.6 and Section 5]. This is an $\mathbf{H}$ -equivariant isomorphism from the variety of unipotent elements of $\mathbf{H}$ to the variety of nilpotent elements in its Lie algebra, chosen to commute with F . Let $\Gamma_\iota^{\text{mod}}$ be the modified class function defined in [70, (11.15)], which is a linear combination of $\Gamma_{u_1}, \dots, \Gamma_{u_d}$. We will prove the required linear independence on $\mathcal{C}^F$ by calculating $\langle D_H(\Gamma_\iota^{\text{mod}}), X_\kappa \rangle_H$ for pairs ι, κ supported on $\mathcal{C}$, as in [40, Lemma 3.5], and using the vanishing assertion in [40, Corollary 3.6(b)].

To carry out Geck–Hézar’s calculation for every odd q , we use Taylor’s expansions of the generalised Gelfand–Graev characters and the modified functions in terms of intersection cohomology trace functions [70, Theorem 11.13 and Lemma 11.16]. These formulas assume that p is *acceptable* for $\mathbf{H}$. Since p is odd, it is *very good* for $\mathbf{H}$, and hence acceptable in Taylor’s sense [70, Definition 6.1 and Lemma 6.3(i)]. Taylor’s acceptability condition also implies Letellier’s condition [51, Assumption 5.0.14], as explained in [70, 6.2]. We will use this implication later to justify the application of Letellier’s Frobenius equivariant induction isomorphism [51, (6.2.11)] in the comparison of Fourier constants. We also use Lusztig’s duality identities [54, Proposition 8.5 and Corollary 8.6]. Because $Z(\mathbf{H})$ is connected, [70, Corollary 13.6] shows that Lusztig’s duality identities hold without a restriction on q .

The equivariant local systems on $\mathcal{C}$ are parametrised by $\text{Irr}(A_{\mathbf{H}}(u_0))$ [41, (4.2)]. Since F acts trivially on $A_{\mathbf{H}}(u_0)$, the coefficient matrix expressing the modified functions in terms of $\Gamma_{u_1}, \dots, \Gamma_{u_d}$ is the character table of $A_{\mathbf{H}}(u_0)$, up to nonzero scalar factors [44, Section 3(A)]. It is therefore invertible. The orthogonality and triangularity relations and the inverse matrix formula needed in the proof of [40, Lemma 3.5] are supplied by [70, Lemma 10.9, (10.11), Theorem 10.14 and (10.20)], without assuming that F is split [70, 10.1]. Using these identities with Taylor’s expansions and Lusztig’s duality identities, the calculation of [40, Lemma 3.5] gives

$$(B.1) \quad \langle D_H(\Gamma_{\iota}^{\text{mod}}), X_{\kappa} \rangle_H = c_{\iota} \delta_{\iota\kappa}, \quad c_{\iota} \neq 0,$$

for pairs ι, κ supported on $\mathcal{C}$.

The proof of [40, Corollary 3.6(b)], using these expansions, shows that

$$(B.2) \quad D_H(\Gamma_{\iota}^{\text{mod}})(v) \neq 0 \implies \mathcal{C} \subseteq \overline{\mathcal{C}}_v$$

for every unipotent element $v \in H$, where $\mathcal{C}_v$ is its $\mathbf{H}$ -conjugacy class. The functions $D_H(\Gamma_{\iota}^{\text{mod}})$ also vanish on nonunipotent elements. By (B.2), they vanish on $\overline{\mathcal{C}}^F \setminus \mathcal{C}^F$. Since X_{κ} vanishes outside $\overline{\mathcal{C}}^F$, only values on $\mathcal{C}^F$ contribute to the scalar product in (B.1). Pairing a linear relation between the restricted modified functions with each X_{κ} forces every coefficient to vanish. The invertible coefficient matrix consequently proves that

$$D_H(\Gamma_{u_1})|_{\mathcal{C}^F}, \dots, D_H(\Gamma_{u_d})|_{\mathcal{C}^F}$$

are linearly independent. The same change of basis gives the vanishing property (B.2) and vanishing on nonunipotent elements for each $D_H(\Gamma_{u_j})$.

We next prove that the restrictions of the characters in $\text{Irr}_{s, \mathcal{F}}(H)$ span the H -invariant functions on $\mathcal{C}^F$. For this purpose, we use the restriction theorem [41, Theorem 4.5], which identifies the relevant restrictions of character sheaves with the irreducible equivariant local systems on $\mathcal{C}$. To apply Geck’s proof of [41, Theorem 4.5] in odd characteristic, we first verify the scalar product and vanishing formulas in [41, (2.4)] and the compatibility of their constants with cuspidal induction in [41, (3.3)(b)]. The large characteristic assumption in [41] enters through these character formulas, as explained in [41, Basic assumptions, p. 140] and [41, (4.3)].

Let $\mathcal{L}$ be the finite collection of groups required in Geck’s reduction: $\mathbf{H}$, the Levi subgroups supporting the cuspidal pairs used in [41, (4.9)], and the adjoint quotients and further supporting Levi subgroups occurring in the proofs of [41, Propositions 5.3 and 5.5]. Recall that a pair is cuspidal if its generalised Springer block consists of that pair alone [41, (3.3)]. Every group in $\mathcal{L}$ has connected centre and is either a torus or of type B modulo its centre. Indeed, a Levi subgroup in type B has components of type A apart from at most one of type B, and a nontrivial component of type A cannot support a cuspidal pair when the centre is connected [41, Remark 3.4 and the proof of Theorem 3.8]. Passing to an adjoint quotient does not change the Dynkin type, and the same restriction on components applies to its supporting Levi subgroups.

The restriction multiplicities are independent of the rational structure [41, p. 151]. We therefore choose a Frobenius endomorphism F' defining a split $\mathbb{F}_{q'}$ -structure, with $q' \equiv 1 \pmod{4}$.

This congruence will give the value 1 for the cuspidal constants in [41, Theorem 3.8], once we have justified the use of its calculation in odd characteristic. It can be ensured by taking an even power of a split Frobenius endomorphism. Choose F' to satisfy [41, (4.7)(a)–(c)] for the chosen semisimple elements in the dual groups and standard Levi subgroups required in Geck's argument. As in the opening paragraph of the proof of [41, Proposition 5.3], we impose these conditions also for the additional semisimple elements used in that proof. Each chosen semisimple element is defined over a finite extension of $\mathbb{F}_p$, so a common power of a split Frobenius endomorphism fixes all these elements. This power permutes the finite sets of character sheaves in the corresponding series. A further common power therefore fixes every required character sheaf, as explained in [41, (4.7)]. Taking a sufficiently large further power makes q' large enough for geometric induction to coincide with Deligne–Lusztig induction [51, p. 115]. The standard Levi subgroups and their adjoint quotients inherit split structures from F' . Since p is odd, it is very good for every group in $\mathcal{L}$, and Taylor's formulas apply by [70, Lemma 6.3(i)]. Thus Letellier's characteristic hypothesis also holds for every group in $\mathcal{L}$. With q' chosen as above, we may use the Frobenius equivariant isomorphism [51, (6.2.11)].

To verify the formulas in [41, (2.4)] used in the proof of Geck's restriction theorem [41, Theorem 4.5], we need the precise scalar product coefficients for every $\mathbf{L} \in \mathcal{L}$. Fix $\mathbf{L} \in \mathcal{L}$, and set $L' = \mathbf{L}^{F'}$. For each F' -stable pair $\iota = (\mathcal{D}, \mathcal{E}_\iota)$, choose the Frobenius isomorphisms with the compatible normalisations of [70, 10.1–10.8]. Write Y_ι for the trace function of $\mathcal{E}_\iota$ and X_ι for the trace function of its intersection cohomology complex, extended by zero outside $\mathcal{D}^{F'}$ and $\overline{\mathcal{D}}^{F'}$, respectively. As before, $\Gamma_\iota^{\text{mod}}$ is the linear combination of generalised Gelfand–Graev characters defined in [70, (11.15)], also given in [41, (2.3)(a)]. Let $\mathcal{I}$ be the generalised Springer block containing ι , induced from a cuspidal pair ι_0 of a standard Levi subgroup $\mathbf{M}$. Set

$$a = |A_{\mathbf{L}}(v)| \quad (v \in \mathcal{D}), \quad b_\iota = \frac{\dim \mathbf{L} - \dim \mathcal{D} - \dim Z(\mathbf{M})}{2}.$$

Let $\zeta_{\mathcal{I}}$ be the fourth root of unity in [70, Proposition 11.5], and set

$$\delta_{\mathcal{I}} = (-1)^{\text{rank}(\mathbf{M}/Z(\mathbf{M}))}, \quad \zeta'_{\mathcal{I}} = \delta_{\mathcal{I}} \zeta_{\mathcal{I}}^{-1},$$

as in [54, (8.4)(a)] and [41, (2.4)].

Taylor's expansions [70, Theorem 11.13 and Lemma 11.16] and Lusztig's duality identities [54, Proposition 8.5 and Corollary 8.6], valid here by [70, Corollary 13.6], allow us to repeat the calculations in [40, Lemma 3.5 and Corollary 3.6(b)] for $(\mathbf{L}, F')$. In particular, $D_{L'}(\Gamma_\iota^{\text{mod}})$ vanishes on nonunipotent elements and satisfies

$$(B.3) \quad D_{L'}(\Gamma_\iota^{\text{mod}})(x) \neq 0 \implies \mathcal{D} \subseteq \overline{\mathcal{C}_x}$$

for unipotent $x \in L'$, where $\mathcal{C}_x$ is its $\mathbf{L}$ -conjugacy class. By (B.3), $D_{L'}(\Gamma_\iota^{\text{mod}})$ vanishes on $\overline{\mathcal{D}}^{F'} \setminus \mathcal{D}^{F'}$. For a pair κ supported on $\mathcal{D}$, we have $X_\kappa = Y_\kappa$ on $\mathcal{D}^{F'}$, and X_κ vanishes outside $\overline{\mathcal{D}}^{F'}$. Replacing X_κ by Y_κ therefore does not change its scalar product with $D_{L'}(\Gamma_\iota^{\text{mod}})$. The calculation of [40, Lemma 3.5] gives

$$(B.4) \quad \langle D_{L'}(\Gamma_\iota^{\text{mod}}), Y_\kappa \rangle_{L'} = a \zeta'_{\mathcal{I}} q'^{-b_\iota} \delta_{\iota\kappa}$$

for F' -stable pairs ι, κ supported on $\mathcal{D}$. Formula (B.4) verifies [41, (2.4)(a)], while (B.3), together with vanishing on nonunipotent elements, provides the required vanishing assertion in [41, (2.4)(c)]. If $\mathbf{M}$ is a torus, then $\zeta'_{\mathcal{I}} = 1$, giving [41, (2.4)(b)].

It remains to compare $\zeta'_{\mathcal{I}}$ with the constant attached to the cuspidal pair ι_0 , computed in $\mathbf{M}$. We prove the compatibility required in [41, (3.3)(b)] using the Frobenius equivariant induction isomorphism [51, (6.2.11)]. Let $\gamma_{\mathbf{L}}$ be the normalised Fourier constant in [70, Lemma 11.3] for the complex induced from ι_0 , and let $\gamma_{\mathbf{M}}$ be the corresponding constant for the complex on the Lie algebra of $\mathbf{M}$ constructed from ι_0 . Thus $\gamma_{\mathbf{L}}$ is the scalar remaining after extracting the factor $q'^{(\dim \mathbf{L} + \dim Z(\mathbf{M}))/2}$, with the analogous convention for $\gamma_{\mathbf{M}}$. Choose compatible Frobenius

structures and an invariant form on the Lie algebra of $\mathbf{L}$, restricted to that of $\mathbf{M}$. Choose an F' -stable parabolic subgroup $\mathbf{P}$ of $\mathbf{L}$ with Levi subgroup $\mathbf{M}$ and unipotent radical $U_{\mathbf{P}}$.

For $w = 1$, the induction isomorphism [51, (6.2.11)] includes a Tate twist by $\dim U_{\mathbf{P}}$, which multiplies the Frobenius trace by $q'^{-\dim U_{\mathbf{P}}}$ [51, Remark 4.4.6]. Since $\dim \mathbf{L} - \dim \mathbf{M} = 2 \dim U_{\mathbf{P}}$, the powers of q' in these normalisations satisfy

$$(B.5) \quad q'^{-\dim U_{\mathbf{P}}} q'^{(\dim \mathbf{L} + \dim Z(\mathbf{M}))/2} = q'^{(\dim \mathbf{M} + \dim Z(\mathbf{M}))/2}.$$

Taking Frobenius traces in [51, (6.2.11)], with the normalisation of [70, Lemma 11.3], compares the Fourier constants for $\mathbf{L}$ and $\mathbf{M}$. Cancelling the powers of q' by (B.5) gives

$$(B.6) \quad \gamma_{\mathbf{L}} = \gamma_{\mathbf{M}}.$$

This comparison also occurs in the proof of [54, Proposition 7.2], with the invariant form restricted to the Lie algebra of the cuspidal Levi subgroup. By [70, Lemma 11.3], $\gamma_{\mathbf{L}}^2 = \nu_{\mathbf{L}}$, where $\nu_{\mathbf{L}}$ is the sign in that lemma. The proof of [70, Proposition 11.5] identifies $\zeta_{\mathcal{I}} = \nu_{\mathbf{L}} \gamma_{\mathbf{L}}^{-1}$. Hence $\zeta_{\mathcal{I}} = \gamma_{\mathbf{L}}$. The same argument in $\mathbf{M}$ gives $\zeta_{\iota_0} = \gamma_{\mathbf{M}}$. The signs attached to $\mathcal{I}$ in $\mathbf{L}$ and to ι_0 in $\mathbf{M}$ satisfy

$$\delta_{\mathcal{I}} = \delta_{\iota_0} = (-1)^{\text{rank}(\mathbf{M}/Z(\mathbf{M}))}$$

by [54, (8.4)(a)]. Combining these equalities with (B.6) and the definition $\zeta' = \delta \zeta^{-1}$ gives

$$(B.7) \quad \zeta'_{\mathcal{I}} = \zeta'_{\iota_0}.$$

This is the compatibility required in [41, (3.3)(b)].

To determine the constants ζ'_{ι_0} by Geck's induction, we first establish the integrality conclusion of [41, Corollary 3.2] for the groups in $\mathcal{L}$. Let $\mathbf{L} \in \mathcal{L}$ be a group other than a torus. For every F' -stable unipotent class $\mathcal{D}$ of $\mathbf{L}$, the theorem of H  zard and Lusztig stated in [68, Theorem 1.1] gives a character $\chi \in \text{Irr}(L')$ with unipotent support $\mathcal{D}$ and generic denominator $n_{\chi} = |A_{\mathbf{L}}(v)|$ for $v \in \mathcal{D}$. The degree polynomial relation for Alvis–Curtis duality gives $n_{\chi^*} = n_{\chi}$ [45, Proposition 3.4.21]. If $v_1, \dots, v_r$ represent the L' -classes in $\mathcal{D}^{F'}$, then [70, (11.15), Lemma 14.15 and Proposition 15.4] give

$$(B.8) \quad \sum_{j=1}^r [A_{\mathbf{L}}(v_j) : A_{\mathbf{L}}(v_j)^{F'}] \langle \chi^*, \Gamma_{v_j} \rangle_{L'} = \frac{|A_{\mathbf{L}}(v)|}{n_{\chi^*}} = 1.$$

The multiplicities are nonnegative integers and the indices are positive integers, so exactly one multiplicity is nonzero, and it and its corresponding index equal 1. Expanding a modified generalised Gelfand–Graev class function indexed by a local system on $\mathcal{D}$ then expresses its scalar product with χ^* as the value of that local system's trace function at the unique representative with nonzero multiplicity. This is the remaining assertion of [41, Proposition 3.1]. For a torus, the only unipotent element is 1 and every irreducible character has generic denominator 1, so the same multiplicity conclusions hold. Together with the identities proved above, this establishes [41, Corollary 3.2] for every group in $\mathcal{L}$.

The induction in the proof of [41, Theorem 3.8] therefore applies to every group in $\mathcal{L}$. For a cuspidal pair ι_0 on $\mathbf{M}$, it gives

$$(B.9) \quad \zeta'_{\iota_0} = \varepsilon^{(\text{rank } \mathbf{M} - \dim Z(\mathbf{M}))/2}, \quad q' \equiv \varepsilon \pmod{4}, \quad \varepsilon \in \{1, -1\}.$$

Our choice $q' \equiv 1 \pmod{4}$ gives $\varepsilon = 1$, so $\zeta'_{\iota_0} = 1$. By (B.7), $\zeta'_{\mathcal{I}} = 1$ for every relevant generalised Springer block. The support properties in [41, (4.3)] follow from [54, Theorem 10.7] by [70, Corollary 13.6]. For this F' , [64, Theorem 3.2] gives the equality of spans stated in [41, Theorem 7.1] for these classical groups with connected centre. For a torus, each rational Lusztig series consists of one linear character, and the trace function of the corresponding character sheaf is a nonzero scalar multiple of it. Thus the same equality of spans holds. This verifies the formulas and comparisons used in Geck's proof of the restriction theorem [41, Theorem 4.5].

We now verify the conditions in [41, (4.4)(a)–(b)] for $(s, \mathcal{F})$. The elementary abelian groups $\mathcal{G}_{\mathcal{F}}$ and $A_{\mathbf{H}}(u_0)$ have the same order and are therefore isomorphic, giving [41, (4.4)(a)]. Isolation

together with this order equality gives [41, (4.4)(b)], by Hézard's result [42, Proposition 2.3]. By the restriction theorem [41, Theorem 4.5], restriction to $\mathcal{C}$, up to shift, gives a bijection between the character sheaves in the family indexed by $(s, \mathcal{F})$ whose restriction to $\mathcal{C}$ is nonzero and the irreducible $\mathbf{H}$ -equivariant local systems on $\mathcal{C}$. The multiplicities in these geometric restrictions are independent of the rational structure, as explained after [41, Theorem 4.5].

We now return to the given Frobenius endomorphism F and use this restriction bijection to show that the restrictions of the characters in $\text{Irr}_{s, \mathcal{F}}(H)$ span the H -invariant functions on $\mathcal{C}^F$. The choice $\mathbf{T}^* \subseteq \mathbf{B}^*$ and the absence of nontrivial graph automorphisms in type C_n imply that F acts trivially on the Weyl group of $\mathbf{H}^*$ and on W_s . Since s is fixed by F , F preserves the family of character sheaves indexed by $(s, \mathcal{F})$. Every irreducible $\mathbf{H}$ -equivariant local system on $\mathcal{C}$ is F -stable because F acts trivially on $A_{\mathbf{H}}(u_0)$. Restriction to $\mathcal{C}$ commutes with pullback by F . The uniqueness in the restriction bijection therefore makes the corresponding character sheaves F -stable, as in the proof of [42, Corollary 3.5].

Since F acts trivially on $A_{\mathbf{H}}(u_0)$, the values at $u_1, \dots, u_d$ of the trace functions of these local systems form the character table of $A_{\mathbf{H}}(u_0)$, up to nonzero scalar multiples of the rows. The same holds for the restrictions to $\mathcal{C}^F$ of the trace functions of the corresponding character sheaves. By Shoji's theorem [64, Theorem 3.2], the trace functions of these character sheaves are nonzero scalar multiples of almost characters lying in the span of $\text{Irr}_{s, \mathcal{F}}(H)$. It follows that there are characters $\chi_1, \dots, \chi_d \in \text{Irr}_{s, \mathcal{F}}(H)$ such that

$$M = (\chi_i(u_j))_{1 \leq i, j \leq d}$$

is nonsingular.

To apply the multiplicity formula to every character in $\text{Irr}_{s, \mathcal{F}}(H)$, we identify its unipotent support and generic denominator. Let W be the Weyl group of $\mathbf{H}^*$ and let E_0 be the unique special character in $\mathcal{F}$. Its truncated induction $j_{W_s}^W(E_0)$ is the unique constituent of $\text{Ind}_{W_s}^W(E_0)$ with b -invariant $b(E_0)$ [69, p. 476]. Under the Springer correspondence, $j_{W_s}^W(E_0)$ corresponds to $\mathcal{C}$ [44, Section 3(C)]. Consequently, every $\rho \in \text{Irr}_{s, \mathcal{F}}(H)$ has unipotent support $\mathcal{C}$ [44, Theorem 3.7]. Since $\mathcal{G}_{\mathcal{F}}$ is abelian, the denominator formula [41, (6.1)(c)] and the degree polynomial relation for Alvis–Curtis duality [45, Proposition 3.4.21] give

$$n_{\rho^*} = n_{\rho} = |\mathcal{G}_{\mathcal{F}}|.$$

By [70, Lemma 14.15], $\mathcal{C}$ is the wave front set of ρ^* .

Conjugation in $\mathbf{H}$ identifies the component groups of u_j and u_0 . Under this identification, the actions of F differ by an inner automorphism of $A_{\mathbf{H}}(u_0)$. Since this group is abelian and F acts trivially on it, F also acts trivially on $A_{\mathbf{H}}(u_j)$. Consequently,

$$[A_{\mathbf{H}}(u_j) : A_{\mathbf{H}}(u_j)^F] = 1 \quad (1 \leq j \leq d).$$

Apply the formula of Lusztig and Geck–Malle in [70, Proposition 15.4] to ρ^* , using the constant local system on $\mathcal{C}$ with trace function identically 1. In the expansion [70, (11.15)], the coefficient of Γ_{u_j} is the index just calculated. Hence

$$(B.10) \quad \sum_{j=1}^d \langle \rho^*, \Gamma_{u_j} \rangle_H = \frac{|A_{\mathbf{H}}(u_0)|}{n_{\rho^*}} = 1.$$

Each multiplicity is a nonnegative integer, so there is a unique index j for which $\langle \rho^*, \Gamma_{u_j} \rangle_H = 1$, and all the other multiplicities vanish. For $1 \leq j \leq d$, define

$$I_j = \{\rho \in \text{Irr}_{s, \mathcal{F}}(H) : \langle \rho^*, \Gamma_{u_j} \rangle_H = 1\}.$$

It remains to prove that every I_j is nonempty.

Suppose that I_r is empty. Then $\langle \rho^*, \Gamma_{u_r} \rangle_H = 0$ for every $\rho \in \text{Irr}_{s, \mathcal{F}}(H)$. Since $D_H(\rho) = \pm \rho^*$, self-adjointness of Alvis–Curtis duality gives

$$\langle \rho, D_H(\Gamma_{u_r}) \rangle_H = \langle D_H(\rho), \Gamma_{u_r} \rangle_H = 0.$$

As proved after (B.2), $D_H(\Gamma_{u_r})$ vanishes on nonunipotent elements. By (B.2), its value at a unipotent element v can be nonzero only if $\mathcal{C} \subseteq \overline{\mathcal{C}_v}$. If $\mathcal{C}_v \neq \mathcal{C}$, this inclusion implies $\dim \mathcal{C}_v > \dim \mathcal{C}$. On such a class, ρ vanishes by [54, Theorem 11.2(iv)], applied to ρ^* . This theorem applies here by [70, Corollary 13.6]. Thus only elements of $\mathcal{C}^F$ contribute to $\langle \rho, D_H(\Gamma_{u_r}) \rangle_H$, and

$$\sum_{j=1}^d \frac{\rho(u_j) \overline{D_H(\Gamma_{u_r})(u_j)}}{|C_H(u_j)|} = 0 \quad (\rho \in \text{Irr}_{s, \mathcal{F}}(H)).$$

Apply these equalities to $\chi_1, \dots, \chi_d$. The nonsingularity of $M = (\chi_i(u_j))_{1 \leq i, j \leq d}$ implies that $D_H(\Gamma_{u_r})(u_j) = 0$ for every j . Its restriction to $\mathcal{C}^F$ is therefore zero, contradicting the linear independence established after (B.1). Hence every I_j is nonempty. Choosing $\rho_j \in I_j$ gives (2.2). $\square$

Proof of Lemma 2.12. We adapt the proof of Lemma 2.11 to the selected type C families. We first verify that the central multiple $\tilde{s}$ in the statement exists and is fixed by F_0 . Condition (P4) of [69, Theorem 2.11] will replace the isolation hypothesis used in Lemma 2.11. By [69, 6.19], the image of s_0 in $\mathbf{H}^*/Z(\mathbf{H}^*) \cong \text{SO}_{2m+1}$ is quasi-isolated, so its square is one [10, Proposition 4.11(a)]. Thus $s_0^2 \in Z(\mathbf{H}^*)$. Since $\mathbf{H}$ has simply connected derived subgroup, $Z(\mathbf{H}^*)$ is a torus [45, Lemma 1.5.22] and contains a square root of s_0^{-2} . This proves that z can be chosen as in the statement. Multiplication by z leaves the centraliser unchanged, so $W_{\tilde{s}} = W_{s_0}$ and the selected family is still $\mathcal{F}$. The Frobenius endomorphism F_0 acts on $\mathbf{T}^*$ by the q_0 -power map. Since q_0 is odd and $\tilde{s}^2 = 1$, $F_0(\tilde{s}) = \tilde{s}$. The torus $\mathbf{T}^*$ is split, so F_0 acts trivially on the Weyl group of $\mathbf{H}^*$ and hence on $W_{\tilde{s}}$.

We next verify the conditions in [41, (4.4)(a)–(b)] for $(\tilde{s}, \mathcal{F})$. By [69, 10.3], we identify $W_{\tilde{s}}$ with $W(\text{D}_a \text{B}_b)$, where $a + b = m$. When $a = 1$, the corresponding factor is a torus with trivial Weyl group, so $W_{\tilde{s}} = W(\text{B}_{m-1})$. Let E_0 be the unique special character in $\mathcal{F}$, using the terminology of [69, p. 476], and let W be the Weyl group of $\mathbf{H}^*$. Hézard's construction [69, Proposition 10.2] gives $j_{W_{\tilde{s}}}^W(E_0) = E_C$, where E_C is the Springer character corresponding to $\mathcal{C}$ and its trivial local system. The construction gives $|\mathcal{G}_{\mathcal{F}}| = |A_{\mathbf{H}}(u)|$ for $u \in \mathcal{C}$ by [69, 7.1 and Proposition 10.2]. The component group $A_{\mathbf{H}}(u)$ is an elementary abelian 2-group by [68, proof of Proposition 2.4], and the same holds for $\mathcal{G}_{\mathcal{F}}$ by [45, Remark 4.2.17]. These two groups are therefore isomorphic, giving [41, (4.4)(a)]. The selected pair satisfies condition (P4) of [69, Theorem 2.11]. For these type C pairs, the verification in [69, 7.4] uses the equality $\tilde{s}^2 = 1$ and does not require $\tilde{s}$ to be isolated. By [69, Definition 2.7], condition (P4) gives

$$\text{Ind}_{W(\text{D}_a \text{B}_b)}^W(E_0) = E_C + \sum_{\substack{E \in \text{Irr}(W) \\ d(E) > b(E_0)}} m_E E, \quad m_E \in \mathbb{Z}_{\geq 0},$$

where $d(E)$ is the dimension of the Springer fibre, the variety of Borel subgroups containing a representative of the unipotent class corresponding to E . This is the induction condition [41, (4.4)(b)].

To apply Geck's restriction theorem [41, Theorem 4.5] in odd characteristic, we verify analogues of (B.3) and (B.4) for $\mathbf{H}$, its standard Levi subgroups supporting cuspidal pairs, their adjoint quotients, and the further supporting Levi subgroups used in [41, (4.9)]. Let $\mathbf{L}$ be any one of these algebraic groups. These groups have connected centre and are either tori or simple modulo their centre of type C. Indeed, a nontrivial component of type A cannot support a cuspidal pair when the centre is connected [41, Remark 3.4 and the proof of Theorem 3.8]. Odd characteristic is very good for these groups and hence acceptable in Taylor's sense by [70, Lemma 6.3(i)]. Choose a sufficiently large and divisible positive integer N such that $F' = F_0^{2N}$ satisfies [41, (4.7)(a)–(c)] and the conditions on the additional objects in the proofs of [41, Propositions 5.3 and 5.5]. As in the proof of Lemma 2.11, take N large enough for geometric induction to coincide with Deligne–Lusztig induction [51, p. 115] for every group used in the proofs of [41, Propositions 5.3 and 5.5]. The field size satisfies $q' = q_0^{2N} \equiv 1 \pmod{4}$. The

calculations in the proof of Lemma 2.11, using [70, Theorem 11.13 and Lemma 11.16], now give the analogues of (B.3) and (B.4) for these groups. The comparison of Fourier constants giving (B.6) and (B.7) also applies. These calculations establish the formulas in [41, (2.4)] and the compatibility of the constants $\zeta'_\mathcal{L}$ with cuspidal induction in [41, (3.3)(b)].

If $\mathbf{L}$ is not a torus, [68, Theorem 1.1] again supplies a character with unipotent support $\mathcal{D}$ and generic denominator $|A_{\mathbf{L}}(v)|$ for every F' -stable unipotent class $\mathcal{D}$ and $v \in \mathcal{D}$. The calculation of (B.8) therefore applies to these groups. In this sum, the unique nonzero multiplicity and its corresponding index both equal 1, as in the proof of Lemma 2.11. Expanding the modified class functions then gives the scalar products in [41, Proposition 3.1]. The torus case was included in that proof. Together with the analogues of (B.3) and (B.4) obtained above, this gives the integrality conclusion of [41, Corollary 3.2] for all these groups. Formula (B.9) consequently applies and gives $\zeta'_{\iota_0} = 1$, since $q' \equiv 1 \pmod{4}$. By (B.7), $\zeta'_\mathcal{L} = 1$ for every relevant generalised Springer block.

It remains to verify the support properties and compare character sheaf trace functions with ordinary characters. The support properties in [41, (4.3)] follow from [54, Theorem 10.7] by [70, Corollary 13.6]. For the groups above other than tori, Shoji's theorem [64, Theorem 3.2] gives the equality of spans in [41, Theorem 7.1], since their centres are connected and their quotients by their centres are simple of classical type. The equality of spans for tori was established in the proof of Lemma 2.11. Having verified the conditions in [41, (4.4)(a)–(b)], we may therefore apply the proof of [41, Theorem 4.5]. Restriction to $\mathcal{C}$, up to shift, gives a bijection between the character sheaves in the family indexed by $(\tilde{s}, \mathcal{F})$ with nonzero restriction to $\mathcal{C}$ and the irreducible $\mathbf{H}$ -equivariant local systems on $\mathcal{C}$.

We now return to F_0 and use this restriction bijection to show that the restrictions of the characters in $\text{Irr}_{\tilde{s}, \mathcal{F}}(H)$ span the H -invariant functions on $\mathcal{C}^{F_0}$. The multiplicities in the restrictions of character sheaves are independent of the rational structure, as explained after [41, Theorem 4.5]. Choose $u_0 \in \mathcal{C}^{F_0}$ as in [68, Proposition 2.4], with F_0 acting trivially on $A_{\mathbf{H}}(u_0)$. Since F_0 fixes $\tilde{s}$ and acts trivially on $W_{\tilde{s}}$, it preserves the chosen family of character sheaves. The trivial action on $A_{\mathbf{H}}(u_0)$ makes the local systems on $\mathcal{C}$ stable under F_0 . Since restriction commutes with pullback by F_0 , the restriction bijection makes the corresponding character sheaves stable under F_0 as well. The values at $u_1, \dots, u_d$ of the local system trace functions form the character table of $A_{\mathbf{H}}(u_0)$, up to multiplication of each row by a nonzero scalar [44, Section 3(A)]. The restriction bijection therefore shows that the corresponding character sheaf trace functions restrict to a basis of the H -invariant functions on $\mathcal{C}^{F_0}$. By Shoji's theorem [64, Theorem 3.2], these trace functions are nonzero scalar multiples of almost characters in the span of $\text{Irr}_{\tilde{s}, \mathcal{F}}(H)$. Thus there are $\chi_1, \dots, \chi_d \in \text{Irr}_{\tilde{s}, \mathcal{F}}(H)$ for which $M = (\chi_i(u_j))_{1 \leq i, j \leq d}$ is nonsingular, as in the proof of [42, Corollary 3.5].

It remains to select the characters with the required multiplicities. The calculations giving (B.1) and (B.2) in the proof of Lemma 2.11 apply to $\mathbf{H}$ with F_0 . They give linear independence of $D_H(\Gamma_{u_1})|_{\mathcal{C}^{F_0}}, \dots, D_H(\Gamma_{u_d})|_{\mathcal{C}^{F_0}}$ by the same change from the modified class functions to the individual generalised Gelfand–Graev characters. Since $j_{W_{\tilde{s}}}^W(E_0) = E_{\mathcal{C}}$, the description in [44, Section 3(C) and Theorem 3.7] shows that every $\rho \in \text{Irr}_{\tilde{s}, \mathcal{F}}(H)$ has unipotent support $\mathcal{C}$. The action of F_0 on $W_{\tilde{s}}$ is trivial and $\mathcal{G}_{\mathcal{F}}$ is abelian, so the generic denominator formula [41, (6.1)(c)] and the degree polynomial relation for Alvis–Curtis duality [45, Proposition 3.4.21] give

$$n_{\rho^*} = n_{\rho} = |\mathcal{G}_{\mathcal{F}}| = |A_{\mathbf{H}}(u_0)|.$$

Changing the rational representative from u_0 to u_j changes the action of F_0 on the component group by an inner automorphism of $A_{\mathbf{H}}(u_0)$. This group is abelian, so F_0 acts trivially on every $A_{\mathbf{H}}(u_j)$ and $[A_{\mathbf{H}}(u_j) : A_{\mathbf{H}}(u_j)^{F_0}] = 1$. By [70, Lemma 14.15], $\mathcal{C}$ is the wave front set of ρ^* . Apply [70, Proposition 15.4] to ρ^* and the constant local system on $\mathcal{C}$ with trace function

identically 1. The expansion [70, (11.15)] now gives

$$\sum_{j=1}^d \langle \rho^*, \Gamma_{u_j} \rangle_H = \frac{|A_{\mathbf{H}}(u_0)|}{n_{\rho^*}} = 1.$$

The multiplicities are nonnegative integers. Thus there is a unique index j for which $\langle \rho^*, \Gamma_{u_j} \rangle_H = 1$, and all the other multiplicities vanish. We must show that every index occurs for some $\rho \in \text{Irr}_{\mathcal{F}}(H)$.

Suppose that $\langle \rho^*, \Gamma_{u_r} \rangle_H = 0$ for every $\rho \in \text{Irr}_{\mathcal{F}}(H)$. Self-adjointness of D_H and $D_H(\rho) = \pm \rho^*$ give $\langle \rho, D_H(\Gamma_{u_r}) \rangle_H = 0$. As in the final argument in the proof of Lemma 2.11, the support calculation corresponding to (B.2) and the pointwise vanishing of ρ on larger unipotent classes show that only elements of $\mathcal{C}^{F_0}$ contribute to $\langle \rho, D_H(\Gamma_{u_r}) \rangle_H$. Taking $\rho = \chi_i$ for $1 \leq i \leq d$ in these equalities and using the nonsingularity of $M = (\chi_i(u_j))_{1 \leq i, j \leq d}$ gives $D_H(\Gamma_{u_r})|_{\mathcal{C}^{F_0}} = 0$, as in the final matrix calculation in the proof of Lemma 2.11. This contradicts the linear independence of the restricted functions. For each j , we may therefore choose ρ_j with $\langle \rho_j^*, \Gamma_{u_j} \rangle_H = 1$. All its other multiplicities vanish, proving the stated identities. $\square$

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OpenAI Codex was used to assist with the preparation of this manuscript (using GPT 5.6 Sol) and the development of the accompanying Lean formalisation (using GPT 5.6 Sol and GPT 6 Astra). The text generated by AI has been reviewed and revised. The formalisation contains more than two thousand Lean source files. This work has also produced a reusable Lean library for the ordinary and modular representation theory of finite groups, which continues to be expanded. The Lean sources and a report describing the formalised results, external assumptions and verification procedures will be made available through the [GitHub page](#) once the manuscript appears on arXiv. The Claude models Opus 5, Fable 5 and Fable 5.1 were also used to assist with the review and revision of the Lean files.

I would like to express my sincere appreciation to those who laid the foundations of the modern representation theory of finite groups and contributed to its development. The advances and corrections presented here build largely on methods and results established before the widespread use of AI. The Lean project initially aimed to develop, from basic foundations, the representation theory needed for arguments concerning Alperin's weight conjecture. A further motivation was to clarify the assumptions and dependencies of commonly used results and to help identify and resolve inaccuracies in the literature. It has been known for decades that inaccuracies in earlier work have caused further errors and misunderstandings in subsequent work. However, it remains unclear how many arguments in the foundational literature might benefit from further clarification or correction. Systematic investigation demands substantial time and effort and can be difficult to sustain alongside other research. We believe that AI assistance now makes such work more practicable and can deepen our understanding of the mathematics involved. AI assistance may also help resolve misunderstandings and correct misattributions in the literature by tracing results and arguments to their original sources. This is a valuable use of AI in mathematics, although the cost of sustained AI assistance remains a significant constraint. After weeks of cumulative working time recorded by Codex for the formalisation project, it became clear that pursuing this aim would still require considerable time to identify and formalise a substantial body of prerequisite theory, even with AI assistance. For example, we estimate that formalising Deligne–Lusztig theory alone would require several people working for at least several months. This illustrates the scale of the work involved in formalising the background needed for modern modular representation theory. The present formalisation therefore checks selected arguments under explicitly stated external assumptions, while work continues on the underlying foundations and on extending the Lean library. The reusability of the library's constructions has been tested in ongoing work on formalising the much longer proofs concerning the iBAW condition for the remaining families of types D and E, for which the problem was previously

largely open. The Lean library is also being used to formalise and check proofs of new results on other problems of current interest in the modular representation theory of finite groups.

Using AI in the modular representation theory of finite groups has also provided insights into how to approach mathematical problems. I regard this as a meaningful exercise because modern modular representation theory rests on many layers of prerequisite theory, with very long chains of arguments connecting its results to the basic foundations of mathematics. In my experience, these very long chains of dependencies make it harder for current AI models to develop and test new theory here than in areas whose arguments lie closer to the basic foundations of logic. In experiments involving hundreds of billions of tokens, I have found that AI models, when tackling some problems directly, repeatedly return to the same approaches without making progress, whereas developing a suitable mathematical architecture, with the intermediate results and their dependencies organised in an effective order, can lead to rapid advances. These experiments suggest that the order in which related conjectures are approached matters. Establishing one key conjecture may make problems that previously resisted AI assistance substantially more accessible to it. When appropriate, I hope to share these observations and strategies for approaching proofs of the remaining major conjectures in modular representation theory after Alperin’s weight conjecture. Writing out all these arguments in full detail could require hundreds of billions of additional tokens and result in more than a thousand pages of proofs, making it an impractical undertaking for one person. Sharing the proposed strategies would allow mathematicians with greater expertise in these conjectures than I have to assess and develop them.

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